

Distribution-independent SQ learning does not imply low dimension complexity

A negative answer to Open Question 2 of Feldman, Kamath and Srebro

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Abstract

Open Question 2 of Feldman, Kamath and Srebro has a negative answer in all three proposed forms and both nondegenerate probabilistic variants. If every product distribution has a monochromatic rectangle of mass at least $1/\rho$, a common statistical-query learner uses $O(\rho(\log |H| + \log(1/\epsilon)))$ queries at tolerance $\Omega(\epsilon/(\rho \log(1/\epsilon)))$. Incidence flips on $2M$ points have $O_\epsilon(\log M)$ -query learners but ordinary dimension $\Omega(M^{1/3}/\log M)$, exact probabilistic dimension $\Omega(M^{1/3}/\log^2 M)$, and averaged probabilistic dimension $\Omega_\eta(M^{1/3}/\log M)$ for fixed $\eta < 1/2$. The learners can be proper, deterministic and polynomial-time in the table; under subexponential PRF security, key-known polynomial-time learners use $O_\epsilon(n^2)$ queries while almost every key has dimension at least $n^c/(30 \log_2 n)$ for any prescribed fixed c . Typical flips require $\Theta_\epsilon(n)$ queries at an accuracy-dependent fixed tolerance, with a sharper lower bound for proper outputs. The benign online-SGD question remains open because the SQ-to-gradient simulation engineers the architecture and initialization and charges the simulated computation, while hidden-table lower bounds concern class-independent learners.

Summary of results

Statement	Status	Reference
$\text{rect}(H) \geq 1/\rho$ gives $O(\rho(\log K + \log(1/\epsilon)))$ queries and tolerance $\Omega(\epsilon/(\rho \log(1/\epsilon)))$.	Unconditional	Theorem 1
Incidence flips have dimension $\Omega(N/\log M)$ and constant inverse rectangle ratio.	Unconditional	Theorem 4
Exact and averaged probabilistic dimension remain $\Omega(N/\log^2 M)$ and $\Omega_\eta(N/\log M)$.	Unconditional	Theorem 5 and Lemma 6
All three proposed SQ-to-dimension implications fail.	Unconditional	Corollary 7
Typical flips require $\Theta_\epsilon(n)$ queries; the leading proper lower bound is $[n - O_\epsilon(1)]/\log_2 \lceil 1/\tau \rceil$.	Unconditional	Theorem 8
Caps give deterministic table-polynomial implementations; Proper RECTIFY returns a member.	Unconditional	Theorems 9 and 10
Succinct evaluation gives a deterministic improper $O(n^2)$ -query learner; regular stars give proper learning.	Unconditional	Theorem 12
For each fixed c , almost every polynomial-length key has dimension at least $n^c/(30 \log_2 n)$.	PRF assumption	Theorem 14
Almost every key has a key-known proper polynomial-time learner.	PRF assumption	Corollary 17
Polynomial advice gives almost-every-advice high dimension and proper $O_\epsilon(\log n)$ -query learning.	Unconditional	Proposition 15
Engineered polynomial-size full-batch gradient models learn every key; high dimension holds for almost every key.	High dimension: PRF	Corollary 18
Bounded product-average communication implies SQ learnability; halfspaces show non-necessity.	Unconditional	Theorems 24 and 32
Hidden-table sample bounds require a table-independent learner.	Unconditional	Proposition 19
Benign online SGD, efficient deterministic selection, succinct $O(n)$ queries, and the optimal (m, τ) tradeoff.	Open	Section 8

The qualifier “typical” means outside the explicit exceptional probabilities in Theorems 4, 5 and 8. The polynomial-time key statements fix c before the asymptotic limit. Exact constants and source-to-proof checks appear in the appendices and accompanying verification ledger.

1 Introduction

Feldman, Kamath and Srebro [17, Open Question 2] ask:

Open Question 2 (SQ Learning vs. Dimension Complexity) Is there a constant C such that for every class $H \subseteq \{\pm 1\}^{\mathcal{X}}$, over any domain $\mathcal{X}$, and any $\epsilon < 1/4$, if there exists an (m, τ) -SQ algorithm s.t. for every input distribution D and every $h^* \in H$, the algorithm returns a predictor $\hat{h}$ with $\mathbb{E}L_{D, h^*}(\hat{h}) \leq \epsilon$ (expectation over the randomness of the algorithm), then $\text{dc}(H) \leq C \cdot m/\tau^2$.

The answer is negative for finite domains, including the proposed relaxations

$$\text{dc}(H) \leq Cm/\tau^2 \text{ (L)}, \quad \text{dc}(H) \leq p(m, 1/\tau) \text{ (P)}, \quad \text{dc}(H) \leq p(m, 1/\tau, n) \text{ (Pn)}, \quad (1.1)$$

where $n = \log_2 |X|$ and C, p are universal. The two nondegenerate probabilistic variants fail as well.

A fixed linear representation must realize every target’s signs simultaneously. A learner can first ask where the marginal places mass and then label an agreed region or eliminate inconsistent targets. Large monochromatic rectangles under every product distribution yield a marginal-free mixture with pointwise coverage. RECTIFY charges either update to the product of the rectangle’s two masses. Random incidence flips preserve the required rectangles while counting forces large sign-rank.

Contributions.

- A common SQ learner with $O(\rho(\log K + \log(1/\epsilon)))$ queries at tolerance $\Omega(\epsilon/(\rho \log(1/\epsilon)))$ whenever $\text{rect}(H) \geq 1/\rho$.
- A refutation of (L), (P), and (Pn) for ordinary and probabilistic dimension, with proper deterministic table-polynomial learners.
- A succinct $O(n^2)$ -query learner and almost-every-key polynomial-time separations under the stated PRF hypothesis.
- Matching $\Theta_\epsilon(n)$ query order for typical flips, including a sharper proper-output lower bound.
- Communication sufficient conditions, spectral and information barriers, and an engineered gradient-learning consequence.

Imported ingredient	Role
Incidence construction and minimax [24]	Large rectangles with many independently variable signs; the mixture device.
Warren’s theorem [5, 33]	Counting strict low-rank sign patterns.
Homogeneous rectangles [4, 24]	The constant 2^{-15} ; Proposition 39 supplies the independent 2^{-18} fallback.
SQ and gradient simulations [2, 15]	Separate comparisons; the main rectangle learner is proved directly.

Table 1: Imported ingredients. The oracle analysis, proper and succinct learners, probabilistic reductions, and query lower bounds are proved here.

Concurrent work. VALG, arXiv:2608.13060v2 (10 September 2026), defines the span of seed-averaged outputs over deterministic complete response rules. Its Assumption 4.42 bounds that dimension by $B(1 + m/\tau^2)^k$; Theorem 4.9 derives a common strict representation under Assumptions 4.39–4.42 [34]. Our separating family shows that this additional polynomial response-rank condition cannot follow from the unrestricted common-learning guarantee. The results have different hypotheses.

Scope. All probabilistic-dimension conclusions use a nondegenerate Boolean tie convention. Table-polynomial time is distinguished from polynomial time in n given a key. High-probability key and advice statements do not select a deterministic good description. The PRF results assume the stated security; model-size exponents may depend on the prescribed degree c . The gradient simulation uses engineered architecture and initialization and charges the simulated computation. Exact-arithmetic tests cover finite identities, endpoint conditions and recorded experiments, not every infinite-instance theorem. Archived formalization drafts are not machine-checked proofs. The uniform hidden-table bounds do not restrict class-dependent architecture choices in OQ1. All remaining research questions are collected in Section 8.

Sections 3–6 give the separation and implementations; Section 7 gives the gradient and information consequences. The communication, spectral, geometric, halfspace and statistical-dimension comparisons, full proofs, and experiments follow in the appendices.

2 Definitions and notation

Let X be finite and nonempty and let $H = \{h_1, \dots, h_K\} \subseteq \{-1, +1\}^X$ be a nonempty indexed class. Repeated rows are permitted. Write

$$L_{D,h}(g) = \Pr_{x \sim D}[g(x) \neq h(x)], \quad \mathbb{E}_D[hg] = 1 - 2L_{D,h}(g).$$

All predictors are Boolean unless a score and its threshold rule are specified. Unsubscripted logarithms in asymptotic bounds may use either base.

Symbol	Meaning
n	$\log_2 X $, the domain encoding length
N, M, K	Grid scale, $M = N^3$, and $K = 2M$ rows and points for incidence flips
ρ, r	Inverse rectangle parameter and $r = 1/\rho$
τ	SQ tolerance
ϵ	Target classification error, $0 < \epsilon < 1/4$
η	Approximation level in probabilistic dimension, $0 \leq \eta < 1/2$
ζ	Effective median-query error, $\epsilon/4096$

Oracle and learner. A query $q : X \times \{-1, +1\} \rightarrow [-1, 1]$ receives any v with $|v - \mathbb{E}_D q(x, h(x))| \leq \tau$ [28]. A valid policy may use the full public query-answer history, including queries that reveal past private choices, but not unrevealed future coins. A depth- m learner makes at most m adaptive queries and satisfies

$$\forall D \forall h \in H \forall \text{ valid policies } O, \quad \mathbb{E}_\omega L_{D,h}(A_\omega^O) \leq \epsilon. \quad (2.1)$$

The seed is independent of the instance. Early stopping can be padded by zero queries. Randomized policies are handled by conditioning on their randomness. Our finite-state algorithms are measurable. Computation is uncharged unless stated; computational results use finite binary answers and reserve half the tolerance for rounding. Deterministic guarantees are pathwise.

Dimension. The dimension complexity $\text{dc}(H)$ is the least d for which one $\phi : X \rightarrow \mathbb{R}^d$ and row weights w_h satisfy

$$h(x) \langle w_h, \phi(x) \rangle > 0 \quad (x \in X, h \in H). \quad (2.2)$$

It is the sign-rank of the truth table: a rank factorization and a common feature representation are the same object [7].

For probabilistic dimension fix $\text{sign}(0) = +1$. A single embedding law, independent of D, h , witnesses $\text{dc}_{1/2}(H) \leq d$ when

$$\forall D, h, \quad \Pr_{\phi}[\exists w : L_{D,h}(\text{sign}\langle w, \phi \rangle) = 0] \geq 1/2.$$

It witnesses $\text{dc}^{\eta}(H) \leq d$ when

$$\forall D, h, \quad \mathbb{E}_{\phi} \inf_w L_{D,h}(\text{sign}\langle w, \phi \rangle) \leq \eta.$$

The source attributes these variants to Chornomaz et al. [10] and Kamath et al. [26]. Its literal raw-score loss assigns zero loss to $w = 0$; the definitions above fix a Boolean tie rule. Finiteness makes the infimum a minimum. Lemma 23 gives the same lower bounds for the opposite fixed tie rule and when zero scores count as errors.

Rectangles. A rectangle $S \times T \subseteq X \times H$ has color c when $h(x) = c$ on it. Define

$$\text{rect}(H) = \inf_{\mu \in \Delta(X), \nu \in \Delta(H)} \max_{S \times T \text{ monochromatic}} \mu(S)\nu(T).$$

Restricting a matrix cannot decrease this ratio, by extending weights by zero. Lemmas 21 and 22 record the finite separation and rational-weight reductions used below.

3 The rectangle theorem and RECTIFY

Theorem 1 (Distribution-independent rectangle learner). *Let X, H be finite and nonempty, $K = |H|$, and suppose $\rho \geq 1$ and $\text{rect}(H) \geq 1/\rho$. For every $0 < \epsilon < 1/4$, one randomized algorithm satisfies (2.1) with*

$$m = O(\rho(\log K + \log(1/\epsilon))), \quad \tau = \Omega\left(\frac{\epsilon}{\rho \log(1/\epsilon)}\right).$$

Query count is a worst-case bound. The learner is explicit from the truth table and a valid lower bound on $\text{rect}(H)$; the general implementation may solve exponential-size linear programs.

Lemma 2 (Exact parameters). *For a finite nonempty H , $\rho \geq 1$, $\text{rect}(H) \geq 1/\rho$, and $0 < \epsilon < 1/4$, Theorem 1 holds with*

$$r = 1/\rho, \quad P = 1 + \lceil 4\rho \ln(8/\epsilon) \rceil, \quad \tau = \epsilon/(24P), \quad (3.1)$$

$$B = \ln K + \ln(8/\epsilon) + 1, \quad R = \lceil 8\rho(B + \ln(2/\epsilon)) \rceil, \quad m = 3R + 1. \quad (3.2)$$

Lemma 3 (A marginal-free rectangle mixture). *Assume H is finite, $0 < r \leq \text{rect}(H)$, and fix nonempty $U \subseteq X$, nonempty $V \subseteq H$, and a distribution ν on V . There is a distribution λ over monochromatic rectangles $S_R \times T_R \subseteq U \times V$ such that*

$$\forall x \in U, \quad \mathbb{E}_{R \sim \lambda}[\mathbf{1}_{S_R}(x)\nu(T_R)] \geq r. \quad (3.3)$$

This distribution can be computed from (H, U, V, ν) without a query about D .

Proof of Lemma 3. For each rectangle R , let $z_R(x) = \mathbf{1}_{S_R}(x)\nu(T_R)$. The rectangle hypothesis gives $\max_R \langle \mu, z_R \rangle \geq r$ for every $\mu \in \Delta(U)$. The finite separation lemma, Lemma 21, gives a convex combination with every coordinate at least r . This is the required law. Rational inputs admit a rational feasible point. $\square$

Algorithm 1: RECTIFY: rectangle peeling and elimination

Input: A table H , a bound $r \leq \text{rect}(H)$, and $0 < \epsilon < 1/4$.

```
1 Set  $U \leftarrow X$ ,  $V \leftarrow H$ , and use  $P, \tau, R$  from Lemma 2;
2 for  $t = 1, \dots, R$  do
3   if  $U = \emptyset$  then
4     return the partial predictor;
5   Query  $\hat{u} \approx D(U)$ ;
6   if  $\hat{u} \leq \epsilon/4$  then
7     return the partial predictor, extended by +1 on  $U$ ;
8   With  $\nu$  uniform on  $V$ , draw  $(S, T, c)$  from Lemma 3;
9   Query  $\hat{s} \approx D(S)$ ;
10  if  $\hat{s} < (r/2)\hat{u}$  then
11    Reject this proposal and continue;
12  Query  $\hat{w} \approx D(S \cap \{h^* \neq c\})$ ;
13  if  $\hat{w} \leq 2\tau$  then
14    Set the predictor to  $c$  on  $S$ ;  $U \leftarrow U \setminus S$ ;
15  else
16     $V \leftarrow V \setminus T$ ;
17 return the partial predictor, extended by +1 on  $U$ ;
```

Proof of the query bound. Fix the instance and an arbitrary valid oracle policy. At a nonstopping round, $u = D(U) \geq \epsilon/8$. Write $a = D(S)/u$ and $b = \nu(T)$. Acceptance forces $a \geq r/4$, and $a \geq 3r/4$ guarantees acceptance at both adverse answer endpoints. A peel has wrong-color mass at most 3τ . An elimination retains the target and increases its normalized weight by $1/(1-b)$.

The cumulative eliminated b -mass is at most $\ln K$. There are at most P peels, and their cumulative a -mass is at most $\ln(8/\epsilon) + 1$. Set $G = a$ for a peel, $G = b$ for an elimination, and $G = 0$ on rejection. Pointwise coverage and the acceptance thresholds give

$$\mathbb{E}[G \mid \text{public history}] \geq r/4, \quad \sum G \leq B$$

until termination. The exponential potential bounds budget exhaustion by $e^{B-rR/8} \leq \epsilon/2$. Normal termination has loss at most $\epsilon/2$. The complete argument, including the oracle endpoints and the final peel of mass one, is in Appendices A and D.

The same theorem gives $m = O(\rho^2 \log(e|X|) + \rho \log(1/\epsilon))$: the rectangle condition bounds VC dimension by $\rho/2$, and the finite trace count bounds $\log K$. The proof is in Appendix D. Figure 1 depicts the two updates.

4 The separating family

Let $N \geq 2$, $M = N^3$ and $K = 2M$. Points and lines are indexed by $[N] \times [2N^2]$. The template F is negative exactly when $y = ax + b$. A monotone flip changes any subset of these incidences to +1, leaving all other signs positive. Write H_A for the row class. The two panels in figure 2 show this operation at $N = 2$.

Theorem 4 (Construction and ordinary dimension). *Every monotone flip A satisfies $\text{VC}(A) \leq 2$ and $\text{rect}(A) \geq 2^{-15}$. The template has $\text{signrank}(F) \leq 6$ and exactly*

$$I = 2N^4 - \left(\frac{N(N+1)}{2} \right)^2 \geq N^4 \tag{4.1}$$

independent flip positions. Set $B_0 = \log_2(4eK)$ and $s_0 = \lfloor I/(4KB_0) \rfloor$. A random A satisfies

$$\Pr[\text{dc}(H_A) \leq s_0] \leq 2^{-I/2}.$$

Every member also satisfies $\text{dc}(H_A) \leq 2N + 1$. In particular most matrices have $\text{dc}(H_A) = \Omega(N/\log M)$. For every $0 < \epsilon < 1/4$, all members have a learner with $m = O_\epsilon(\log M)$ and $\tau = \Omega_\epsilon(1)$, by Theorem 1 with $\rho = 2^{15}$. Independently of the sharp rectangle input, the self-contained cap proof in Appendix V supplies $\rho = 2^{18}$ and the same asymptotic learning bound.

The proof is in Appendix F.

Theorem 5 (Approximate counting and exact probabilistic dimension). *Fix $0 \leq \eta < 1/2$ and write*

$$h_2(\eta) = -\eta \log_2 \eta - (1 - \eta) \log_2(1 - \eta), \quad c_\eta = 1 - h_2(\eta) > 0, \quad s_\eta = \left\lfloor \frac{c_\eta I}{4KB_0} \right\rfloor,$$

with $0 \log 0 = 0$. For the random incidence-flip matrix,

$$\Pr[\text{dc}^\eta(H_A) < s_\eta] \leq 2^{-c_\eta I/2}. \quad (4.2)$$

Consequently most A have $\text{dc}_{1/2}(H_A) = \Omega(N/\log^2 M)$ and $\text{dc}^\eta(H_A) = \Omega_\eta(N/\log M)$.

The proof is in Appendix F.1.

Lemma 6 (From exact probabilistic to strict dimension). *For every finite nonempty class H with $K = |H|$,*

$$\text{dc}(H) \leq (\lceil \log_2 K \rceil + 1) \text{dc}_{1/2}(H) + 1. \quad (4.3)$$

The proof is in Appendix F.2.

Corollary 7 (Negative answers to the dimension implications). *For every fixed $0 < \epsilon < 1/4$, none of (L), (P) and (Pn) holds universally with $\text{dc}(H)$ or $\text{dc}_{1/2}(H)$ on the left. For every fixed $C > 0$, choose*

$$0 < \epsilon < \min\{1/4, 1/(2C)\}, \quad \eta = C\epsilon.$$

The same three conclusions fail with $\text{dc}^{C\epsilon}(H)$ on the left, under each nondegenerate tie convention of Lemma 23. The counterexamples have proper deterministic, table-polynomial learners with $m = O_\epsilon(\log M)$ and $\tau = \Omega_\epsilon(1)$.

The corollary follows by fixing $\eta = C\epsilon < 1/2$ and intersecting the counting events; see Appendix F.3. The template has six features, but its I independent incidence bits survive the monotone operation. For averaged probabilistic dimension, the counting proof weights each nonempty row by its number of template incidences and tests it on that support. A uniform loss over all entries would hide these sparse bits.

Theorem 8 (Optimal query order for random incidence flips). *Put $\chi = 1 - h_2(3/8) > 0$. Suppose $N \geq 25$ and $\chi N \geq 6 \log_2 N$. With probability at least $1 - 2^{-2N^3} - N^3 e^{-N/32}$ over independent fair flips, the following holds simultaneously for all $0 < \epsilon < 1/4$ and $0 < \tau < 1$. Every randomized depth- m SQ learner with expected error at most ϵ on every target-line instance (D_ℓ, h_ℓ) , $\ell \in [N] \times [N^2]$, obeys*

$$m \geq \frac{\log_2 N + \log_2(\chi(1 - 8\epsilon/3)/8)}{\log_2 \lceil 1/\tau \rceil}.$$

Here D_ℓ is uniform on the line's N template points, and the guarantee must hold against every valid oracle. If the learner is proper, then

$$m \geq \frac{\log_2(2M) - 1 + \log_2(1 - 3\epsilon)}{\log_2 \lceil 1/\tau \rceil}.$$

For every fixed ϵ , there is a fixed tolerance $\tau_\epsilon > 0$ at which both proper and unrestricted query complexity are $\Theta_\epsilon(n)$ for matrices in this event. The $O_\epsilon(n^2)$ succinct learner is within a factor n in query order.

The proof is in Appendix G. It counts outputs that fit multiple target-line instances before applying the response-grid bound. The lower bound concerns typical random flips; the all-positive member has a zero-query learner.

5 Efficient, proper and succinct learners

5.1 Caps and proper learning

A strict low-rank template supplies geometric rectangles without enumerating every rectangle. The no-negative- 2×2 condition ensures that any negative template rectangle has a singleton side.

Theorem 9 (Pointwise cap mixture and table-polynomial implementation). *Fix $d \geq 1$. Let F have a strict rank- d representation and no negative 2×2 , and let A be any monotone flip of its negative entries. Perturb the row vectors once into linear general position without changing F . At any nonempty survivor set, use the uniform row prior ν . There is a cap sampler, depending on the surviving rows but not on the unknown marginal, such that*

$$\forall x \in U, \quad \mathbb{E}[\mathbf{1}_S(x)\nu(T)] \geq \frac{c_d}{8d}, \quad c_d = \Pr_{u \sim \text{Unif}(S^{d-1})}[u_1 \geq \sqrt{1 - 1/(2d)}]. \quad (5.1)$$

Here $c_1 = 1/2$. Thus Theorem 1 applies with $\rho = 8d/c_d$ in the ideal real-arithmetic model.

For each fixed d , given the explicit table of A and rational template vectors of bit length at most b , there are randomized and deterministic implementations, polynomial in $|X|, |H|, b, 1/\epsilon$, for every $0 < \epsilon < 1/4$, with

$$m = O_d(\log |H| + \log(1/\epsilon)), \quad \tau = \Omega_d(\epsilon/\log(1/\epsilon)).$$

Queries are intersections of stored cap tests, or such tests split by one row label or one point's row labels. The running time is polynomial in the supplied table size.

The proof is in Appendix I.1.

The sampler perturbs the template once, uses radial isotropy for the uniform surviving-row prior, and selects the heavier of two threshold sides. A negative rectangle is split by its singleton side's labels in A . The finite net enumerates a constant number of outcomes for fixed dimension. The complete procedure and geometric figure are in Appendix H.

5.2 Proper completion

Theorem 10 (Proper RECTIFY). *For every finite nonempty class H and $0 < \epsilon < 1/4$, suppose H contains the all-positive hypothesis and has no negative 2×2 . Given a rectangle sampler of coverage r , it has a distribution-independent proper SQ learner of expected error at most ϵ , using*

$$m_{\text{proper}} \leq m_{\text{RECTIFY}}(\epsilon/4) + \lceil \log_2 |X| \rceil + \lceil \log_2 K \rceil + \lceil 8/\epsilon \rceil + 4 \quad (5.2)$$

queries at the tolerance of RECTIFY with accuracy $\epsilon/4$. For the incidence flips,

$$m_{\text{proper}} = O(\log M + \epsilon^{-1} + \log(1/\epsilon)), \quad \tau = \Omega(\epsilon/\log(1/\epsilon)).$$

The cap implementation is polynomial-time in the explicit table. Its finite-net deterministic variant is also proper and has the same asymptotic query bound, with constants depending on the template dimension.

A failed test of a negative rectangle's singleton row forces its target-negative mass onto the unique intersection point. Outside that point, the surviving rows' negative supports are disjoint. This gives Theorem 10; see Appendix I.2.

Algorithm 2: Proper RECTIFY

```
1 Run RECTIFY at accuracy  $\epsilon/4$ ; interrupt at the first negative peel;
2 if none occurs then
3   return the all-positive member;
4 Use the singleton point, or test the singleton row at loss threshold  $\epsilon/2$ ;
5 if the row passes then
6   return that row;
7 Locate its unique heavy negative target point  $p$  by binary search;
8 Query negative target mass outside  $p$ ;
9 if the answer exceeds  $\epsilon/4$  then
10  | Binary-search the rows negative at  $p$  by their disjoint outside negative supports;
11 else
12  | Test up to  $\lceil 8/\epsilon \rceil + 1$  distinct such rows and return one whose loss answer is at most  $\epsilon/2$ ;
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5.3 The succinct order-median learner

For every query, reserve half the raw tolerance for a finite answer encoding. With raw tolerance $\epsilon/8192$ and $\zeta = \epsilon/4096$, effective error is at most ζ . Let $\mu^-(S) = D(S \cap \{h^* = -1\})$ and $\beta = \mu^-(X)$.

Lemma 11 (Noisy integer medians). *Suppose $\zeta \geq 0$, $\beta > 0$, $|\hat{\beta} - \beta| \leq \zeta$, and $\beta \geq 512\zeta$. Let f be integer-valued in $[L, H]$ on the negative support. Binary search using the query $\mu^-\{f \leq t\}$ and threshold $\hat{\beta}/2$ returns $v \in [L, H]$ with*

$$\mu^-\{f \leq v\} \geq \beta/2 - 3\zeta/2, \quad \mu^-\{f < v\} < \beta/2 + 3\zeta/2.$$

If the answer $\hat{m}$ to $m = \mu^-\{f = v\}$ is at most $\hat{\beta}/8$, then

$$\beta/2 - 3\zeta/2 \leq \mu^-\{f \leq v\} < 5\beta/8 + 21\zeta/8. \quad (5.3)$$

Otherwise $m > \beta/9$.

The proof is in Appendix J.1.

For $p = (p_x, p_y)$ the template star has slopes $1 \leq a \leq t(p) := \min\{N, \lfloor (p_y - 1)/p_x \rfloor\}$ and intercept $p_y - ap_x$. Call A *star-regular* when every star of size at least $64\lceil n \rceil$ has at least one quarter of its rows negative at its center.

Theorem 12 (Succinct median learner). *Let $0 < \epsilon < 1/4$. Suppose entries of a real-grid incidence flip A can be evaluated in time $E(n)$, where $n = \log_2(2N^3)$. There is a deterministic improper distribution-independent SQ learner of error at most ϵ , using $O(n^2)$ queries at raw tolerance $\epsilon/8192$, with total computation polynomial in $(n, 1/\epsilon, E(n))$ and an efficiently evaluable output. If A is star-regular, there is a proper randomized learner with*

$$m = O(n^2 + 1/\epsilon + \log(1/\epsilon)), \quad \tau = \epsilon/8192,$$

with the same polynomial-time property and expected error at most ϵ against every valid oracle.

At a trial slope a , the residual $y - ax = (a^* - a)x + b^*$ increases or decreases along the target. A fixed balanced prefix and a residual-median set therefore have intersection mass near at least $\beta/2$ or at most $\beta/4$. Their oracle answers distinguish the two cases at threshold $3\hat{\beta}/8$. A heavy residual atom instead identifies the line or one negative point. Each slope stage uses $O(\log N)$ median queries; there are $O(\log N)$ stages. The full procedure and proof are in Appendix J.

6 Conditional explicit families

Cryptographic assumption. Fix a constant $\delta > 0$, decreasing it to be less than one if necessary. Assume a polynomial-time bit-output pseudorandom function family $\{f_s\}_{s \in \{0,1\}^\sigma}$, defined on inputs long enough to encode a grid row and point, such that every oracle algorithm running in time at most 2^{σ^δ} has negligible distinguishing advantage between f_s for uniform s and a uniform random function. Negligibility is in σ . We use this assumption directly. The one-way-function, pseudorandom-generator and pseudorandom-function constructions of Goldreich et al. [20], Håstad et al. [23] provide background for this assumption. The quantitative security hypothesis above is assumed directly; the lower bounds below do not require a separate quantitative reduction from one-way functions.

For $n = 3b + 1$ and $N = 2^b$, encode a row–point pair by

$$E_b(a, b', x, y) = \text{bin}_b(a - 1) \parallel \text{bin}_{2b+1}(b' - 1) \parallel \text{bin}_b(x - 1) \parallel \text{bin}_{2b+1}(y - 1).$$

This injection has length $6b + 2 = 2n$; append zero padding to the prescribed PRF input length. Every subgrid query uses this same ambient encoding E_b , rather than an encoding at the smaller scale. Off incidences set $A_s = +1$; on incidences set $A_s = 2f_s(E_b(a, b', x, y)) - 1$.

Lemma 13 (Small subgrid counting and decision cost). *At grid scale $k \geq 2$, put $K_k = 2k^3$ and*

$$I_k = 2k^4 - \left(\frac{k(k+1)}{2} \right)^2, \quad D_k = \left\lfloor \frac{k}{24 \log_2 k} \right\rfloor.$$

Under independent fair flips,

$$\Pr[\text{signrank}(A) \leq D_k] \leq 2^{-k^4/2}.$$

Whether $\text{signrank}(A) \leq D_k$ can be decided after reading at most $I_k \leq 2k^4$ oracle bits, in time $2^{O(k^3 D_k \log k)} = 2^{O(k^4)}$ when $D_k \geq 1$. When $D_k = 0$, the event is empty and is decided immediately.

The proof is in Appendix K.1.

Theorem 14 (Conditional almost-every-key explicit lower bounds). *Assume the stated PRF security for a fixed $\delta > 0$. Fix $c \geq 1$, $0 < \epsilon < 1/4$, $n = 3b + 1$, and put*

$$\sigma = n^{\lceil (4c+2)/\delta \rceil}.$$

For every key, entries of A_s and hypotheses in H_{A_s} are evaluable in $\text{poly}_c(n)$ time. Every key has RECTIFY and proper learners with $O_\epsilon(n)$ queries and tolerance $\Omega_\epsilon(1)$. For all sufficiently large admissible n , outside a negligible fraction of keys,

$$\boxed{\text{dc}(H_{A_s}) \geq \frac{n^c}{30 \log_2 n}}.$$

Here the learner with $O(n)$ queries may use table-polynomial computation. A succinct polynomial-time implementation with $O(n^2)$ queries is established in Section 5.3.

The proof is in Appendix K.2.

Proposition 15 (Unconditional polynomial advice). *Fix $c \geq 1$ and $0 < \epsilon < 1/4$. For all sufficiently large $n = 3b + 1$, there is a family of incidence-flip classes on 2^n points indexed by advice strings of length $O_c(n^{4c})$. Every entry is evaluable in $\text{poly}_c(n)$ time. For all but a $2^{-\Omega_c(n^{4c})}$ fraction of advice strings,*

$$\text{dc}(H) \geq \frac{n^c}{30 \log_2 n}.$$

Every advice string admits a proper deterministic learner running in $\text{poly}_{c,\epsilon}(n)$ time, with $O_{c,\epsilon}(\log n)$ queries at tolerance $\Omega_\epsilon(1)$.

The proof is in Appendix K.5.

Lemma 16 (Regular stars for almost every key). *Use the star-regularity definition in Section 5.3: every template star with at least $64\lceil n \rceil$ rows has at least one quarter of its rows negative at its center. This property is independent of ϵ ; the algorithm tests it only for stars larger than $C_0 = \max\{64\lceil n \rceil, \lceil 128/\epsilon \rceil\}$. For independent fair incidence flips, the probability of failure of star-regularity is at most e^{-3n} , where $n = \log_2(2N^3)$. Under the PRF assumption and parameters of Theorem 14, failure occurs for only a negligible fraction of keys.*

The proof is in Appendix K.3.

Corollary 17 (Conditional polynomial-time, seed-known separation). *Under the PRF assumption, for every fixed $c \geq 1$ and $0 < \epsilon < 1/4$, all but a negligible fraction of keys in Theorem 14 define classes with*

$$\text{dc}(H_{A_s}) \geq \frac{n^c}{30 \log_2 n}, \quad m = O_\epsilon(n^2), \quad \tau = \Omega_\epsilon(1),$$

and a proper learner running in $\text{poly}_{c,\epsilon}(n)$ time given the key. Every key has the corresponding deterministic improper polynomial-time learner, independently of star-regularity.

The proof is in Appendix K.4.

For the $O_\epsilon(n)$ -query table learner, choosing c above a prescribed polynomial degree refutes that degree in $(m, 1/\tau, n)$. For the succinct $O_\epsilon(n^2)$ -query learner, choose c above twice the degree. Taking $k = n^{\log_2 n}$ and $\sigma = \lceil k^{6/\delta} \rceil$ gives superpolynomial dimension with quasipolynomial descriptions by the same reduction.

7 Gradient learning and class information

The full-batch simulation of Abbe, Kamath, Malach, Sandon and Srebro [2, Theorem 3d] converts a time-bounded SQ learner into an engineered network. It uses their two-stage-ramp activation, a prescribed initialization map, square loss, one reused sample, and coordinatewise clipped, rounded gradients. Its guarantee places a supremum over admissible gradient sequences inside the expectation over the sample and initialization. Appendix O states the imported model and every parameter used below.

Corollary 18 (Engineered differentiable learning with high dimension). *Assume the PRF hypothesis of Theorem 14. Fix $c \geq 1$ and $0 < \epsilon < 1/4$. For every sufficiently large admissible encoding length n and every key s , the class H_{A_s} , encoded on $\{0, 1\}^n$, has an engineered neural network and initialization of the form specified in Appendix O, each describable and evaluable in $\text{poly}_{c,\epsilon}(n)$ time, with $\text{poly}_{c,\epsilon}(n)$ parameters. After $O(n^2)$ full-batch updates using $\text{poly}_{c,\epsilon}(n)$ examples and the finite-precision gradient rule (O.1), thresholding the final output at $1/2$ has expected classification error at most ϵ for every marginal and every target in H_{A_s} . For all but a negligible fraction of keys,*

$$\text{dc}(H_{A_s}) \geq \frac{n^c}{30 \log_2 n}.$$

The model and initialization may depend on s , but are fixed before the marginal and target.

The proof is in Appendix O.1.

The source learner is the deterministic improper algorithm of Theorem 12 at accuracy $\epsilon/8$. Its polynomial query-evaluation time, rather than table enumeration, enters the model-size bound. After recoding to $\{0, 1\}$, square loss is half classification loss. Simulation slack $\epsilon/16$ gives square loss at most $\epsilon/8$; threshold errors cost at least $1/8$ each.

The prescribed online process. Open Question 1 fixes a fully connected bias-free ReLU network, independent layer- i Gaussian weights of variance $1/n_{i-1}$, logistic loss, one step size, T fresh-sample updates, and the sign of the tail-sum output. The architecture, step size and horizon may depend on the class. Its expectation includes initialization and sampling. The full definition is retained in Appendix N; the remaining question is listed in Section 8.

Information about the class.

Proposition 19 (Unseen independent labels). *Fix the full-length template line $\ell_0 : y = x + 1$ and the uniform distribution on its N points. Choose A by independent fair incidence flips. Let a single learner, chosen independently of A , receive T independent labeled samples from (D, h_{ℓ_0}) and no further information about A . Then*

$$\mathbb{E}_{A, \text{sample}, \text{coins}} L_{D, h_{\ell_0}}(\hat{h}) \geq \frac{1}{2}(1 - 1/N)^T \geq \frac{1}{2}(1 - T/N).$$

Thus a guarantee of average error at most ϵ , or a guarantee for every flip table, requires $T \geq (1 - 2\epsilon)N$. If $\epsilon < 1/4$ and $S \geq 2$, such a uniformly successful S -parameter learner satisfies $TS \geq \text{dc}(H_A)/3$ for every member of the flip family.

The proof is in Appendix R.3.

Proposition 20 (Hidden-key computational hardness). *Under the PRF assumption of Theorem 14, no probabilistic polynomial-time learner independent of s , with a polynomial-time evaluable output and only polynomially many labeled samples from the uniform distribution on ℓ_0 , has average error at most $1/2 - \gamma(n)$ for uniform s , where $\gamma(n) \geq 1/\text{poly}(n)$. More precisely, a learner with $q(n)$ samples and advantage $\gamma(n)$ gives a PRF distinguisher of advantage at least*

$$\gamma(n) - \frac{q(n)}{2N}.$$

The proof is in Appendix P.2.

The unseen-label bound conditions on the sample indices and observed labels; every unseen incidence bit remains fair. The PRF reduction adds one fresh test point, whose training collision probability is at most $q(n)/N$. Both propositions fix the learner independently of the table or key. Class-dependent architecture choices in OQ1 have a different quantifier.

8 Open problems

Explicit selection. Find one efficiently entry-computable family with polynomial-length descriptions, polynomial inverse rectangle ratio, and dimension exceeding every polynomial in n . The existing key and advice families give each prescribed degree separately. Efficient deterministic selection follows the direction of Hatami, Hatami, Pires, Tao and Zhao’s Problem 4.1 [24]. The classical certificate limitation and the source-length counterexample are retained in Appendix Q.

Benign online SGD and charged resources. Determine whether the exact Gaussian-initialized, bias-free logistic-loss process in Appendix N, with class-dependent architecture and hyperparameters, implies $\text{dc}(H) \leq CTS$ or a polynomial bound. A negative answer requires both a small compatible network and a distribution-independent analysis of these dynamics for a class of high dimension. Separately, determine what dimension bound follows when preprocessing, query evaluation, generated memory and expanded description are charged in a finite-bit model.

A characterization beyond rectangle size. Identify a resource characterizing distribution-independent SQ learning. Cube halfspaces and parities both have exponentially small rectangles, but only the latter have an exponential SQ lower bound. The communication and RSD sufficient conditions are in Appendices L and S.

Query and tolerance tradeoffs. Projective planes satisfy $\rho = \Theta(q)$ and $m/\tau^2 = \Omega(\rho)$, whereas the rectangle theorem gives $O_\epsilon(\rho^3 \log K)$. Determine the optimal tradeoff. For succinct incidence evaluation, close the remaining query gap between the $\Theta_\epsilon(n)$ order in Theorem 8 and the $O_\epsilon(n^2)$ order-median learner, including the possible reuse of residual medians across slope stages.

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References

- [1] Emmanuel Abbe, Enric Boix-Adserà, Matthew Brennan, Guy Bresler, and Dheeraj Nagaraj. The staircase property: how hierarchical structure can guide deep learning. In *Advances in Neural Information Processing Systems*, 2021. URL <https://arxiv.org/abs/2108.10573>.
- [2] Emmanuel Abbe, Pritish Kamath, Eran Malach, Colin Sandon, and Nathan Srebro. On the power of differentiable learning versus PAC and SQ learning. In *Advances in Neural Information Processing Systems*, volume 34, pages 24340–24351, 2021. URL <https://arxiv.org/abs/2108.04190>. Full version: arXiv:2108.04190v2.
- [3] Emmanuel Abbe, Enric Boix-Adserà, and Theodor Misiakiewicz. The merged-staircase property: a necessary and nearly sufficient condition for SGD learning of sparse functions on two-layer neural networks. In *Proceedings of the Thirty-Fifth Conference on Learning Theory*, volume 178 of *Proceedings of Machine Learning Research*, pages 4782–4887, 2022. URL <https://proceedings.mlr.press/v178/abbe22a.html>. Preprint: arXiv:2202.08658.
- [4] Noga Alon, János Pach, Rom Pinchasi, Radoš Radoičić, and Micha Sharir. Crossing patterns of semi-algebraic sets. *Journal of Combinatorial Theory, Series A*, 111(2):310–326, 2005. doi: 10.1016/j.jcta.2004.12.008. URL <https://doi.org/10.1016/j.jcta.2004.12.008>.
- [5] Noga Alon, Shay Moran, and Amir Yehudayoff. Sign rank versus VC dimension. In *Conference on Learning Theory*, volume 49 of *Proceedings of Machine Learning Research*, pages 47–80, 2016. URL <https://arxiv.org/abs/1503.07648>. Full version: arXiv:1503.07648.
- [6] Maria-Florina Balcan and Vitaly Feldman. Statistical active learning algorithms. In *Advances in Neural Information Processing Systems*, volume 26, 2013. URL https://proceedings.neurips.cc/paper_files/paper/2013/hash/84117275be999ff55a987b9381e01f96-Abstract.html. Extended version: arXiv:1307.3102v4; Appendix A refers to that version.
- [7] Shai Ben-David, Nadav Eiron, and Hans Ulrich Simon. Limitations of learning via embeddings in euclidean half spaces. *Journal of Machine Learning Research*, 3:441–461, 2002. URL <https://www.jmlr.org/papers/v3/bendavid02a.html>.
- [8] Riju Bindua, Hamed Hatami, Hasti Karimi, and Robert Robere. Lower bounds for approximate sign rank. arXiv:2605.01038v2, 2026. URL <https://arxiv.org/abs/2605.01038>.
- [9] Ari Blondal, Hamed Hatami, Pooya Hatami, Chavdar Lalov, and Sivan Treiatak. Sign-rank, index, and list replicability: connections and separations. arXiv:2606.18236v1, 2026. URL <https://arxiv.org/abs/2606.18236>.
- [10] Bogdan Chornomaz, Shay Moran, and Tom Waknine. On reductions and representations of learning problems in euclidean spaces, 2025. URL <https://arxiv.org/abs/2411.10784>. STOC 2025; preprint arXiv:2411.10784.
- [11] Ilias Diakonikolas, Christos Tzamos, and Daniel M. Kane. A strongly polynomial algorithm for approximate Forster transforms and its application to halfspace learning. arXiv:2212.03008, 2022. URL <https://arxiv.org/abs/2212.03008>.
- [12] John Dunagan and Santosh Vempala. A simple polynomial-time rescaling algorithm for solving linear programs. In *Proceedings of the ACM Symposium on Theory of Computing*, pages 315–320, 2004.
- [13] Vitaly Feldman. Evolvability from learning algorithms. In *Proceedings of the ACM Symposium on Theory of Computing*, pages 619–628, 2008.

- [14] Vitaly Feldman. Distribution-independent evolvability of linear threshold functions. In *Conference on Learning Theory*, volume 19 of *Proceedings of Machine Learning Research*, pages 253–272, 2011. URL <https://proceedings.mlr.press/v19/feldman11b.html>.
- [15] Vitaly Feldman. A general characterization of the statistical query complexity. In *Conference on Learning Theory*, volume 65 of *Proceedings of Machine Learning Research*, pages 785–830, 2017. URL <https://proceedings.mlr.press/v65/feldman17c.html>.
- [16] Vitaly Feldman, Cristóbal Guzmán, and Santosh Vempala. Statistical query algorithms for mean vector estimation and stochastic convex optimization. *Mathematics of Operations Research*, 46(3):912–945, 2021. doi: 10.1287/moor.2020.1111. URL <https://doi.org/10.1287/moor.2020.1111>. Preprint: arXiv:1512.09170.
- [17] Vitaly Feldman, Pritish Kamath, and Nathan Srebro. Invited open problem: Is the power of deep learning over linear models inherently distribution dependent? In *Proceedings of the Thirty-Ninth Conference on Learning Theory*, volume 336 of *Proceedings of Machine Learning Research*, pages 7117–7122, 2026. URL <https://proceedings.mlr.press/v336/feldman26a.html>.
- [18] Jürgen Forster. A linear lower bound on the unbounded error probabilistic communication complexity. *Journal of Computer and System Sciences*, 65(4):612–625, 2002. doi: 10.1016/S0022-0000(02)00019-3. URL [https://doi.org/10.1016/S0022-0000\(02\)00019-3](https://doi.org/10.1016/S0022-0000(02)00019-3).
- [19] Florian Frick, Kaave Hosseini, and Aliaksei Vasileuski. A $\mathbb{Z}_2$ -topological framework for sign-rank lower bounds. arXiv:2604.01510v2, 2026. URL <https://arxiv.org/abs/2604.01510>.
- [20] Oded Goldreich, Shafi Goldwasser, and Silvio Micali. How to construct random functions. *Journal of the ACM*, 33(4):792–807, 1986. doi: 10.1145/6490.6503. URL <https://doi.org/10.1145/6490.6503>.
- [21] Alon Gonen, Sivan Sabato, and Shai Shalev-Shwartz. Efficient active learning of halfspaces: an aggressive approach. *Journal of Machine Learning Research*, 14:2583–2615, 2013. URL <https://jmlr.org/papers/volume14/gonen13a/gonen13a.pdf>.
- [22] Mika Göös, Nathaniel Harms, Valentin Imbach, and Dmitry Sokolov. Sign-rank of k -hamming distance is constant. arXiv:2506.12022v2, 2025. URL <https://arxiv.org/abs/2506.12022>.
- [23] Johan Håstad, Russell Impagliazzo, Leonid A. Levin, and Michael Luby. A pseudorandom generator from any one-way function. *SIAM Journal on Computing*, 28(4):1364–1396, 1999. doi: 10.1137/S0097539793244708. URL <https://doi.org/10.1137/S0097539793244708>.
- [24] Hamed Hatami, Pooya Hatami, William Pires, Ran Tao, and Rosie Zhao. Lower bound methods for sign-rank and their limitations. *Electronic Colloquium on Computational Complexity*, TR22-079, 2022. URL <https://eccc.weizmann.ac.il/report/2022/079/>.
- [25] Hamed Hatami, Kaave Hosseini, and Xiang Meng. A Borsuk–Ulam lower bound for sign-rank and its applications. In *Proceedings of the ACM Symposium on Theory of Computing*, pages 463–471, 2023. URL <https://eccc.weizmann.ac.il/report/2022/130/>. Preprint: ECCC TR22-130.
- [26] Pritish Kamath, Omar Montasser, and Nathan Srebro. Approximate is good enough: probabilistic variants of dimensional and margin complexity. In *Conference on Learning Theory*, volume 125 of *Proceedings of Machine Learning Research*, pages 2236–2262, 2020. URL <https://proceedings.mlr.press/v125/kamath20b.html>.

- [27] Ari Karchmer and Eran Malach. The power of random features and the limits of distribution-free gradient descent. In *International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 2025. URL <https://proceedings.mlr.press/v267/karchmer25a.html>.
- [28] Michael Kearns. Efficient noise-tolerant learning from statistical queries. *Journal of the ACM*, 45(6):983–1006, 1998. doi: 10.1145/293347.293351. URL <https://doi.org/10.1145/293347.293351>.
- [29] Assia Mahboubi and Cyril Cohen. Formal proofs in real algebraic geometry: from ordered fields to quantifier elimination. arXiv:1201.3731, 2012. URL <https://arxiv.org/abs/1201.3731>.
- [30] Eran Malach, Pritish Kamath, Emmanuel Abbe, and Nathan Srebro. Quantifying the benefit of using differentiable learning over tangent kernels. In *International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 7379–7389, 2021. URL <https://proceedings.mlr.press/v139/malach21a.html>.
- [31] James Renegar. On the computational complexity and geometry of the first-order theory of the reals. I–III. *Journal of Symbolic Computation*, 13(3):255–352, 1992.
- [32] Nicolai Vorobjov. Lecture notes on complexity of quantifier elimination over the reals. arXiv:2112.00456, 2021. URL <https://arxiv.org/abs/2112.00456>.
- [33] Hugh E. Warren. Lower bounds for approximation by nonlinear manifolds. *Transactions of the American Mathematical Society*, 133:167–178, 1968.
- [34] Dechen Zhang, Xuan Tang, Xinxiang Yin, Xingwu Chen, Jian Qian, and Difan Zou. VALG: An agentic system for ML theory research and demonstrations on COLT 2026 open problems. arXiv:2608.13060v2, 2026. URL <https://arxiv.org/abs/2608.13060v2>. Version 2, 10 September 2026.

A Oracle endpoints, conventions and constants

The inequalities below prove Theorem 1 for arbitrary valid oracle responses. In the table below $u = D(U)$, $s = D(S)$, $a = s/u$, $b = \nu(T)$, $w = D(S \cap \{h^* \neq c\})$, and hats denote oracle answers. “Pathwise” means that no random-oracle assumption is used. The conditioning for mixture expectations includes the returned $\hat{u}$ and all earlier history, but precedes the fresh rectangle draw.

	Inequality or invariant	Adverse valid answer or justification	Randomness used
R1	$\tau \leq \epsilon/8, \tau \leq r\epsilon/64$	Direct substitution of P ; no reply.	None
R2	Nonstop $u > \epsilon/4 - \tau \geq \epsilon/8$	$\hat{u} = u + \tau$ can keep the smallest u active.	Pathwise
R3	$\tau/u \leq r/8$	R1–R2.	None
R4	Acceptance gives $s \geq ru/2 - (1 + r/2)\tau$	$\hat{s} = s + \tau, \hat{u} = u - \tau$ favor accepting a small set.	Pathwise
R5	Accepted $a \geq 5r/16 \geq r/4$	Substitute R3 into R4.	None
R6	$a \geq 3r/4$ gives $\hat{s} - r\hat{u}/2 \geq ru/16 > 0$	$\hat{s} = s - \tau, \hat{u} = u + \tau$ oppose acceptance.	Pathwise
R7	Target in T implies $w = 0, \hat{w} \leq \tau < 2\tau$	Maximal reply $\hat{w} = w + \tau$.	Pathwise
R8	$\sum_{\text{elim}} b \leq \sum_{\text{elim}} -\ln(1 - b) \leq \ln K$	Target retained; normalized target weight grows from $1/K$ to at most one.	Pathwise
R9	Before peel j : $\epsilon/8 \leq (1 - r/4)^{j-1} \leq e^{-(j-1)r/4}$	R2, R5, and $\ln(1 - z) \leq -z$.	Pathwise
R10	Number of peels at most P	Take logarithms in R9.	None
R11	A peel has $w \leq 3\tau$; total wrong mass $\leq 3P\tau = \epsilon/8$	$\hat{w} = w - \tau$ permits the largest wrong mass. Peeled sets are disjoint.	Pathwise
R12	Normal stop: $u \leq \epsilon/4 + \tau$; loss $\leq \epsilon/2$	$\hat{u} = u - \tau$ permits the largest residual mass. Empty residual has mass zero.	Pathwise
R13	On acceptance, $G = a$ or b , so $G \geq ab$	Both admissible actions obey this, irrespective of the mismatch reply.	Pathwise
R14	$\mathbb{E}[ab \mid \text{past}] \geq r$	Integrate pointwise mixture coverage against D conditioned on U .	Fresh learner draw
R15	$\mathbb{E}G \geq r - (3r/4)\mathbb{E}b \geq r/4$	Rejection is confined to $a < 3r/4$ by R6; $b \leq 1$.	Fresh learner draw
R16	$\sum G \leq \ln K + \ln(8/\epsilon) + 1 = B$ before budget termination	R8; peel ratios telescope excluding the last, whose $a \leq 1$. No logarithm of zero.	Pathwise
R17	$e^{-g} \leq 1 - (1 - e^{-1})g$ for $0 \leq g \leq 1$	Convexity chord. After normal stopping put $G = 1$.	None
R18	$\mathbb{E}[e^{-G} \mid \text{past}] \leq e^{-r/8}; \mathbb{E}e^{-\sum G} \leq e^{-rR/8}$	R15, R17 and conditional expectation induction.	Learner draws; oracle arbitrary
R19	$\Pr[\text{budget failure}] \leq e^{B-rR/8} \leq \epsilon/2$	R16 and Markov’s inequality.	Learner draws; oracle arbitrary
R20	$\mathbb{E}L \leq \epsilon/2 + \Pr[\text{budget failure}] \leq \epsilon$	R12 and $0 \leq L \leq 1$.	Learner draws; oracle arbitrary

A randomized oracle can be included by conditioning on its past private coins; each displayed response bound still holds for every permitted answer. No step averages away an adversarial error.

The drift proof allows the oracle to choose either permissible action after seeing the current query. It does not give an oracle access to future learner randomness.

Proof of Lemma 23 and the zero-incidence convention. There are exactly $N(N+1)/2$ empty template rows: row (a, b) has no incidence precisely when $b > 2N^2 - a$. They are all-positive rows of every flip. Their weights k_h/I in Theorem 5 are zero; no uniform distribution on an empty set is formed. For nonempty rows these weights sum to one. For a fixed embedding, loss minimization ranges over finitely many attained sign patterns, so the infimum is a minimum. The weighted incidence mistake count is an integer divided by I ; some embedding has value at most its expectation bound. The same embedding is used for every row, with row-specific weights, exactly as required by the definition.

The working convention is $\text{sign}(0) = +1$. A positive row-specific bias in one common added coordinate turns zero scores positive without changing any negative score. Thus the strict rank in the counting argument is $d + 1$, and the event is $\text{dc}^\eta < s_\eta$, not $\text{dc}^\eta \leq s_\eta$. If the sign convention is $\text{sign}(0) = -1$, use a negative bias. If zero predictions are counted as errors, replacing them by $+1$ can only reduce the number of mistakes, so the same lower bound still follows. Exact probabilistic dimension uses a full-support marginal; zero loss means that every point is correct. These observations make the conclusions insensitive to either fixed tie rule, without weakening the universal quantifiers over D and h . $\square$

B Endpoint and randomness ledgers for the later results

Every row below is pathwise unless its randomness column says otherwise. Endpoint pairs may be mutually incompatible across rows; using them separately only strengthens each bound. Bounds involving imported simulation randomness retain the supremum inside the expectation. The first column uses local row identifiers; the electronic ledger also records the previous statement numbers.

B.1 Lemma 11: Integer medians

ID	Inequality	Adverse answer or justification	Randomness
1	$\zeta \geq 0, \beta > 0, \beta \geq 512\zeta$.	Explicit parameter domain; no answer used.	None.
2	$\widehat{F}(t) \geq \widehat{\beta}/2 \Rightarrow F(t) \geq \beta/2 - 3\zeta/2$.	$\widehat{F} = F + \zeta, \widehat{\beta} = \beta - \zeta$ minimize the certified mass.	Pathwise.
3	$\widehat{F}(t) < \widehat{\beta}/2 \Rightarrow F(t) < \beta/2 + 3\zeta/2$.	$\widehat{F} = F - \zeta, \widehat{\beta} = \beta + \zeta$ maximize it.	Pathwise.
4	$F(v) \geq \beta/2 - 3\zeta/2, F(v-1) < \beta/2 + 3\zeta/2$.	Binary search retains a certified lower and upper endpoint; replies need not be monotone.	Pathwise.
5	$\widehat{m} \leq \widehat{\beta}/8 \Rightarrow m \leq \beta/8 + 9\zeta/8$.	$\widehat{m} = m - \zeta, \widehat{\beta} = \beta + \zeta$.	Pathwise.
6	$F(v) < 5\beta/8 + 21\zeta/8$.	Add the preceding atom bound to $F(v-1) < \beta/2 + 3\zeta/2$.	None.
7	$\widehat{m} > \widehat{\beta}/8 \Rightarrow m > \beta/8 - 9\zeta/8 > \beta/9$.	$\widehat{m} = m + \zeta, \widehat{\beta} = \beta - \zeta$; the last step uses $\beta > 81\zeta$.	Pathwise.

B.2 Theorem 12: Succinct learner

ID	Inequality	Adverse answer or justification	Randomness
1	$\zeta = \epsilon/4096$; raw and rounding errors are at most $\zeta/2$ each.	Triangle inequality; finite answer representation is part of the computational interface.	None.

ID	Inequality	Adverse answer or justification	Randomness
2	$\widehat{\beta} \leq \epsilon/4 \Rightarrow \beta \leq \epsilon/4 + \zeta$.	$\widehat{\beta} = \beta - \zeta$.	Pathwise.
3	$\widehat{\beta} > \epsilon/4 \Rightarrow \beta > 1023\zeta > 512\zeta$.	$\widehat{\beta} = \beta + \zeta$.	Pathwise.
4	$\mu^-(P \cap Q) \geq \beta/2 - 3\zeta/2$ for two prefixes.	Intersection is the smaller prefix; apply both median lower bounds.	None.
5	$\mu^-(P \cap Q) \leq \beta/4 + 21\zeta/4$ for a prefix and a suffix.	Their intersection mass is $\max(0, \mu^-(P) + \mu^-(Q) - \beta)$.	None.
6	$\mu^-(\widehat{P \cap Q}) - 3\widehat{\beta}/8 \geq (\beta - 23\zeta)/8 > 0$ in the prefix case.	Intersection answer minus ζ ; total answer plus ζ .	Pathwise.
7	$3\widehat{\beta}/8 - \mu^-(\widehat{P \cap Q}) \geq (\beta - 53\zeta)/8 > 0$ in the suffix case.	Intersection answer plus ζ ; total answer minus ζ .	Pathwise.
8	$\widehat{\mu^-(P)} > \widehat{\beta}/8$ for the constant residual.	Lower observed mass is $\beta/2 - 5\zeta/2$; compare with $(\beta + \zeta)/8$.	Pathwise.
9	$\widehat{\mu^-(Q \setminus P)} > \widehat{\beta}/8$ for the constant residual.	Even a difference of two answers is at least $3\beta/8 - 37\zeta/8$; $\beta > 19\zeta$ suffices.	Pathwise.
10	A wrong-slope residual atom is a singleton; one split answer is at most $2\zeta < \widehat{\beta}/8$.	Allow two endpoint errors if the split is obtained by subtraction; $\beta > 17\zeta$.	Pathwise.
11	$\zeta < \widehat{\beta}/32 < \beta/9 - \zeta$.	Use $\widehat{\beta} \in [\beta - \zeta, \beta + \zeta]$ and $\beta \geq 512\zeta$; separates zero from a heavy atom.	Pathwise.
12	Large outside negative mass is $> \epsilon/4 - \zeta$; star-half queries are zero or that mass.	Every other line through the known point misses the target away from it; threshold $\epsilon/8$ remains separated by ζ .	Pathwise.
13	Small outside negative mass is $\leq \epsilon/4 + \zeta$.	The answer can underestimate by ζ ; the one-point predictor has this error.	Pathwise.
14	$C_0 = \max(64\lceil n \rceil, \lceil 128/\epsilon \rceil)$; a smaller enumerated star contains the target.	Target loss answer is at most $\zeta < \epsilon/2$.	None.
15	At most $8/\epsilon$ star rows have outside negative mass $> \epsilon/8$.	Their outside negative supports are disjoint and total mass is at most one.	None.
16	At least $t/4 - 8/\epsilon \geq t/8$ rows pass; their loss is at most $3\epsilon/8 + \zeta$.	On a regular star with $t > C_0$; answer at most $3\epsilon/8 + 2\zeta < \epsilon/2$.	None.
17	Each row has sampling probability $\geq 1/(2t)$; success per trial is $\geq 1/16$.	Use $\lceil \log_2 t \rceil$ bits reduced modulo t ; the good-row count is pathwise.	Fresh learner bits.
18	$J = \lceil 16 \ln(4/\epsilon) \rceil$; failure $\leq (15/16)^J \leq \epsilon/4$.	Independent trials; every good row passes every permitted loss answer.	Learner bits.
19	$\mathbb{E}L \leq \epsilon/2 + \zeta + \epsilon/4 < \epsilon$.	Any passed test certifies loss $\leq \epsilon/2 + \zeta$; failure output loss is at most one.	Learner bits only.
20	$m = O(n^2 + 1/\epsilon + \log(1/\epsilon))$.	At most $O(n)$ slope stages and $O(n)$ binary queries per median; all integer tests use $O(n)$ -bit coordinates.	None.

B.3 Lemma 13: Counting on a subgrid

ID	Inequality	Adverse answer or justification	Randomness
1	$K_k = 2k^3, I_k = 2k^4 - [k(k+1)/2]^2$.	Sum $2k^2 - ax$ over $a, x \in [k]$; no oracle.	None.
2	$D_k = \lfloor k/(24 \log_2 k) \rfloor \leq K_k/2$.	For $D_k = 0$ the nonpositive-rank event is empty.	None.
3	$D_k \geq 1 \Rightarrow k \geq 32 \Rightarrow \log_2(8ek^3) \leq 4 \log_2 k$.	The implication is only a sufficient range check; it does not locate the first positive D_k .	None.
4	$\log_2 \#\{\text{signrank} \leq D_k\} \leq 2K_k D_k 4 \log_2 k \leq 2k^4/3$.	Warren applies to strict patterns of degree two in $2K_k D_k$ variables.	None.
5	$k \geq 3 \Rightarrow I_k \geq 3k^4/2$.	$(k+1)^2/(4k^2) \leq 1/2$ for $k \geq 3$.	None.

ID	Inequality	Adverse answer or justification	Randomness
6	$\Pr[\text{signrank} \leq D_k] \leq 2^{2k^4/3-I_k} \leq 2^{-5k^4/6} \leq 2^{-k^4/2}$.	All I_k free positions are independent uniform bits.	Random flips.
7	$\text{signrank}(A) \leq D_k \iff \exists U, V : A_{ij} \sum_s U_{is} V_{sj} \geq 1$.	Scale a strict factorization by its finite minimum signed score; coefficients are $-1, 0, 1$.	None.
8	$(K_k^2)^{O(K_k D_k)} = 2^{O(k^3 D_k \log k)} = 2^{O(k^4)}$.	Use the cited singly exponential existential-real decision bound with degree two and bounded coefficient length.	None.

B.4 Theorem 14: Pseudorandom restriction

ID	Inequality	Adverse answer or justification	Randomness
1	$n = 3b + 1, N = 2^b, k = \lceil cn^c \rceil \leq N$ eventually.	Fix c, δ before the asymptotic limit.	None.
2	$E_b(a, b_0, x, y)$ has $6b + 2 = 2n$ bits and is injective.	Concatenate zero-based coordinates in widths $b, 2b + 1, b, 2b + 1$; keep this ambient encoding in every restriction.	None.
3	Exactly $I_k \leq 2k^4$ function values determine the restricted matrix.	Query only incidences. Uniform oracle values at these distinct ambient inputs are independent fair bits.	Function oracle, not SQ.
4	$2^{C_c n^{4c}} \leq 2^{n^{4c+2}} \leq 2^{\sigma^\delta}$ eventually.	$\sigma = n^p, p = \lceil (4c + 2)/\delta \rceil$; the n^2 slack absorbs the fixed decision-procedure constant and reading cost.	None.
5	$\Pr_s[\text{signrank}(A_s _k) \leq D_k] \leq 2^{-k^4/2} + \text{negl}(\sigma)$.	Use the low-rank decision as the acceptance event; nonnegligible excess would distinguish from a random function.	Uniform key / random function.
6	$\text{negl}(n^p) = \text{negl}(n)$.	For each desired exponent q , use the security estimate with exponent q/p ; p is fixed.	None.
7	$\text{dc}(H_{A_s}) \geq D_k + 1 > k/(24 \log_2 k)$.	Restriction cannot increase sign-rank; use the complement of the low-rank event.	Good-key event.
8	$\log_2 k \leq c \log_2 n + \log_2(2c) \leq (5/4)c \log_2 n$.	For all sufficiently large n ; the extra factor c in k is retained.	None.
9	$k/(24 \log_2 k) \geq n^c/(30 \log_2 n)$.	Substitute $k \geq cn^c$ and the preceding logarithmic arithmetic estimate.	None.

B.5 Corollary 17: Joint good-key event

ID	Inequality	Adverse answer or justification	Randomness
1	$\Pr[\text{Bin}(t, 1/2) < t/4] \leq e^{-t/16}$.	Multiplicative Chernoff with mean $t/2$ and relative deviation $1/2$.	Random incidence bits.
2	$t \geq 64 \lceil n \rceil \Rightarrow 2^n e^{-t/16} \leq e^{-3n}$.	Union bound over all points; the regularity event is independent of ϵ .	Random flips.
3	Star-regularity is decidable in $2^{O(n)} \leq 2^{\sigma^\delta}$ time eventually.	Enumerate incidences using the same ambient encoding; transfer the bad-star event by PRF security.	Uniform key / function.
4	$\Pr_s[\text{bad rank or bad stars}] = \text{negl}(n)$.	Union bound, not independence of the events.	Uniform key.
5	$m = O_\epsilon(n^2), 1/\tau = O_\epsilon(1), \text{TIME} = \text{poly}_{c,\epsilon}(n)$.	Apply the proper succinct algorithm on regular keys; its deterministic improper version works for every key.	Learner bits for proper output.

B.6 Theorem 24: Product-average communication

ID	Inequality	Adverse answer or justification	Randomness
1	$\Pr[L \leq 2c] \geq 1/2$.	Markov on the cost of an exact deterministic protocol under the chosen product measure.	Product-distributed inputs.
2	$\#\{\text{leaves of depth} \leq 2c\} \leq 2^{\lfloor 2c \rfloor}$.	Prefix-free transcripts can be injected into strings of length $\lfloor 2c \rfloor$.	None.
3	$\max_Q(\mu \times \nu)(Q) \geq 2^{-\lfloor 2c \rfloor - 1} \geq 2^{-2c-1}$.	One shallow monochromatic leaf carries the average shallow mass. At $c = 0$, all positive-mass inputs reach a zero-cost leaf.	Product inputs.
4	$\Pr[\text{continue after } j \text{ stages}] \leq (1-r)^j$.	Conditioning a product measure on a protocol cell preserves product form; each selected rectangle has conditional mass $\geq r$.	Product inputs.
5	$\mathbb{E}[\text{bits}] \leq 2 \sum_{j \geq 0} (1-r)^j = 2/r$.	Two membership bits per stage; complete zero-mass cells by a finite exact protocol.	Product inputs.
6	$\rho = 2^{2c+1} = 2 \cdot 4^c$.	Substitute into the rectangle learner; its arbitrary-policy guarantee is unchanged.	RECTIFY draws.

B.7 Theorem 32: Cube halfspaces

ID	Inequality	Adverse answer or justification	Randomness
1	$\text{dc}(H_n) \leq n$ for odd n .	Identity embedding; dot products are odd integers and nonzero.	None.
2	$\langle w, x \rangle < 0 \Rightarrow d_H(w, x) \geq (n+1)/2$.	Use $\langle w, x \rangle = n - 2d_H(w, x)$.	None.
3	$\langle w, x \rangle > 0 \Rightarrow d_H(-w, x) \geq (n+1)/2$.	Negate one side only; it is a measure-preserving bijection of the uniform cube.	None.
4	$\alpha \leq e^{-2s^2/n}$, with $s = \mathbb{E}d_H(Z, A)$.	Distance to A changes by at most one per coordinate. Apply the lower bounded-differences tail at s .	Independent uniform cube coordinates.
5	$s \leq t \Rightarrow \beta \leq e^{-2(t-s)^2/n}$.	Every point of the other set has distance at least t ; apply the upper tail.	Uniform cube.
6	$\alpha\beta \leq e^{-2[s^2 + (t-s)^2]/n} \leq e^{-t^2/n} \leq e^{-n/4}$.	$s^2 + (t-s)^2 \geq t^2/2$ and $t = (n+1)/2$. If $s \geq t$, use $\alpha \leq e^{-2t^2/n}$ instead.	None.
7	$F(v^*) = 0$, $\ g\ \leq 1$, $\ \hat{g} - g\ \leq \tau$.	$\gamma = 1/n$; coordinate query errors at either endpoint are divided by $\sqrt{n}$.	Pathwise SQ answers.
8	$F(\bar{v}) \leq 1/(2aT) + a(1+\tau)^2/2 + 2\tau$.	Projected subgradient distance telescope; perturbation inner product is at most 2τ .	Pathwise.
9	$T \geq 100/(\epsilon^2\gamma^2)$, $a = T^{-1/2}$, $\tau = \epsilon\gamma/8$.	$(1+\tau)^2 < 4$ implies $F(\bar{v}) \leq 5/(2\sqrt{T}) + 2\tau \leq \epsilon\gamma/2$.	None.
10	$L \leq F(\bar{v})/\gamma \leq \epsilon/2$, $m = nT = O(n^3/\epsilon^2)$.	Every thresholding error incurs hinge cost at least γ .	Deterministic learner.

B.8 Corollary 18: Engineered full-batch simulation

ID	Inequality	Adverse answer or justification	Randomness
1	$\sup_O L_{01}(A^O) \leq \epsilon/8$.	Use the deterministic improper learner, valid for every key. No expected-to-pathwise conversion is made.	None.

ID	Inequality	Adverse answer or justification	Randomness
2	$L_{\text{square}}(A) \leq \epsilon/16.$	For Boolean labels and predictions, half squared error is half classification error.	Pathwise.
3	$\lambda = \tau_0/16, T' = 2k, m_s \lambda^2 > C(k \log(1/\lambda) + \log(1/\delta)).$	Imported full-batch Theorem 3d: one sample is reused, with λ -grid gradients within $3\lambda/4$ of the clipped empirical gradient.	Sample and initialization.
4	$p = \text{poly}(k, 1/\tau_0, r, \text{TIME}, 1/\delta).$	TIME includes the succinct learner and query evaluations. Set $r = 1$ with an ignored bit, not an assumed $r = 0$ case.	Ignored initialization bit.
5	$\mathbb{E}_{S,R} \sup_{g_1, \dots, g_{T'}} L_{\text{square}} \leq \epsilon/16 + \delta = \epsilon/8.$	$\delta = \epsilon/16$. Retain the imported supremum inside the expectation, including its dependence on the whole sample.	Sample / initialization; arbitrary admissible gradients.
6	$L_{01}(\mathbf{1}_{f \geq 1/2}) \leq 8L_{\text{square}}(f).$	A threshold error gives $ f - y \geq 1/2$ and half squared error at least $1/8$.	Pointwise.
7	$\mathbb{E} L_{01} \leq \epsilon, p, m_s = \text{poly}_{c,\epsilon}(n), T' = O(n^2).$	Only the time bound of the succinct learner and the imported polynomial are used; no polynomial in query count alone is substituted.	Sample / initialization.

B.9 Proposition 19: Unseen independent labels

ID	Inequality	Adverse answer or justification	Randomness
1	$\Pr[x \text{ unseen}] = (1 - 1/N)^T.$	The sample indices are iid uniform on a full template line.	Sample indices.
2	$\mathbb{E}[L \mid \text{indices, observed labels, learner coins}] \geq \#\{\text{unseen}\}/(2N).$	Each unseen label remains a fair independent bit because the learner was fixed independently of the table.	Unseen random flip bits.
3	$\mathbb{E} L \geq \frac{1}{2}(1 - 1/N)^T \geq \frac{1}{2}(1 - T/N).$	Average the preceding identity; Bernoulli inequality.	Table, samples, learner coins.
4	$\mathbb{E} L \leq \epsilon \Rightarrow T \geq (1 - 2\epsilon)N.$	This is a uniform-over-tables or average-table premise, not a claim for each individually selected table.	None.
5	$\epsilon < 1/4, S \geq 2 \Rightarrow TS \geq N \geq \text{dc}(H_A)/3.$	$\text{dc}(H_A) \leq 2N + 1 \leq 3N$. Keep the table-independent learner premise.	None.

C Finite separation, product weights and tie conventions

Lemma 21 (Finite minimax and pointwise coverage). *Let $\mathcal{A}$ be a nonempty finite action set and Z a nonempty finite set of instance types. Let $g_a \in \mathbb{R}^Z$ and $r \in \mathbb{R}$. The following statements are equivalent:*

$$\forall \mu \in \Delta(Z), \max_{a \in \mathcal{A}} \mathbb{E}_\mu g_a \geq r; \quad \exists \lambda \in \Delta(\mathcal{A}), \forall z \in Z, \mathbb{E}_\lambda g_a(z) \geq r.$$

The result also applies when an arbitrary instance set induces only finitely many payoff vectors $(g_a(z))_{a \in \mathcal{A}}$.

Proof. The reverse implication follows by averaging. For the forward direction, let $C = \text{conv}\{g_a : a \in \mathcal{A}\}$ and $Q = r\mathbf{1} + \mathbb{R}_{\geq 0}^Z$. If disjoint, their minimum distance is positive and attained: C is compact, and a bounded-distance minimizing sequence has bounded Q -components. Let (c, z) attain it and put $v = z - c \neq 0$. Differentiation along feasible segments gives

$$\langle v, c' - c \rangle \leq 0 \quad (c' \in C), \quad \langle v, z' - z \rangle \geq 0 \quad (z' \in Q).$$

Taking $z' = z + te_x$ yields $v_x \geq 0$. Moreover,

$$\max_{c' \in C} \langle v, c' \rangle \leq \langle v, c \rangle < \langle v, z \rangle = \min_{z' \in Q} \langle v, z' \rangle = r \sum_x v_x.$$

Normalize v to a probability distribution, contradicting the first condition. Hence $C \cap Q \neq \emptyset$, and its convex coefficients give λ . For finitely many payoff types, retain one representative of each type and apply the finite result. $\square$

Lemma 22 (From uniform rectangles to product weights). *Suppose a property of finite matrices is preserved by repeating rows and columns. Suppose every matrix with that property has a monochromatic rectangle with row fraction at least a and column fraction at least b . Then every such matrix has rectangle ratio at least ab under arbitrary product distributions.*

Proof. For strictly positive rational weights, repeat rows and columns according to common-denominator multiplicities. Projecting a monochromatic rectangle back to the original indices preserves its signs and can only increase its weighted side masses. For arbitrary weights, approximate by strictly positive rational distributions. The largest rectangle mass is the maximum of finitely many continuous products of linear functions of the weights, hence is continuous. Passing to the limit gives the assertion. $\square$

Lemma 23 (Tie conventions and strict realization). *For any finite family of predictions obtained from a d -dimensional embedding with the convention $\text{sign}(0) = +1$ or -1 , the same Boolean prediction matrix has a strict realization in dimension at most $d + 1$. If all zero scores are counted as errors, replacing their predictions by $+1$ cannot increase loss, for any marginal and target.*

The proof, including the attained-infimum and zero-incidence cases, appears in Appendix A.

D Proof of the rectangle theorem

Proof of Theorem 1 and Lemma 2. Fix D, h^* and any valid oracle policy. First,

$$\tau \leq \epsilon/8, \quad \tau \leq r\epsilon/64. \tag{D.1}$$

Indeed $P \geq 4\rho \ln(8/\epsilon) > 4\rho \ln 32$, and $24 \cdot 4 \ln 32 > 64$. At a nonstopping round, write

$$u = D(U) > \epsilon/4 - \tau \geq \epsilon/8, \quad a = D(S)/u, \quad b = \nu(T), \quad w = D(S \cap \{h^* \neq c\}).$$

Thus $\tau/u \leq r/8$. If the proposal is accepted, validity implies

$$D(S) \geq (r/2)u - (1 + r/2)\tau \geq (5r/16)u \geq (r/4)u.$$

Conversely, every rectangle with $a \geq 3r/4$ is accepted, since

$$\hat{s} - (r/2)\hat{u} \geq (r/4)u - (1 + r/2)\tau \geq (r/16)u > 0.$$

These inequalities hold at the extreme valid oracle answers, not only for exact answers.

If $h^* \in T$, monochromaticity gives $w = 0$, hence $\hat{w} \leq \tau < 2\tau$. Therefore the algorithm never eliminates the target. In particular, V remains nonempty. On an elimination, its normalized target weight increases by a factor $1/(1 - b)$. Because $-\ln(1 - b) \geq b$ for $0 \leq b < 1$, summing gives

$$\sum_{\text{eliminations}} b \leq \ln K. \tag{D.2}$$

For completeness, $\ln t \leq t - 1$ for $t > 0$ follows by minimizing $t - 1 - \ln t$, whose derivative is $1 - 1/t$. Taking $t = 1 - b$ proves the displayed inequality. The target weight starts at $1/K$ and never exceeds one.

Each peel multiplies u by at most $1 - r/4$. Immediately before every peel, $u \geq \epsilon/8$. Before the j th peel,

$$\epsilon/8 \leq (1 - r/4)^{j-1} \leq e^{-(j-1)r/4}.$$

Hence there are at most P peels. Their combined wrong-color mass is at most $3P\tau = \epsilon/8$, since a peeling answer implies $w \leq 3\tau$. At a normal stop the residual mass is at most $\epsilon/4 + \tau$, or zero. The error at such a stop is at most $\epsilon/2$.

It remains to bound the probability of reaching the round limit. Define a round's progress by $G = a$ for a peel, $G = b$ for an elimination, and $G = 0$ for a rejected proposal. Condition on all public history through the current stopping answer $\hat{u}$, but before the fresh rectangle draw. This includes every earlier query and response; conditional on that history the next rectangle has the prescribed mixture law. Always $0 \leq G \leq 1$. For every accepted rectangle, either action has $G \geq ab$. All rectangles with $a \geq 3r/4$ are accepted. Averaging (3.3) under D restricted to U gives $\mathbb{E}[ab] \geq r$. Consequently,

$$\mathbb{E}[G \mid \text{history}] \geq \mathbb{E}[ab] - \mathbb{E}[ab\mathbf{1}_{a < 3r/4}] \geq r - (3r/4)\mathbb{E}b \geq r/4. \quad (\text{D.3})$$

This holds for either oracle-induced action on every accepted rectangle. The oracle policy was fixed at the start of the proof and may react to each query; no supremum over policies is moved inside an expectation.

On any trajectory of nonstopping rounds, the sum of all peeling a values except the last is at most the logarithm of the reciprocal residual mass immediately before the last peel. Indeed $a \leq -\ln(1 - a)$ and the residual ratios telescope. This sum is at most $\ln(8/\epsilon)$; the last a is at most one. If there are no peels the sum is zero. Together with (D.2), this proves

$$\sum G \leq B. \quad (\text{D.4})$$

There is no logarithm of zero: a peel with $a = 1$ must be the last peel.

After any normal termination, extend the sequence by setting $G = 1$. Let $p = r/4$. For $0 \leq g \leq 1$, convexity places e^{-g} below the chord between its endpoint values:

$$e^{-g} \leq 1 - (1 - e^{-1})g.$$

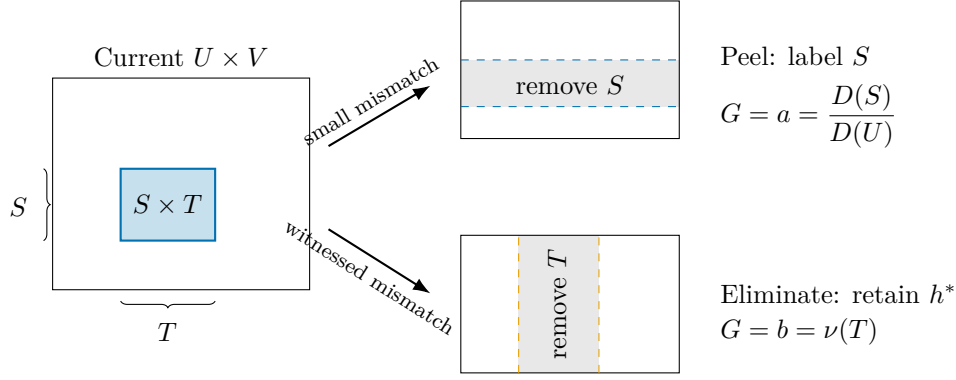
Since $1 - e^{-1} > 1/2$, conditional expectation and (D.3) imply

$$\mathbb{E}[e^{-G} \mid \text{history}] \leq e^{-p/2}.$$

Induction gives $\mathbb{E}e^{-\sum_{t=1}^R G_t} \leq e^{-pR/2}$. On the event of no normal termination by the budget, (D.4) holds. Markov's inequality therefore gives

$$\Pr[\text{budget termination}] \leq e^{B-pR/2} \leq \epsilon/2.$$

A point that becomes empty after the final peel already has the normal-stop error bound; counting it as budget termination only weakens this estimate. Finally, error is always at most one, so expected error is at most $\epsilon/2 + \epsilon/2 = \epsilon$. There are at most three queries per round and at most one extra stopping check. The big- O and Ω bounds follow directly from the ceilings and $\rho \geq 1$, $\epsilon < 1/4$. $\square$



Either accepted action has $G \geq ab$; pointwise coverage forces $\mathbb{E}[G] \geq r/4$.

Figure 1: One RECTIFY proposal. The horizontal axis indexes surviving hypotheses and the vertical axis unpeeled points. A nearly monochromatic region is labeled and removed; a disagreement removes the rectangle’s hypotheses. The two actions are charged to different potentials, but both dominate the product ab (Theorem 1).

Domain-size consequence of Theorem 1. The dependence on $\log K$ can always be bounded in terms of the domain and ρ . If H shatters $v \geq 1$ points, put the uniform marginal on those points and the uniform prior on one representative of each of their 2^v label patterns. A nonempty monochromatic rectangle using s of the points has mass at most $s/(v2^s) \leq 1/(2v)$, because $s \leq 2^{s-1}$ for every positive integer s (by induction). Therefore

$$\text{VC}(H) \leq \rho/2.$$

After removing duplicate rows, the finite counting inequality $K \leq \sum_{i=0}^v \binom{|X|}{i}$ follows by induction on $|X|$: split restrictions on the last point into their union A and intersection B . Their sizes sum to K , the VC dimension of A is at most v , and that of B is at most $v - 1$, since shattering a set in B also permits either label on the last point. Pascal’s identity completes the induction, with the empty domain as base case. For $v = 0$, all rows are identical and $K = 1$. For $1 \leq v \leq Q = |X|$, setting $t = v/Q \leq 1$ gives

$$t^v \sum_{i=0}^v \binom{Q}{i} \leq (1+t)^Q \leq e^v, \quad \ln K \leq v \ln(eQ/v) \leq (\rho/2) \ln(eQ).$$

Consequently the same learner, with no statistical-dimension conversion, has

$$m = O(\rho^2 \ln(e|X|) + \rho \ln(1/\epsilon)), \quad \tau = \Omega(\epsilon/(\rho \ln(1/\epsilon))).$$

For bounded ρ , this is logarithmic in the domain size at fixed accuracy. The argument is a proved consequence, not a conjectured converse characterizing SQ complexity.

E The two updates in RECTIFY

F The grid construction and counting bound

Proof of Theorem 4. Define

$$u_p = (x^2, xy, y^2, x, y, 1), \quad v_\ell = (a^2, -2a, 1, 2ab, -2b, b^2 - 1/2).$$

Their dot product is $(ax + b - y)^2 - 1/2$. It is negative on incidences and positive on nonincidences because all coordinates are integers. Thus the six-dimensional strict factorization proves $\text{signrank}(F) \leq$

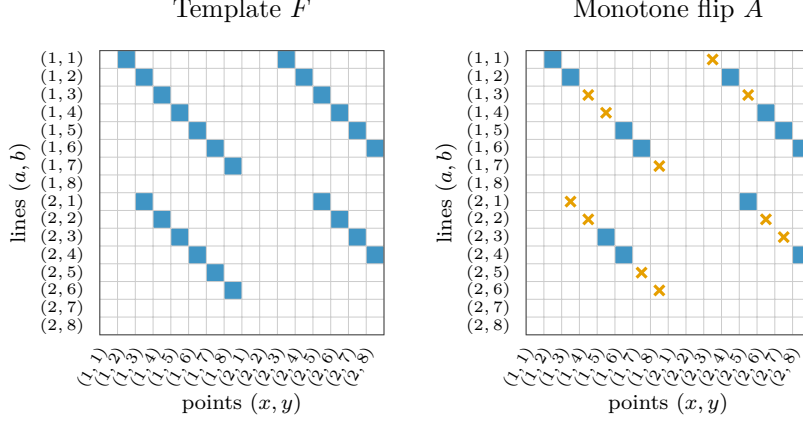


Figure 2: The construction at $N = 2$. Blue cells are negative incidences $y = ax + b$; white cells are positive. The right panel changes a fixed subset of the negative cells to positive, marked by orange crosses. Rows and columns retain their geometric indices. Larger scales supply the counting separation in Theorem 4.

6. Both vectors are nonzero and may be normalized to unit vectors. Every row of A has at most N negative points. Enumerate all domain points by $t \in [K]$ and use $(1, t, \dots, t^{2N})$. For a row with negative index set S , the polynomial $\prod_{a \in S} (t - a)^2 - 1/2$ is negative exactly on S and positive at every other integer domain point. The empty product is one. This proves $\text{dc}(H_A) \leq 2N + 1$ for every member.

For fixed a, x , exactly $2N^2 - ax$ values of $b \in [2N^2]$ give an incidence with $y \in [2N^2]$. Summing proves (4.1). The lower bound uses $(N + 1)/2 \leq N$. If two distinct affine lines share points (x, y) and (x', y') , subtraction gives $(a - c)(x - x') = 0$. If $a = c$, their shared point forces equal intercepts, a contradiction. Otherwise $x = x'$ and then $y = y'$. Hence F , and therefore A , has no all-negative 2×2 . Shattering three columns would require one row negative on all three and a distinct row negative on two of them, which creates such a submatrix. Therefore $\text{VC}(A) \leq 2$.

The homogeneous-rectangle theorem of Alon et al. [4], in the formulation checked in Hatami et al. [24, Theorem 1.9], states that an $a \times b$ sign matrix of sign-rank at most d contains a monochromatic rectangle with each side at least the corresponding side length divided by 2^{d+1} . Apply Lemma 22, since sign-rank is unchanged by repetition. It follows that

$$\text{rect}(F) \geq 2^{-2d-2}. \quad (\text{F.1})$$

Fix a product distribution and such a rectangle of F . If its color is positive, it survives every monotone flip. If its color is negative, one side is a singleton, by the absence of a negative 2×2 . Partition the other side according to its sign in A ; one part retains at least half its mass. Thus

$$\text{signrank}(F) \leq d, \quad F \text{ has no negative } 2 \times 2 \implies \text{rect}(A) \geq 2^{-(2d+3)}. \quad (\text{F.2})$$

For $d = 6$ this is 2^{-15} .

For the counting bound, we use the strict-pattern formulation of Warren's theorem [33] given in Alon et al. [5, Theorem 21 and Lemma 22]: $s \geq v$ real polynomials of degree at most k in v variables have at most $(4eks/v)^v$ strict sign patterns. A rank-at-most- d $K \times K$ matrix factors as a $K \times d$ matrix times a $d \times K$ matrix. Its K^2 entries are degree-two polynomials in $2Kd$ variables. Therefore, for $1 \leq d \leq K/2$,

$$\#\{B \in \{-1, +1\}^{K \times K} : \text{signrank}(B) \leq d\} \leq (4eK/d)^{2Kd} \leq 2^{2KdB_0}. \quad (\text{F.3})$$

The random family has exactly 2^I equiprobable matrices. Taking $d = s_0$ bounds the probability by $2^{2Ks_0B_0-I} \leq 2^{-I/2}$. The range $s_0 \leq K/2$ follows from $I \leq K^2$ and $B_0 > 1$; when $s_0 = 0$ the event is empty. Finally $I \geq N^4$ and $K = 2N^3$ give the asymptotic lower bound. Apply Theorem 1 for the learner. $\square$

F.1 Proof of Theorem 5

Proof of Theorem 5. Let J_ℓ be the set of template incidences in row ℓ , with $k_\ell = |J_\ell|$. For each nonempty J_ℓ , take D_ℓ uniform on it. If a law on d -dimensional embeddings witnesses $\text{dc}^\eta(H_A) \leq d$, then

$$\mathbb{E}_\phi \sum_{\ell: k_\ell > 0} \frac{k_\ell}{I} \min_w L_{D_\ell, h_\ell}(\text{sign}\langle w, \phi \rangle) \leq \eta.$$

For a fixed embedding, only finitely many Boolean predictions are possible, so each infimum is attained as a minimum of finitely many loss values. There exists one embedding with the displayed weighted sum at most η . Otherwise its positive excess over η would have strictly positive expectation. Choose one minimizing weight vector for each row. Their prediction matrix makes at most ηI mistakes on the incidence positions.

With $\text{sign}(0) = +1$, this prediction matrix has strict sign-rank at most $d + 1$, not necessarily at most d . Add the constant feature one. In each row add a strictly positive bias smaller than the absolute value of every negative score in that row, if there are negative scores. This retains all signs and turns every zero into a positive score. Finiteness makes the bias choice possible. Rows with no incidence may use any weight vector.

For any fixed sign matrix B , the number of possible incidence flip patterns within ηI mistakes is at most

$$\sum_{j=0}^{\lfloor \eta I \rfloor} \binom{I}{j} \leq 2^{h_2(\eta)I}.$$

For $0 < \eta < 1/2$, this follows from

$$1 \geq \sum_{j \leq \eta I} \binom{I}{j} \eta^j (1-\eta)^{I-j} \geq 2^{-h_2(\eta)I} \sum_{j \leq \eta I} \binom{I}{j};$$

the second inequality uses $j \leq \eta I$ and $\eta/(1-\eta) < 1$. For $\eta = 0$ the sum is one. Combining with (F.3), the probability that any rank-at-most- s_η sign matrix makes at most ηI incidence errors is at most

$$2^{2Ks_\eta B_0 + h_2(\eta)I - I} \leq 2^{-c_\eta I/2}.$$

An embedding of dimension below s_η would produce just such a strict sign matrix after adding the constant coordinate. If $s_\eta = 0$, the event in question is empty. This proves (4.2). Combining it with Theorem 4 and Lemma 6 gives the two asymptotic dimension bounds in the statement; the sum of the two exceptional probabilities tends to zero. $\square$

F.2 Proof of Lemma 6

Proof of Lemma 6. Choose a full-support distribution on X . Zero loss under this distribution means correct labels at every point. Draw $t = \lceil \log_2 K \rceil + 1$ independent embeddings from a law witnessing $\text{dc}_{1/2}(H) = d$. Each fixed target fails in all t embeddings with probability at most 2^{-t} . A union bound gives probability at most $K2^{-t} < 1$ that some target fails everywhere. Hence one collection works for all targets. Concatenate its t embeddings; each row selects a successful block. Adding one constant coordinate as above makes the representation strict. This proves (4.3). $\square$

Resource	Incidence flips on $2M$ points, $M = N^3$
VC dimension; rectangle ratio	$VC \leq 2$; $\text{rect} \geq 2^{-15}$.
Strict dimension, most flips	$\Omega(M^{1/3}/\log M) \leq \text{dc} \leq 2M^{1/3} + 1$.
Exact probabilistic dimension	$\text{dc}_{1/2} = \Omega(M^{1/3}/\log^2 M)$.
Averaged dimension	$\text{dc}^\eta = \Omega_\eta(M^{1/3}/\log M)$ for fixed $\eta < 1/2$.
Table-known learners	$m = O_\epsilon(\log M)$, $\tau = \Omega_\epsilon(1)$; proper and deterministic; table-polynomial time.
Key-known median learner	$m = O_\epsilon(n^2)$, $\tau = \Omega_\epsilon(1)$; deterministic improper for every key; proper for regular stars.
Conditional high dimension	$\text{dc} \geq n^c/(30 \log_2 n)$ for almost every PRF key, each fixed c .
Refuted conclusions	(L), (P), (Pn) for the three nondegenerate dimensions; every prescribed degree for the corresponding succinct families.

Table 12: The separating family’s parameters. The probabilistic lower bounds concern each prescribed fixed $\eta < 1/2$ and hold with exponentially small failure probability. A polynomial-time evaluator with a supplied key is distinguished from deterministic selection of a good key.

F.3 Proof of Corollary 7

Proof of Corollary 7. Fix the required constant accuracies and intersect the high-probability events of Theorems 4 and 5. Their failure probability is at most $2^{-I/2} + 2^{-(1-h_2(\eta))I/2} < 1$ for all sufficiently large N . Thus one matrix satisfies the two lower bounds simultaneously. Apply Lemma 6 for exact probabilistic dimension, and Theorems 9 and 10 for the proper deterministic learner. Since $M = N^3$ and $n = \log_2(2M)$, the dimension lower bounds grow exponentially in n , up to powers of $\log M$, whereas $m = O_\epsilon(n)$ and $1/\tau = O_\epsilon(1)$. No fixed polynomial in the proposed variables can dominate these lower bounds. $\square$

G Query lower bounds on target-line marginals

Proof of Theorem 8. Let $Q = 2N^3$ be the number of points and let $\mathcal{L}_0 = [N] \times [N^2]$. These $L = N^3$ lines each have exactly N template points in the domain. All their incidence bits are independent, including bits assigned by different rows at a shared point.

Put $\eta_0 = 3/8$ and $\chi = 1 - h_2(\eta_0) > 0$. Fix any Boolean function g on the Q points. For each $\ell \in \mathcal{L}_0$, the number of errors of g against its random row on the N template points is $\text{Bin}(N, 1/2)$. The entropy estimate used in Theorem 5 gives

$$\Pr[L_{D_\ell, h_\ell}(g) \leq \eta_0] \leq 2^{-\chi N}.$$

For this fixed g , the events for distinct rows are independent. Let $s = \lceil 8N^2/\chi \rceil$. If $s > L$, the assertion that no g fits s rows is automatic. Otherwise, a union bound over the 2^Q functions and the s -subsets of rows gives

$$\begin{aligned} \Pr[\exists g : \#\{\ell : L_{D_\ell, h_\ell}(g) \leq \eta_0\} \geq s] &\leq 2^Q \binom{L}{s} 2^{-\chi N s} \\ &\leq 2^{Q+s \log_2 L - \chi N s} \leq 2^{Q-\chi N s/2} \leq 2^{-Q}. \end{aligned}$$

The third inequality uses $\chi N \geq 6 \log_2 N = 2 \log_2 L$ and the fourth uses $s \geq 8N^2/\chi$. On the complementary event every output is good for at most $s - 1 < 8N^2/\chi$ of the target-line instances. This event is simultaneous over all outputs, including outputs chosen with full knowledge of A .

For the proper bound, require in addition that every row of $\mathcal{L}_0$ has at least $3N/8$ negative incidences. The probability that this fails is at most $Le^{-N/32}$ by the bounded-differences estimate in

Algorithm 3: Cap sampler from a strict template

Input: Unpeeled points U , surviving rows V , and the uniform prior ν on V .

- 1 Use the fixed general-position perturbation and place V in radial-isotropic position in effective dimension j ;
 - 2 Set $\alpha = (2j)^{-1/2}$, $\beta = \sqrt{1 - \alpha^2}$; draw u uniformly on S^{j-1} ;
 - 3 Set $S \leftarrow U \cap \{x : \langle u, z_x \rangle \geq \beta\}$ and $T^\pm \leftarrow \{h : \pm \langle u, v_h \rangle > \alpha\}$;
 - 4 Choose the larger of T^+, T^- under ν ;
 - 5 **if** *the positive side is chosen* **then**
 - 6 **return** $(S, T^+, +1)$;
 - 7 The negative rectangle has a singleton side. Split the opposite side according to its labels in A , choose either color with probability $1/2$, and return that monochromatic rectangle;
-

Appendix U. If g is a different row of the full class, the two template lines share at most one point. Thus

$$L_{D_\ell, h_\ell}(g) \geq 3/8 - 1/N > 1/3 \quad (N \geq 25).$$

No proper output is $1/3$ -good for more than one of these L targets. Duplicate hypotheses cannot cause an exception: a selected target has more than one negative point, whereas two distinct template rows cannot share two negative points.

Fix a matrix on both events. Let $B_\tau = \lceil 1/\tau \rceil \geq 2$, and partition $[-1, 1]$ into B_τ equal intervals, each represented by its midpoint. Answer every query by the midpoint for its true expectation. The error is at most $1/B_\tau \leq \tau$. This is a fixed valid oracle for each instance, independent of the learner's private seed. For each fixed seed, the depth- m response tree has at most B_τ^m leaves, after padding early stops. Query functions can depend on the seed; their number adds no branching once the seed is fixed.

For unrestricted outputs, at most $B_\tau^m(s-1)$ of the L instances have loss at most η_0 for that seed. Averaging this bound over the seed is legitimate because the matrix event holds for every possible output. On the other hand, the expected-error guarantee on each instance and Markov's inequality imply

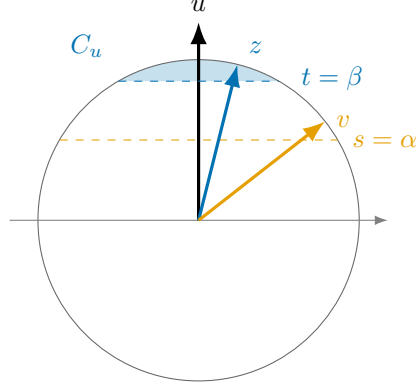
$$\Pr_\omega[L_{D_\ell, h_\ell}(A_\omega^O) \leq \eta_0] \geq 1 - \epsilon/\eta_0 = 1 - 8\epsilon/3.$$

Averaging over targets and comparing yields

$$B_\tau^m \geq \frac{(1 - 8\epsilon/3)L}{s-1} > \frac{\chi(1 - 8\epsilon/3)N}{8}.$$

For proper outputs, repeat the argument with threshold $1/3$ and one good target per leaf to get $B_\tau^m \geq (1 - 3\epsilon)N^3$. Taking base-two logarithms proves both bounds. The union of the two exceptional matrix events has probability at most $2^{-2N^3} + N^3e^{-N/32}$. These events can also be intersected with any fixed-accuracy dimension event in Theorems 4 and 5. $\square$

Why the proper coefficient does not transfer to arbitrary outputs. For a fixed slope a , define $g_a(x, y)$ by the label of the row $(a, y - ax)$ at (x, y) whenever that intercept is valid, and set it to $+1$ otherwise. A point belongs to at most one line of that slope. Therefore g_a has zero loss for every one of the N^2 full-length rows of slope a , for every flip matrix. Independent row bits do not force these instances to have distinct good improper outputs. The unrestricted argument above counts the maximum number of good targets per output instead.



$$\begin{aligned}
t &= \langle u, z \rangle \geq \beta = \sqrt{1 - \alpha^2} \\
s &= \langle u, v \rangle > \alpha \\
\langle z, v \rangle &\geq ts - \sqrt{1 - t^2} \sqrt{1 - s^2} > 0 \\
&\text{because } s^2 + t^2 > 1.
\end{aligned}$$

The point cap and the row threshold differ. A negative row side is handled by a singleton-side split.

Figure 3: Cap geometry in a two-dimensional section. The drawing uses $\alpha = 1/2$ and $\beta = \sqrt{3}/2$. Blue denotes point vectors in the cap; the orange row vector satisfies the weaker row-side threshold. The strict inequality proves the common template sign used in Theorem 9.

H Cap sampler and geometry

I Proofs for cap mixtures and proper learning

I.1 Proof of Theorem 9

Proof of Theorem 9. We first make the once-only perturbation precise. Number the rows $j = 1, \dots, K$. Replace a row vector v_j by

$$v_j(t) = v_j + t(1, j, \dots, j^{d-1}).$$

Choose a positive rational upper bound t_0 so that every dot product retains its strict sign for $0 < t < t_0$. The finitely many original rational dot products and the perturbation dot products give such a bound by direct comparison. For each d -subset of rows, its determinant is a polynomial in t of degree d whose leading coefficient is a nonzero Vandermonde determinant. At most $d \binom{K}{d}$ values are forbidden. Testing that many plus one rational grid values in $(0, t_0)$ finds a valid value, with polynomial bit length and polynomial work for fixed d . When $K < d$, no perturbation is needed: use the evaluation-coordinate construction below for every survivor set. For $K \geq d$, the same perturbation works for every later survivor set.

Suppose there are $k > d$ surviving rows. Any proper j -dimensional subspace contains at most j of them, strictly less than kj/d . The radial-isotropy proof of Theorem 27 therefore applies; its convex formulation is also proved in Appendix T. Transform and normalize the rows to vectors $v_h \in S^{d-1}$ with

$$\mathbb{E}_\nu v_h v_h^\top = I_d/d.$$

Transform points contragrediently and normalize them to z_x . All template signs remain strict. If $k \leq d$, retain *all* rows and replace the factorization by evaluation coordinates: let $\tilde{z}_x = (\langle v_h, u_x \rangle)_{h \in V}$ and use row vectors $e_h \in \mathbb{R}^k$. Every coordinate of $\tilde{z}_x$ is nonzero by strictness, so its normalization is defined and preserves every sign. For the uniform surviving-row prior, $k^{-1} \sum_{h \in V} e_h e_h^\top = I_k/k$. This is a new factorization, not an invertible map of the original ambient space. No target is discarded.

Write the effective dimension as $j \leq d$, put $\alpha^2 = 1/(2j)$ and $\beta = \sqrt{1 - \alpha^2}$, and sample u uniformly on S^{j-1} . Define

$$C_u = \{x : \langle u, z_x \rangle \geq \beta\}, \quad T_u^\pm = \{h : \pm \langle u, v_h \rangle > \alpha\}.$$

For $Z = \langle u, v_h \rangle^2 \in [0, 1]$, isotropy gives $\mathbb{E}_\nu Z = 1/j$. Since

$$\mathbb{E} Z \leq \alpha^2 + \Pr[Z > \alpha^2],$$

we have $\nu(T_u^+) + \nu(T_u^-) \geq 1/(2j)$. Choose the larger side, of mass at least $1/(4j)$.

The geometry is strict. If $s = \langle u, v \rangle > \alpha$ and $t = \langle u, z \rangle \geq \beta$, then

$$\langle z, v \rangle \geq ts - \sqrt{1-t^2}\sqrt{1-s^2} > 0,$$

because $s, t > 0$ and $s^2 + t^2 > 1$. Replacing v by $-v$ proves the negative-side assertion. Thus $C_u \times T_u^+$ is positive in F , and $C_u \times T_u^-$ is negative in F .

A positive rectangle survives in A . A negative rectangle has a singleton side. If its row side is $\{h\}$, partition $C_u \cap U$ according to $h(x)$ in A and choose either color with probability $1/2$. If its point side is $\{x\}$, partition the row side according to $h(x)$ and choose either color with probability $1/2$. Empty outcomes are allowed. The singleton-point case is essential and is not replaced by a singleton-row assumption. Conditional on $x \in C_u$, its expected weighted coverage after this split is at least half the chosen side's mass, hence at least $1/(8j)$. Rotational invariance gives probability c_j of cap membership. Therefore the pointwise bound is $c_j/(8j) \geq c_d/(8d)$. To verify the last inequality, couple the first coordinate of uniform spheres by normalizing an increasing list of independent standard Gaussians. Its absolute value decreases as dimension grows, while $\sqrt{1-1/(2j)}$ increases. Symmetry supplies the factor $1/2$, including the case $j = 1$.

For finite-bit computation, Appendix T gives a deterministic rational transform with normalized covariance at least $3I_j/(4j)$. The preceding second-moment argument then gives total side mass at least $1/(4j)$, selected-side mass at least $1/(8j)$, and split coverage at least $1/(16j)$ whenever a point belongs to the cap.

Use the finite direction set

$$W_j = \left\{ \frac{g}{\|g\|} : 0 \neq g \in (16j)^{-1}\mathbb{Z}^j \cap [-1, 1]^j \right\}, \quad J_j = |W_j| \leq (32j+1)^j.$$

Repetitions of a direction cause no difficulty. Rounding a unit vector coordinatewise produces g with $\|g - z\| \leq 1/(32\sqrt{j})$; normalization changes it by at most twice this amount. Thus $\langle z, g/\|g\| \rangle \geq 1 - 1/(512j) > \beta$, where the last inequality follows from $\beta \leq 1 - 1/(4j)$. Every point is in at least one cap. Uniformly sampling W_j and the split color yields coverage at least $1/(16jJ_j)$. The common rational lower bound

$$r_d = \frac{1}{16d(32d+1)^d} \tag{I.1}$$

works in all effective dimensions. All membership comparisons are rational: for example $\langle z, g/\|g\| \rangle \geq \beta$ means that the unnormalized dot product is positive and

$$\langle z_{\text{raw}}, g \rangle^2 \geq (1 - 1/(2j))\|z_{\text{raw}}\|^2\|g\|^2.$$

The row-side tests use the analogous strict squared comparison. Consequently no exact sampling of a continuous direction is required. Theorem 1 with r_d proves the randomized finite-bit bound.

For determinism, enumerate the at most $2J_j$ outcomes and query each $D(S)$, using the same tolerance and stopping check as Theorem 1 with r_d . Choose the outcome maximizing $b\hat{s}$, where $b = \nu(T)$. Coverage implies that some outcome has $bs \geq r_d u$. The chosen one has

$$bs \geq r_d u - 2\tau, \quad ab \geq r_d - 2\tau/u \geq 3r_d/4.$$

Thus both a and b are at least $3r_d/4$. Query its mismatch mass and peel or eliminate exactly as before. The target survives, and all error bounds remain valid. Each stage has progress at least $3r_d/4$, whereas the deterministic total-progress bound is B . There cannot be more than $\lceil 4B/(3r_d) \rceil + 1$ nonstopping stages. There are $O_d(1)$ queries per stage. This proves the deterministic claim, without fixing a successful private seed after seeing D . $\square$

I.2 Proof of Theorem 10

Proof of Theorem 10. Set $\theta = \epsilon/4$ and run RECTIFY at accuracy θ , stopping at its first proposed negative peel. Until then all assigned labels, and its fallback, are positive. If it stops normally without a negative peel, output the all-positive member of H . The error is at most $\theta/2$. Its budget-failure probability is at most $\theta/2$.

Suppose a negative rectangle $S \times T$ reaches a peeling answer. The acceptance calculation in Theorem 1 and $u > \theta/8$ give

$$D(S) > 5r\theta/128.$$

For $P_\theta = 1 + \lceil 4r^{-1} \ln(8/\theta) \rceil$ and $\tau = \theta/(24P_\theta)$,

$$\frac{D(S)}{\tau} > \frac{15}{16}rP_\theta > \frac{15}{4} \ln(8/\theta) > 18,$$

since $\theta < 1/16$. A peeling answer implies $D(S \cap \{h^* = +1\}) \leq 3\tau$. Hence the negative mass in S exceeds 15τ .

If $S = \{p\}$, then $h^*(p) = -1$ and p is known. Otherwise the negative rectangle has a single row h_0 . Query its full loss, accepting it when the answer is at most $\epsilon/2$. An accepted candidate has true loss at most $\epsilon/2 + \tau$. If it is rejected, $h_0 \neq h^*$, since the target's answer would be at most τ . The two rows share at most one negative point. Therefore the negative measure in S is concentrated at a unique point p and has mass greater than 15τ . Locate p by binary search over an arbitrary ordering of S : for a half $B \subseteq S$, query $D(B \cap \{h^* = -1\})$. The answer is at most τ when $p \notin B$, and greater than 14τ when $p \in B$. Threshold 4τ safely determines the half. This takes at most $\lceil \log_2 |X| \rceil$ queries.

Now let $V_p = \{h : h(p) = -1\}$. Its negative supports away from p are pairwise disjoint; otherwise two rows share two negative points. Query the target's negative mass away from p ,

$$\beta = D(\{x \neq p : h^*(x) = -1\}).$$

If the answer exceeds $\epsilon/4$, then $\beta > \epsilon/4 - \tau$. For any half $W \subseteq V_p$, define

$$Q_W(x, y) = \mathbf{1}_{\{y=-1\}} \mathbf{1}_{\{x \neq p\}} \mathbf{1}_{\{x \in \bigcup_{h \in W} \{h=-1\}\}}.$$

Its expectation is exactly β if $h^* \in W$ and exactly zero otherwise. Thus threshold $\epsilon/8$ distinguishes the two cases, since $\tau < \epsilon/16$. Binary search finds the target row in at most $\lceil \log_2 K \rceil$ queries. If two indexed rows coincide and contain only p as a negative point, this branch cannot occur; in the positive-tail branch the target row is unique.

If the answer for β is at most $\epsilon/4$, then $\beta \leq \epsilon/4 + \tau$. Test up to

$$J = \lceil 8/\epsilon \rceil + 1$$

distinct members of V_p , again accepting an estimated full loss at most $\epsilon/2$. If fewer than J members exist, these tests include the target. Otherwise their negative masses outside p sum to at most one, by disjointness. One tested row h has such mass at most $1/J < \epsilon/8$. It agrees at p , and therefore

$$L_{D, h^*}(h) \leq \beta + D(\{x \neq p : h(x) = -1\}) < 3\epsilon/8 + \tau.$$

Its answer is less than $\epsilon/2$, because $2\tau < \epsilon/8$. Some test consequently passes. Every accepted row has loss at most $\epsilon/2 + \tau$.

All nonfailure outcomes are proper and have loss at most $\epsilon/2 + \tau$. Any budget failure can output the all-positive row, with loss at most one. Expected loss is bounded by

$$\epsilon/2 + \tau + \theta/2 < \epsilon.$$

The extra query counts give (5.2). Using the deterministic net implementation removes the budget-failure event and leaves the same proper completion. Its computation consists of cap tests, row-label tests, and unions of explicitly listed row supports, all polynomial in the table size. $\square$

Algorithm 4: Order-median learner

Input: An entry evaluator for A , the scale N , and $0 < \epsilon < 1/4$.

- 1 Set $\zeta = \epsilon/4096$. Query the total negative mass β ;
- 2 **if** $\hat{\beta} \leq \epsilon/4$ **then**
- 3 **return** the all-positive row;
- 4 Find a noisy median of the negative x -marginal;
- 5 **if** *its atom is heavy* **then**
- 6 Locate its negative point and use the star completion;
- 7 Fix the balanced prefix P at the light median; maintain a slope interval containing a^* ;
- 8 **while** *the slope interval has more than one element* **do**
- 9 At its midpoint a , find a noisy median v of $f_a(x, y) = y - ax$;
- 10 **if** *the median atom is heavy* **then**
- 11 Its intersections with P and P^c either identify $(a^*, b^*) = (a, v)$ or locate one negative point for the star completion;
- 12 Query $\mu^-(P \cap \{f_a \leq v\})$;
- 13 Compare its answer to $3\hat{\beta}/8$ and retain the correct strict half of the slope interval;
- 14 Find the constant residual b^* and return h_{a^*, b^*} ;

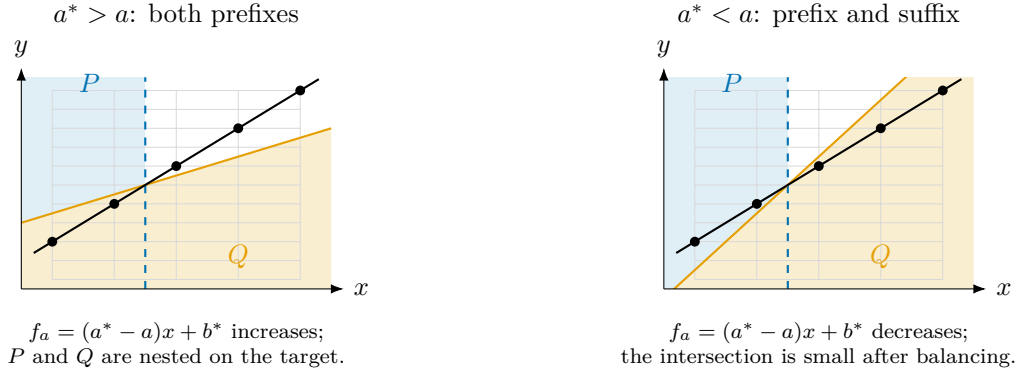


Figure 4: The order test behind Theorem 12. The target is $y = 2x + 1$. The blue region is an x -prefix; orange denotes $y - ax \leq v$. On the target, the orange region becomes a prefix when the trial slope is too small and a suffix when it is too large. The proof uses noisy median mass bounds to separate these cases by a constant fraction of the negative mass.

J The order-median algorithm and proof

J.1 Proof of Lemma 11

Proof of Lemma 11. Start with the bracket $[L - 1, H]$, regarding the lower endpoint as a negative comparison and the upper as a positive comparison. The corresponding true masses are zero and β , so both satisfy the claimed endpoint inequalities. At an interior query, an answer at least $\hat{\beta}/2$ implies true mass at least $\beta/2 - 3\zeta/2$; an answer below it implies true mass below $\beta/2 + 3\zeta/2$. Preserve the respective endpoint inequalities during binary search. They remain valid even if different valid answers do not form a monotone sequence. At termination the endpoints are $v - 1, v$, proving the first two assertions.

In the light-atom case, $m \leq \hat{\beta}/8 + \zeta \leq \beta/8 + 9\zeta/8$. Add this to the upper bound on $\mu^-\{f < v\}$ to obtain (5.3). In the other case $m > \hat{\beta}/8 - \zeta \geq \beta/8 - 9\zeta/8 > \beta/9$, since $\beta \geq 512\zeta$. $\square$

J.2 Proof of Theorem 12

Proof of Theorem 12. Query β , obtaining $\hat{\beta}$. If $\hat{\beta} \leq \epsilon/4$, return the all-positive row $(a, b) = (1, 2N^2)$. Its loss is at most $\epsilon/4 + \zeta$. Otherwise

$$\beta > \epsilon/4 - \zeta > \epsilon/8, \quad \beta > 1023\zeta > 512\zeta.$$

All subsequent negative-measure queries are unconditioned indicators and have error at most ζ .

First take a noisy median v_x of x . If its atom is heavy in Lemma 11, it consists of one negative point: a nonvertical target line has at most one point with a given x coordinate. Binary queries $\mu^-\{x = v_x, y \leq t\}$ locate its y coordinate. Each such query has true mass either zero or more than $\beta/9$. Threshold $\hat{\beta}/32$ distinguishes these cases despite error ζ : it exceeds ζ and is less than $\beta/9 - \zeta$. We then use the heavy-point completion described below.

Otherwise fix the prefix $P = \{x \leq v_x\}$, whose mass satisfies (5.3). Maintain an integer interval containing the unknown slope $a^* \in [1, N]$. At its midpoint a , take a noisy median v of

$$f_a(x, y) = y - ax, \quad 1 - N^2 \leq f_a(x, y) \leq 2N^2.$$

On the target line this is $(a^* - a)x + b^*$. Query the mass m of its median atom.

Suppose that atom is heavy. Query its intersection mass l with P . If both $\hat{l} \geq \hat{\beta}/8$ and $\hat{m} - \hat{l} \geq \hat{\beta}/8$, then $a = a^*$ and $v = b^*$. Indeed, for $a \neq a^*$, f_a is injective on the target line, so its atom is a single point and lies wholly on one side of P . One of the true split masses is zero. Its directly observed value is at most ζ , or its difference of two observed values at most 2ζ , both less than $\hat{\beta}/8$. Conversely, when $a = a^*$, the atom has mass β and the median value is exactly b^* . The two true split masses are at least $\beta/2 - 3\zeta/2$ and $3\beta/8 - 21\zeta/8$ by (5.3). Their observed lower bounds exceed $\hat{\beta}/8$, using $\beta > 512\zeta$. Thus the test accepts precisely the constant-residual case. Return the exact target row in that case. If the split test fails, the heavy atom is a single negative point; locate its x coordinate by binary queries restricted to $\{f_a = v\}$, then put $y = ax + v$ and use the heavy-point completion.

It remains to handle a light residual atom. The set $Q = \{f_a \leq v\}$ also satisfies (5.3). If $a^* > a$, the restrictions of P, Q to the target line are prefixes of the same order. Consequently

$$\mu^-(P \cap Q) \geq \beta/2 - 3\zeta/2.$$

If $a^* < a$, they are a prefix and a suffix. Their union covers the entire negative support whenever they overlap. Thus, whether or not they overlap,

$$\mu^-(P \cap Q) \leq \max\{0, \mu^-(P) + \mu^-(Q) - \beta\} \leq \beta/4 + 21\zeta/4.$$

Query this intersection and compare its answer to $3\hat{\beta}/8$. In the first case the answer is at least $\beta/2 - 5\zeta/2 > 3\hat{\beta}/8$; in the second it is at most $\beta/4 + 25\zeta/4 < 3\hat{\beta}/8$. Both inequalities follow from $\beta > 512\zeta$. Equality of slopes is impossible in the light case, because the entire negative mass would be one residual atom. Retain the appropriate strict half of the slope interval. After $O(\log N)$ stages only a^* remains. A median query search for its constant residual gives b^* , so return h_{a^*, b^*} . Each slope stage uses $O(\log N)$ queries. This is the all-light algorithm.

Heavy-point completion. We have located an actual negative point p . Query the negative tail

$$\beta_p = \mu^-(X \setminus \{p\}).$$

If its answer exceeds $\epsilon/4$, then $\beta_p > \epsilon/4 - \zeta$. For a half-interval of slopes in the star at p , query negative mass away from p on the union of those lines. Every negative point other than p determines the same unique target slope, so this query is exactly zero or β_p . Threshold $\epsilon/8$ distinguishes the

cases. Binary search identifies the target in $O(\log N)$ queries. The union is succinctly evaluable: for a query point $q \neq p$, test $q_x \neq p_x$, divisibility of $q_y - p_y$ by $q_x - p_x$, and membership of the quotient in the chosen slope interval. This requires no enumeration and no evaluation of flips.

If the tail answer is at most $\epsilon/4$, the improper learner returns the point predictor that is negative only at p . Its loss is $\beta_p \leq \epsilon/4 + \zeta$. This proves the deterministic improper statement for every evaluable flip matrix.

For proper output, put

$$C_0 = \max\{64\lceil\log_2(2N^3)\rceil, \lceil 128/\epsilon \rceil\}.$$

If $t(p) \leq C_0$, enumerate its slopes, discard rows positive at p by an entry evaluation, and query the full loss of each remaining row until an answer is at most $\epsilon/2$. The target is among them and must pass. A passing row has actual loss at most $\epsilon/2 + \zeta$.

If $t(p) > C_0$, star-regularity supplies at least $t(p)/4$ rows negative at p . The sets of negative points of different star rows outside p are disjoint. At most $8/\epsilon$ rows can therefore have D -mass greater than $\epsilon/8$ on their outside negative sets. At least $t(p)/4 - 8/\epsilon \geq t(p)/8$ rows are negative at p and have outside negative mass at most $\epsilon/8$. Every such row has loss at most

$$\beta_p + \epsilon/8 \leq 3\epsilon/8 + \zeta,$$

and its loss answer is at most $3\epsilon/8 + 2\zeta < \epsilon/2$, regardless of the oracle. Sample

$$J = \lceil 16 \ln(4/\epsilon) \rceil$$

independent star slopes as follows: draw $q = \lceil \log_2 t(p) \rceil$ unbiased bits, interpret them as an integer, reduce modulo $t(p)$, and add one. Every slope has probability at least $1/(2t(p))$, so a guaranteed-passing row has probability at least $1/16$ in each trial. Evaluate its label at p and test negative rows as above. The chance that no guaranteed-passing row is sampled is at most $(1 - 1/16)^J \leq \epsilon/4$. On that event return the all-positive row. All other outputs have error at most $\epsilon/2 + \zeta$. The expected loss is at most $\epsilon/2 + \zeta + \epsilon/4 < \epsilon$.

There are $O(n^2)$ median queries, $O(n)$ locating and star-search queries, and at most $O(n + 1/\epsilon + \log(1/\epsilon))$ candidate tests. Every predicate uses polynomially many n -bit integer operations and at most one entry evaluation. The output is either a row index with its evaluation algorithm or, in the improper case, a point predictor. The finite answer rounding specified before Lemma 11 completes the bit-complexity assertion. $\square$

K Proofs for the explicit families

K.1 Proof of Lemma 13

Proof of Lemma 13. When $D_k \geq 1$, we have $k \geq 32$ and $1 \leq D_k \leq k \leq K_k/2$, so the range condition in (F.3) holds. The implication $D_k \geq 1 \Rightarrow k \geq 32$ supplies the logarithmic estimate below. Thus

$$\log_2(4eK_k/D_k) \leq \log_2(8ek^3) \leq 4\log_2 k.$$

The strict-pattern bound (F.3) therefore gives at most

$$2^{2K_k D_k \cdot 4\log_2 k} \leq 2^{(2/3)k^4}$$

low-rank tables. For $k \geq 3$, $(k+1)^2/(4k^2) \leq 4/9 < 1/2$, so $I_k \geq (3/2)k^4$. Dividing the preceding count by 2^{I_k} gives at most $2^{-5k^4/6} \leq 2^{-k^4/2}$. If $D_k = 0$, no nonempty strict sign matrix has that rank.

For the decision algorithm, query the incidence bits and fill the known off-incidence entries. The condition is equivalent to feasibility of

$$A_{ij} \sum_{s=1}^{D_k} U_{is} V_{sj} \geq 1 \quad (i, j \in [K_k]).$$

Indeed a strict factorization can be scaled to have all signed scores at least one, since there are finitely many positive signed scores. The converse is immediate. There are $2K_k D_k$ real variables and K_k^2 degree-two polynomial inequalities with bounded integer coefficients. We use the singly exponential existential-real consistency bound in the formulation of Vorobjov [32], attributed there to Renegar [31]: $(sd)^{O(v)} L^{O(1)}$ bit operations for s polynomials of degree at most d , v variables, and coefficient bit length L . Substitution gives $2^{O(k^3 D_k \log k)}$. The resulting decision procedure is applied only to the polynomial-size subgrid. $\square$

K.2 Proof of Theorem 14

Proof of Theorem 14. The evaluation claim follows from integer arithmetic and one PRF evaluation. The rectangle and proper-learning bounds hold for every monotone flip, without any randomness assumption on its entries.

Take the subgrid scale $k = \lceil cn^c \rceil$, which is at most N for sufficiently large n . Restrict both the full row set and point set to $[k] \times [2k^2]$. The restricted template is the scale- k construction. Each queried bit is evaluated at the ambient encoding E_b defined in Section 6; the restriction never re-encodes an input at scale k . A uniform oracle assigns independent fair values to these distinct strings. An oracle distinguisher reads its incidence bits and accepts precisely when its sign-rank is at most D_k . Its time is

$$2^{O_c(n^{4c})} \leq 2^{n^{4c+2}} \leq 2^{\sigma^\delta}$$

for sufficiently large n . Under a uniform function its acceptance probability is at most $2^{-k^4/2}$ by Lemma 13. Under the PRF it is therefore at most $2^{-k^4/2} + \text{negl}(\sigma)$, which is negligible in n : for $p = \lceil (4c+2)/\delta \rceil$ and any $q > 0$, a bound smaller than $\sigma^{-q/p}$ is smaller than n^{-q} . Equivalently, a nonnegligible bad-key probability along admissible n would give nonnegligible advantage along the key lengths $\sigma = n^p$. Restricting a sign realization cannot increase its rank. Consequently, except on that event, the full dimension is at least $D_k + 1 > k/(24 \log_2 k)$.

For large n , $k \geq cn^c$ and $\log_2 k \leq c \log_2 n + \log_2(2c) \leq (5/4)c \log_2 n$. Hence

$$D_k + 1 > \frac{k}{24 \log_2 k} \geq \frac{n^c}{30 \log_2 n}.$$

This proves the displayed constant. The choice $k = n^c$ alone would instead give $n^c/(24c \log_2 n)$; its factor c cannot be discarded. Also $N = 2^{(n-1)/3}$, rather than $2^{n/3}$, respects the fixed definition $n = \log_2 |X|$. $\square$

K.3 Proof of Lemma 16

Proof of Lemma 16. At a fixed point with t incident rows, the number of negatives is a sum of t independent fair bits. For $\lambda = \ln 3$,

$$\Pr[Z < t/4] \leq e^{\lambda t/4} \left(\frac{1 + e^{-\lambda}}{2} \right)^t = \left(\frac{2}{3^{3/4}} \right)^t \leq e^{-t/16}.$$

For the last elementary inequality, $\ln(3^{3/4}/2) > 1/16$; for example $3^3/2^4 = 27/16 > e^{1/4}$, since $e^{1/4} \leq (1 - 1/4)^{-1} = 4/3 < 27/16$. If $t \geq 64 \lceil n \rceil$, the failure probability is at most e^{-4n} . There are at most 2^n points, so a union bound gives $e^{-(4 - \ln 2)n} \leq e^{-3n}$.

A distinguisher can read every incidence bit and test all stars in time $2^{O(n)}$. This is below 2^{σ^δ} for the stated key length. The random-function bound plus PRF indistinguishability proves the claim. Polynomial growth of σ in n makes e^{-3n} negligible in the key length as well. $\square$

K.4 Proof of Corollary 17

Proof of Corollary 17. Intersect the high-rank key event from Theorem 14 with the star-regular event from Lemma 16. Their two negligible exceptional probabilities have negligible sum. Apply Theorem 12 and the polynomial-time entry evaluator. $\square$

K.5 Proof of Proposition 15

Proof of Proposition 15. Take $k = \lceil cn^c \rceil$ and store the I_k independent incidence bits of the scale- k subgrid. Set every other template incidence positive. The proof of Theorem 14, with independent bits in place of the PRF, gives the lower bound and exceptional probability from Lemma 13. Entry evaluation checks subgrid membership and, only there, reads the relevant advice bit.

Collapse all points outside the subgrid into one positive point. The effective domain has $2k^3 + 1$ points and the effective class at most $2k^3 + 1$ rows; every row outside the subgrid is all-positive. A strict template representation in dimension seven is obtained from the original six coordinates by adjoining a coordinate that is zero on old points, one on the added point, and one on all row vectors. The added point vector has its other coordinates zero. An all-positive row already occurs in the subgrid, so the collapsed rows require no additional row sign pattern. Adding it preserves the no-negative- 2×2 property.

Apply the deterministic cap and proper constructions to this polynomial-size effective table. Every effective SQ is simulated by composing its query with the known collapsing map, so its expectation under an arbitrary original marginal is exactly the expectation under the pushforward marginal. The proper output extends by +1 outside the subgrid and belongs to the original class. The query count is $O_\epsilon(\log k) = O_{c,\epsilon}(\log n)$, and the table computation is polynomial in n for fixed c . $\square$

L Sufficient conditions and their limits

The rectangle condition admits a communication interpretation, but neither condition characterizes SQ learnability. Spectral certificates give a separate sufficient condition for hardness. Cube halfspaces and parities distinguish these directions.

L.1 Product-distribution communication

Let $\text{AC}_\times(H)$ be the supremum over product distributions $\mu \times \nu$ of the minimum expected bit cost of an exact deterministic communication protocol for $(x, h) \mapsto h(x)$. The protocol may depend on the two distributions. It must compute the function on every pair, including pairs of zero probability. Leaves of a binary deterministic protocol are monochromatic rectangles.

Theorem 24 (Communication as a sufficient condition). *For every finite class,*

$$\text{rect}(H) \geq 2^{-2\text{AC}_\times(H)-1}, \quad \text{AC}_\times(H) \leq 2 \text{rect}(H)^{-1}.$$

In particular, for every $0 < \epsilon < 1/4$, if $\text{AC}_\times(H) \leq c$, one distribution-independent SQ learner satisfies (2.1) using

$$m = O(4^c(\log |H| + \log(1/\epsilon))), \quad \tau = \Omega\left(\frac{\epsilon}{4^c \log(1/\epsilon)}\right).$$

Proof: see Appendix M.1.

The cube halfspaces in Theorem 32 show non-necessity here too. Under the uniform product, their largest rectangle has mass at most $e^{-n/4}$. The first proof therefore gives expected communication at least $n/(8 \ln 2) - 1/2$, even though they have a distribution-independent polynomial-query learner. This direct consequence needs no Gap-Hamming reduction. Bounded product-average communication and a rectangle ratio bounded below are equivalent up to the displayed nonlinear conversion; neither characterizes SQ learning.

L.2 Spectral certificates and why they miss this

Lemma 25 (Finite energy and threshold counting). *Let P_0 be a probability distribution on a finite set Z , and let $r_1, \dots, r_K \in L^2(P_0)$. Set*

$$\Gamma_{ij} = \mathbb{E}_{P_0} r_i r_j, \quad \Lambda = \|\Gamma\|_{\text{op}}.$$

For every $a : Z \rightarrow \mathbb{R}$,

$$\sum_{j=1}^K (\mathbb{E}_{P_0} r_j a)^2 \leq \Lambda \mathbb{E}_{P_0} a^2. \quad (\text{L.1})$$

In particular, if $|a| \leq 1$ and $\tau > 0$, then

$$|\{j : |\mathbb{E}_{P_0} r_j a| > \tau\}| \tau^2 \leq \Lambda. \quad (\text{L.2})$$

More generally, for any finite list of real numbers c_i , $\tau \geq 0$, and $\sum_i c_i^2 \leq \Lambda$, one has $|\{i : \tau < |c_i|\}| \tau^2 \leq \Lambda$.

Proof: see Appendix M.2.

Theorem 26 (Joint-distribution SQ barrier). *Let μ be a full-support distribution on X and let $P_0 = \mu \otimes \text{Unif}\{-1, +1\}$. For $j \in [K]$, let P_j be the labeled distribution arising from some D_j and $h_j \in \mathcal{H}$. Put*

$$r_j = \frac{P_j}{P_0} - 1, \quad \Gamma_{ij} = \mathbb{E}_{P_0} r_i r_j, \quad \Lambda = \|\Gamma\|_{\text{op}}, \quad R = K/\Lambda.$$

Any randomized adaptive (m, τ) -SQ learner satisfying (2.1) obeys

$$1 - 2\epsilon \leq \sqrt{\frac{\Lambda}{K}} + \frac{m\Lambda}{K\tau^2}, \quad \text{equivalently} \quad \frac{m}{\tau^2} \geq (1 - 2\epsilon)R - \sqrt{R}. \quad (\text{L.3})$$

Consequently, for $\epsilon < 1/4$,

$$R \leq 4 + 4m/\tau^2. \quad (\text{L.4})$$

Proof: see Appendix M.3.

The stated fixed-marginal barrier. Set $D_j = \mu$ for every j . Then $r_j(x, y) = y h_j(x)$ and

$$\Gamma_{ij} = G_{ij} = \mathbb{E}_{\mu} [h_i h_j].$$

For $R = K/\|G\|_{\text{op}} \geq 16$, (L.3) implies

$$m \geq \frac{1}{4} R \tau^2. \quad (\text{L.5})$$

Indeed, $1 - 2\epsilon > 1/2$ and $\sqrt{R} \leq R/4$. The cutoff matters: a uniformly positive lower bound proportional to $R\tau^2$ cannot hold for all tiny families, since a singleton needs no queries. The next theorem supplies the accompanying dimension lower bound.

L.2.1 Spectral sign-rank bounds

We use the spectral method associated with Forster [18], but include its geometric step rather than treating the published theorem as a proof.

Theorem 27 (Spectral sign-rank bounds). *If $A \in \mathbb{R}^{N \times K}$ satisfies $|A_{xj}| \geq 1$ for all entries, then*

$$\text{signrank}(\text{sign } A) \geq \frac{\sqrt{NK}}{\|A\|_{\text{op}}}. \quad (\text{L.6})$$

If $M \in \{-1, +1\}^{N \times K}$ and $G = M^\top \text{diag}(\mu)M$ for any distribution μ on its rows, then

$$\text{signrank}(M) \geq \sqrt{K/\|G\|_{\text{op}}}. \quad (\text{L.7})$$

Proof: see Appendix M.4.

Theorems 26 and 27 prove the valid conditional assertion in the proposed spectral barrier: a certificate with ratio R yields dimension lower bound $\sqrt{R}$ and query lower bound $\Omega(R\tau^2)$ when R is large. They do not prove that all explicit sign-rank lower bounds admit such an ordinary label-Gram certificate. The following family disproves that extra assertion in precisely this form.

L.2.2 Projective planes

Theorem 28 (Projective-plane incidence classes). *For each prime $q \geq 2$, let X_q be the points of the projective plane over $\mathbb{F}_q$, and let $\mathcal{H}_q$ have one hypothesis per line, positive on that line and negative elsewhere. Set $N = q^2 + q + 1$ and $k = q + 1$. Then:*

$$\frac{q+1}{\sqrt{q}} \leq \text{dc}(\mathcal{H}_q) \leq 2q+3, \quad (\text{L.8})$$

$$\frac{|J|}{\|[\mathbb{E}_\mu h_i h_j]_{i,j \in J}\|_{\text{op}}} \leq 16 \quad \text{for every distribution } \mu \text{ and nonempty } J \subseteq \mathcal{H}_q. \quad (\text{L.9})$$

There is a distribution-independent N -query tolerance- τ learner of loss at most τ . For each fixed D and $0 < \epsilon < 1/4$, there is a learner allowed to depend on D , with tolerance $\epsilon/2$ and

$$m_D \leq 1 + s + \binom{s}{2}, \quad s = \left\lceil \frac{4 \log N + 2}{\epsilon} \right\rceil,$$

that has loss at most ϵ for every target. Here $\log$ is natural. Nevertheless every single learner satisfying (2.1), with $m \geq 1$, $0 < \tau \leq 1$, and $\epsilon < 1/4$, satisfies

$$\text{dc}(\mathcal{H}_q) \leq 19m/\tau^2. \quad (\text{L.10})$$

Proof: see Appendix M.5.

Why projective planes do not separate. The family has unbounded dimension while all the proposed ordinary label-Gram ratios remain bounded. This is a genuine failure of the stated universality premise, not a counterexample to OQ2. The efficient learner needed for such a separation is impossible here: the target-dependent balanced marginals produce a joint-distribution ratio of order q . The lower-bound witness changes entry magnitudes, which the ordinary label Gram matrix cannot capture. This also verifies an explicit separation between fixed-distribution and distribution-independent SQ learning: $m_D = O(n^2/\epsilon^2)$ at tolerance $\epsilon/2$, whereas for $q \geq 16$ any common learner needs $m/\tau^2 \geq q/4$. Since $N \leq 2q^2$, the latter is at least $2^{(n-1)/2}/4$. The dimension lower bound is at least $2^{(n-1)/4}$. None of these fixed- D learners can be substituted for the single algorithm in OQ2.

L.2.3 Transcript compression

Theorem 29 (Marginal-branching and approximate probabilistic bounds). *Let $H \subseteq \{-1, +1\}^X$ be finite and nonempty, $m \geq 1$ an integer, $0 < \tau \leq 1$, and $0 < \epsilon < 1/4$. Put $B = \lceil 2/\tau \rceil + 1$. Suppose a depth-at-most- m deterministic (m, τ) -SQ learner satisfies (2.1) for this H and ϵ . Decompose each query as*

$$q(x, y) = a(x) + yb(x), \quad a(x) = \frac{1}{2}(q(x, +1) + q(x, -1)), \quad b(x) = \frac{1}{2}(q(x, +1) - q(x, -1)).$$

Construct its marginal-response tree as follows. If a is constant, retain only the answer equal to that constant. Otherwise retain the B equally spaced answers in $[-1, 1]$. Let u be the maximum number of nonconstant- a nodes on a retained root-to-leaf path. Then

$$\text{dc}(\mathcal{H}) \leq (m + 1)B^u. \quad (\text{L.11})$$

In particular, a deterministic correlational-only learner has $\text{dc}(\mathcal{H}) \leq m + 1$. For an arbitrary randomized (m, τ) -SQ learner with worst-case query budget m satisfying (2.1) under the same hypotheses,

$$\text{dc}^\epsilon(\mathcal{H}) \leq B^m. \quad (\text{L.12})$$

Proof: see Appendix M.6.

The deterministic argument does not derandomize the learner: fixing its seed need not preserve its expected guarantee. The probabilistic bound is exponential, not polynomial, and gives no exact $\text{dc}_{1/2}$ bound.

Proposition 30 (Affine-line quantitative comparison). *For the p^2 affine nonvertical lines on $\mathbb{F}_p^2$, the joint-distribution ratio in Theorem 26 can be chosen as*

$$R = \frac{2p}{1 + (p - 1)^{-2}} \geq p.$$

Thus every common learner of constant expected error $\epsilon < 1/4$ has $m/\tau^2 = \Omega(p)$. More generally, if on every instance it achieves correlation at least $2\gamma > 0$ with probability at least α , then

$$m \geq \tau^2 \left(\alpha R - \frac{1}{4\gamma^2} \right).$$

At Feldman's weak-learning parameters $\tau = 1/t$, $\gamma > 1/t$, $t = (p/32)^{1/4}$ and $\alpha = 2/3$, this gives $m = \Omega(\sqrt{p})$; at constant tolerance and constant advantage it gives $m = \Omega(p)$.

Proof: see Appendix M.7.

L.3 Rectangle geometry: tight examples and non-necessity

Theorem 31 (Global projective-plane rectangle ratio). *For the point-line incidence class of $\text{PG}(2, q)$, over any prime-power order $q \geq 2$ for which the plane is defined,*

$$\frac{1}{8(q + 1)} \leq \text{rect}(H_q) \leq \frac{q}{(q + 1)^2}.$$

Thus the infimum over all product distributions is $\Theta(1/q)$, not merely the maximum rectangle under the uniform product measure.

Proof: see Appendix M.8.

What is and is not tight. Theorem 26 and the balanced target-dependent marginals of Theorem 28 give $m/\tau^2 = \Omega(q) = \Omega(\rho)$ for this family. At a fixed positive tolerance this yields $m = \Omega(\rho)$. Theorem 1, however, uses $\tau = \Theta_\epsilon(1/\rho)$ and has $m/\tau^2 = O_\epsilon(\rho^3 \log K)$. Thus the geometry is sharp and a linear inverse-rectangle dependence is unavoidable at fixed tolerance, but the joint (m, τ) dependence of Theorem 1 is *not* established as tight up to logarithms. Lowering tolerance cannot be ignored in a tightness claim.

Theorem 32 (Small rectangle ratio does not prevent SQ learning). *For odd n , let*

$$X_n = \{-1, +1\}^n, \quad H_n = \{x \mapsto \text{sign}\langle w, x \rangle : w \in \{-1, +1\}^n\}.$$

Then $\text{dc}(H_n) \leq n$ and $\text{rect}(H_n) \leq e^{-n/4}$. For every $0 < \epsilon < 1/4$, one deterministic distribution-independent SQ learner satisfies (2.1) with $m = O(n^3/\epsilon^2)$ and $\tau = \epsilon/(8n)$.

Proof: see Appendix M.9.

For contrast, all parities on $\{-1, +1\}^n$ have an orthonormal label Gram matrix under the uniform marginal: the product of distinct characters is a nonconstant character, whose mean cancels by flipping one coordinate. Theorem 26 therefore gives $m/\tau^2 = \Omega(2^n)$ at any fixed accuracy below $1/4$. The same Walsh orthogonality bounds every uniform monochromatic rectangle mass by 2^{-n} . Small rectangles can therefore occur on either the easy or the hard side of SQ learning.

For an ordinary Gram matrix $G = \mathbb{E}_\mu h h^\top$ on K rows, let $R = K/\|G\|_{\text{op}}$. A monochromatic rectangle $S \times T$ satisfies

$$\mu(S)|T|^2 \leq \mathbb{E}_\mu \left(\sum_{h \in T} h(x) \right)^2 \leq \|G\|_{\text{op}}|T|.$$

Thus its mass under μ times the uniform row prior is at most $1/R$. A large ordinary certificate supplies both small rectangles and the SQ lower bound; small rectangles alone supply neither converse. This distinction is depicted in figure 5.

Why the spectral certificates miss the counterexample. The ordinary-Gram estimate above shows why large rectangles exclude a large fixed-marginal certificate. The balanced projective-plane certificate is a joint-distribution certificate, not one of these bounded ordinary-Gram ratios. For the flip family, Theorems 26 and 1 give $\mathfrak{J}(H_A) = O_\epsilon(\log M)$ while $\text{dc}(H_A) = \Omega(N/\log M)$. This is the promised failure of joint-certificate completeness.

Theorem 33 (Marginal adaptation, under a small-query budget). *Let H_A be a flip class at scale N with $M = N^3$, let $m \geq 1$ be an integer, and let $0 < \tau \leq 1$ and $0 < \epsilon < 1/4$. Every deterministic (m, τ) -SQ learner with worst-case query budget m satisfying (2.1) obeys*

$$u \geq \frac{\ln \text{dc}(H_A) - \ln(m+1)}{\ln(\lceil 2/\tau \rceil + 1)},$$

where u is the marginal-branching depth defined in Theorem 29. In particular, if $\text{dc}(H_A) \geq c_0 N/\log M$ for an absolute $c_0 > 0$, then for any fixed $C > 0$, $m \leq (\log M)^C$ and all sufficiently large N ,

$$u = \Omega_C \left(\frac{\log N}{\log(2 + 1/\tau)} \right).$$

At fixed positive tolerance, the $\Omega(\log N)$ requirement matches the $O_d(\log M)$ marginal-query upper bound of the deterministic finite-net learner, at fixed accuracy.

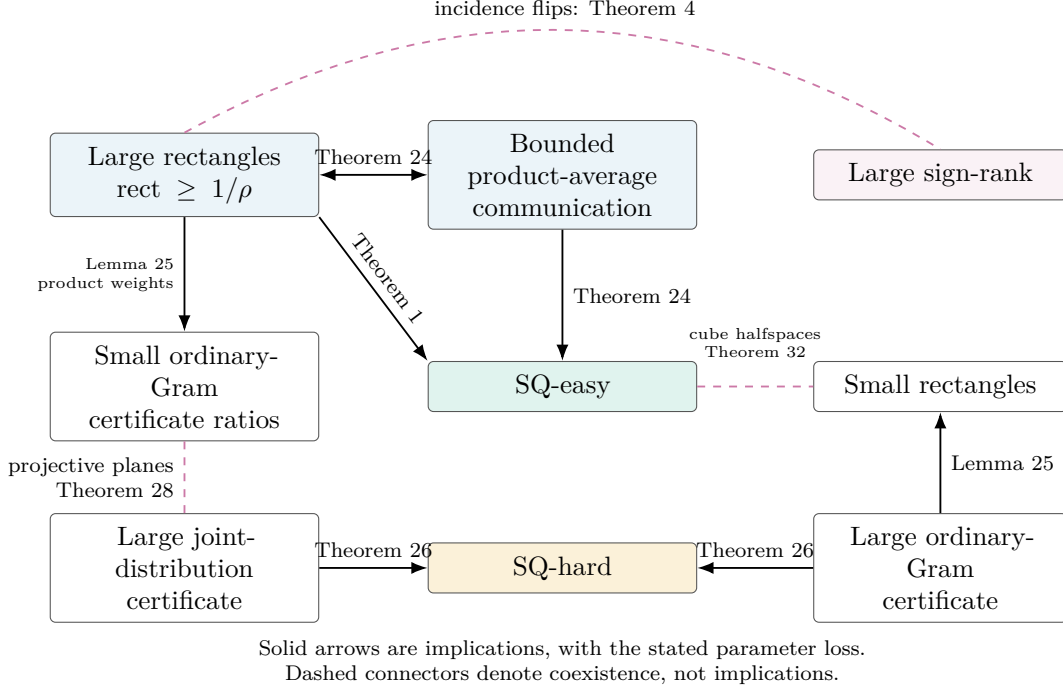


Figure 5: Which resources separate. Small rectangles alone do not imply SQ hardness: cube halfspaces are a counterexample. An ordinary spectral certificate separately forces small rectangles and SQ hardness. Projective planes require the stronger joint-distribution certificate, while incidence flips combine large rectangles with large sign-rank.

Proof: see Appendix M.10.

The query-budget qualification cannot be dropped: querying the correlation with every row is a deterministic correlational learner with $u = 0$ and $m = K$. Randomization is not intrinsically necessary for the separation, because the finite-net enumeration is deterministic. What is necessary for a small-query deterministic learner here is adaptation to marginal information. At inverse-polylogarithmic tolerance the proved lower bound loses a $\log \log M$ factor; no sharper matching statement is made.

L.4 Approximate deterministic embeddings

Corollary 34 (Published approximate rank as an SQ sufficient condition). *Fix $0 < \eta < 1/2$ and $0 < \epsilon < 1/4$. Suppose a total sign matrix has approximate sign-rank at most d in the deterministic-embedding sense of Bindu et al. [8]. Put $\bar{d} = \max\{2, d\}$ and*

$$\rho_\eta = \bar{d}^{C_\eta(\bar{d} + \bar{d}^2)},$$

where C_η is a sufficiently large constant valid in their rectangle theorem. Then

$$\text{rect}(H) \geq \rho_\eta^{-1}, \quad m = O(\rho_\eta(\log K + \log(1/\epsilon))), \quad \tau = \Omega\left(\frac{\epsilon}{\rho_\eta \log(1/\epsilon)}\right)$$

for one distribution-independent learner. Equivalently $\rho_\eta = \bar{d}^{O_\eta(\bar{d}^2)}$. The implicit constant depends only on η .

Proof: see Appendix M.11.

For comparison with randomized SQ dimension, see Appendix S. The construction of RECTIFY is independent of that characterization.

Class	Rectangle ratio	Dimension	Bounds on m/τ^2 and certificate
Incidence flips	$\geq 2^{-15}$	$\Omega(N/\log N)$ for most flips	$O_\epsilon(\log N)$ upper; counting proves high rank.
Projective planes	$\Theta(1/q)$	$[(q+1)/\sqrt{q}, 2q+3]$	$\Omega_\epsilon(q)$ lower; $O_\epsilon(q^3 \log q)$ rectangle upper; joint certificate.
Affine lines over $\mathbb{F}_p$	$\Theta(1/p)$	$\leq 2p+1$	$\Omega_\epsilon(p)$ lower; $O_\epsilon(p^2)$ exhaustive upper; joint certificate.
Cube halfspaces, odd n	$\leq e^{-n/4}$	$\leq n$	$O(n^5/\epsilon^4)$ upper via bounded-margin optimization.
All cube parities	$\leq 2^{-n}$	$\geq 2^{n/2}$	$\Omega_\epsilon(2^n)$ lower and $O_\epsilon(2^n)$ upper; orthogonal Gram matrix.

Table 13: Proved comparison bounds. Query lower bounds refer to fixed $\epsilon < 1/4$ and the adversarial SQ model. Entries give bounds established here, rather than a claim of globally optimal complexity. The affine rectangle estimate follows from the degree proof for Theorem 31 and the affine centered incidence norm.

Comparison of the example families. Table 13 collects the bounds. For affine lines, the degree argument in Appendix M.8 gives $\text{rect} \geq 1/(8p)$. The centered affine incidence matrix has norm $\sqrt{p}$ by Proposition 30; under uniform product weights the same bilinear argument gives rectangle mass at most $1/p$. Each affine row has p exceptional labels, so the moment-curve construction of Theorem 4 gives dimension at most $2p+1$. Exhaustively testing row correlations gives the displayed affine upper bound. For parities, the uniform Gram matrix is the identity and the spectral theorem gives dimension at least $2^{n/2}$; exhaustive testing matches the query lower bound at fixed accuracy.

M Spectral, communication and representation proofs

M.1 Proof of Theorem 24

Proof of Theorem 24. Fix a product distribution and a protocol of expected cost at most $c > 0$. Markov’s inequality gives probability at least $1/2$ to leaves of depth at most $2c$. There are at most $2^{\lfloor 2c \rfloor}$ such leaves: their transcript words are prefix-free, and each can be extended to a distinct word of length $\lfloor 2c \rfloor$. One of their rectangles has mass at least $2^{-\lfloor 2c \rfloor - 1} \geq 2^{-2c-1}$. A zero-cost protocol has a monochromatic support rectangle of mass one, so the same claimed weaker bound holds. Taking the infimum over product distributions proves the first inequality. The finite domain permits a minimum expected-cost exact protocol: redundant repeated-state branches can be removed, bounding depth by a finite function of the domain sizes. Alternatively, use costs within arbitrarily small additive slack and pass to the limit.

For the converse put $r = \text{rect}(H)$. At any current rectangular cell, condition the original product distribution on that cell. Its positive-mass conditional law is still product, and the restricted matrix still has rectangle ratio at least r . Fix a monochromatic rectangle of conditional mass at least r . The parties exchange their two membership bits. If both are one, output its color. Otherwise continue in the indicated one of the three remaining rectangular cells. Every nonempty continuing cell is a proper subcell: a positive-mass rectangle has both sides nonempty, and either a row or point has been excluded. Thus this exact protocol terminates after finitely many stages on every positive-support pair. Zero-mass cells can be completed by any finite exact protocol, without expected cost.

Conditional on reaching a stage with positive probability, its termination probability is at least r . By induction its survival probability after t stages is at most $(1-r)^t$. Summing the tail probabilities, the expected number of stages is at most $1/r$, and each costs at most two bits. This proves the second inequality. This converse is the protocol of Hatami et al. [24, Proposition 3.14], included here with the conditioning and termination details. Substitute $\rho = 2^{2c+1}$ into Theorem 1 for the SQ bounds. $\square$

M.2 Proof of Lemma 25

Proof of Lemma 25. Let $T : \mathbb{R}^K \rightarrow L^2(P_0)$ send b to $\sum_j b_j r_j$. Then $T^*T = \Gamma$. For every unit vector b ,

$$|\langle b, T^*a \rangle| = |\langle Tb, a \rangle| \leq \|Tb\|_2 \|a\|_2 \leq \sqrt{\Lambda} \|a\|_2.$$

Taking the supremum over b and squaring proves (L.1). Each selected summand in (L.2) is greater than τ^2 ; discarding nonnegative unselected summands proves the result. The same argument, with weak inequalities, proves the final statement even when $\tau = 0$. $\square$

M.3 Proof of Theorem 26

Proof of Theorem 26. First $\Lambda > 0$. Indeed, with $a_j(x, y) = y h_j(x)$ we have $\|a_j\|_{L^2(P_0)} = 1$ and $\langle r_j, a_j \rangle = \mathbb{E}_{P_j} a_j - \mathbb{E}_{P_0} a_j = 1$. Thus $\|r_j\|_2 \geq 1$ and $\Lambda \geq \Gamma_{jj} \geq 1$.

For a fixed private seed, run the learner on the reference responses $\mathbb{E}_{P_0} q$. Write the resulting queries as $q_1, \dots, q_m$ and the leaf output as g_0 . This is a counterfactual tree path; the reference need not be realizable. For target j , choose the following memoryless oracle, which does not know the private seed:

$$O_j(q) = \begin{cases} \mathbb{E}_{P_0} q, & |\mathbb{E}_{P_j} q - \mathbb{E}_{P_0} q| \leq \tau, \\ \mathbb{E}_{P_j} q, & \text{otherwise.} \end{cases}$$

It is valid for P_j on every query. If its run first departs from the reference transcript, that departure occurs at one of the reference queries with $|\langle r_j, q_t \rangle| > \tau$. Lemma 25 and a union bound give, for this fixed seed,

$$\frac{1}{K} |\{j : \text{the transcript departs from the reference}\}| \leq \frac{m\Lambda}{K\tau^2}.$$

For the reference leaf, define $c_j = \mathbb{E}_{P_j}[y g_0(x)]$. Since $\mathbb{E}_{P_0}[y g_0(x)] = 0$, Lemma 25 gives

$$\frac{1}{K} \sum_j |c_j| \leq \sqrt{\frac{1}{K} \sum_j c_j^2} \leq \sqrt{\Lambda/K}.$$

Let s_j be the correlation of the actual output under O_j . On a nondeparting transcript it equals c_j ; on a departing transcript it is at most 1. Therefore $s_j \leq |c_j| + \mathbf{1}_{\text{departure for } j}$. Average this inequality over j and then over the private seed. The learner's guarantee gives an average correlation at least $1 - 2\epsilon$. This proves (L.3). Notice that we did not choose a different oracle after seeing the seed.

Finally write $M = m/\tau^2$. From (L.3), $R/2 \leq \sqrt{R} + M$. The square $(\sqrt{R} - 2)^2 \geq 0$ gives $\sqrt{R} \leq R/4 + 1$. Combining yields (L.4). $\square$

M.4 Proof of Theorem 27

Proof of Theorem 27. Let $d = \text{signrank}(M)$, with $M = \text{sign } A$ in the first assertion. Choose a strict representation $M_{xj} = \text{sign}\langle z_x, v_j \rangle$ in $\mathbb{R}^d$. If $d \geq K$, both conclusions are immediate: $\|A\|_{\text{op}} \geq \sqrt{N}$ from any column, while $G_{jj} = 1$ implies $\|G\|_{\text{op}} \geq 1$. We henceforth have $1 \leq d < K$.

Perturb the finitely many v_j within sufficiently small open neighborhoods so that signs are unchanged and every d of them are independent. Such a perturbation exists: for each d -subset the determinant is a nonzero polynomial in the vector coordinates, and a finite union of its zero sets has empty interior. To justify this elementary fact, a nonzero univariate polynomial has finitely many roots; induction on the number of variables, by viewing the polynomial as a polynomial in the last variable, shows that a multivariate polynomial cannot vanish on a nonempty open box. Apply this to the product of the finitely many determinant polynomials.

We prove that there is an invertible T such that the normalized vectors $u_j = Tv_j/\|Tv_j\|$ satisfy

$$\sum_{j=1}^K u_j u_j^\top = (K/d)I_d. \quad (\text{M.1})$$

For every $1 \leq r \leq d$, there is a common $\delta > 0$ such that at most $d - r$ of the v_j have projection norm less than δ onto any r -dimensional subspace. In fact, any $d - r + 1$ of the vectors are independent, so cannot all belong to the orthogonal complement of an r -dimensional subspace. Their maximum projection norm is therefore positive for every such subspace. Its minimum is positive by compactness of the space of orthogonal projection matrices of rank r ; this space is closed and bounded. Take the minimum over finitely many subsets and r .

For positive definite Q with $\det Q = 1$, minimize

$$F(Q) = \sum_{j=1}^K \log(v_j^\top Q v_j).$$

Write the decreasing eigenvalues of Q as $e^{s_1} \geq \dots \geq e^{s_d}$, where $\sum_i s_i = 0$. For each r , at least $K - d + r$ of the quadratic forms are at least $\delta^2 e^{s_r}$. Order the K log quadratic forms increasingly. For the first d entries use respectively the bounds $\log \delta^2 + s_d, \dots, \log \delta^2 + s_1$; each of the remaining $K - d$ entries is at least $\log \delta^2 + s_1$. Thus

$$F(Q) \geq K \log \delta^2 + (K - d)s_1.$$

Since $K > d$, every sublevel set has bounded largest eigenvalue. The determinant condition then bounds the smallest eigenvalue away from zero. Sublevel sets are compact, so a minimizer Q exists. For a symmetric traceless matrix B , the path $Q(t) = Q^{1/2} \exp(tB) Q^{1/2}$ retains determinant one. Its derivative at zero gives

$$0 = \sum_j \frac{v_j^\top Q^{1/2} B Q^{1/2} v_j}{v_j^\top Q v_j} = \text{tr} \left(B \sum_j u_j u_j^\top \right), \quad u_j = \frac{Q^{1/2} v_j}{\|Q^{1/2} v_j\|}.$$

A symmetric matrix orthogonal to every traceless symmetric matrix is scalar: subtract its scalar trace part and take that difference as B . Taking traces gives (M.1), with $T = Q^{1/2}$.

Replace each row vector by the normalization of $T^{-\top} z_x$, denoted e_x , so $\|e_x\| = 1$ and all signs are preserved. Let U be the matrix with columns u_j . For every row of A ,

$$\langle e_x, U A_{x,\cdot}^\top \rangle = \sum_j |A_{xj}| |\langle e_x, u_j \rangle| \geq \sum_j \langle e_x, u_j \rangle^2 = K/d.$$

Here $|\langle e_x, u_j \rangle| \leq 1$ was used. Consequently,

$$NK^2/d^2 \leq \|U A^\top\|_F^2 \leq \|A\|_{\text{op}}^2 \|U\|_F^2 = K \|A\|_{\text{op}}^2.$$

This proves (L.6). For M , use the same row inequality, square, and average under μ :

$$K^2/d^2 \leq \sum_x \mu(x) \|U M_{x,\cdot}^\top\|^2 = \text{tr}(U G U^\top) \leq \|G\|_{\text{op}} \text{tr}(U U^\top) = K \|G\|_{\text{op}}.$$

The last inequality follows by diagonalizing the positive semidefinite G , whose eigenvalues lie between zero and its operator norm. This proves (L.7). $\square$

M.5 Proof of Theorem 28

Proof of Theorem 28. Points are one-dimensional subspaces of $\mathbb{F}_q^3$; lines are kernels of nonzero linear functionals, modulo nonzero scalar multiples. There are $(q^3 - 1)/(q - 1) = N$ of each. A line contains $(q^2 - 1)/(q - 1) = k$ points. Two distinct points span a unique two-dimensional subspace, hence lie on one common line. Two distinct kernels intersect in one one-dimensional subspace. These facts also show each point lies on k lines.

Let P be the $N \times N$ incidence matrix. Then

$$P\mathbf{1} = k\mathbf{1}, \quad P^\top P = qI + \mathbf{1}\mathbf{1}^\top.$$

Define $A = (N/k)P - \mathbf{1}\mathbf{1}^\top$. Its positive entries equal $N/k - 1 = q^2/(q + 1) \geq 1$, and its negative entries equal -1 . Direct multiplication gives

$$A\mathbf{1} = 0, \quad A^\top A = (N/k)^2 q \left(I - \frac{\mathbf{1}\mathbf{1}^\top}{N} \right).$$

Indeed, the coefficient of $\mathbf{1}\mathbf{1}^\top$ before simplification is $N^2/k^2 - N = -Nq/k^2$, since $k^2 = N + q$. Thus $\|A\|_{\text{op}} = (N/k)\sqrt{q}$, and Theorem 27 gives the lower bound in (L.8).

For the upper bound, enumerate the points by distinct integers $1, \dots, N$ and use the common embedding $\phi(t) = (1, t, \dots, t^{2k})$. For a positive set S of size k , the polynomial

$$p_S(t) = \frac{1}{2} - \prod_{a \in S} (t - a)^2$$

is positive on S and negative on all other domain points: away from S the product is a positive integer. Its coefficient vector is the required weight vector. Hence $\text{dc} \leq 2k + 1$.

Now prove (L.9). Write $K = |J|$, and restrict P to the selected columns. If $K < 16$, the Gram matrix has diagonal entries one, so its norm is at least one and the result follows. Suppose $K \geq 16$ and let r_x be the number of selected lines containing point x . If $r_x < K/4$ for every x , take the unit vector $a = \mathbf{1}/\sqrt{K}$. At each point,

$$|\sum_{j \in J} a_j h_j(x)| = |2r_x - K|/\sqrt{K} > \sqrt{K}/2.$$

Its Gram Rayleigh quotient is at least $K/4$. Otherwise choose x_0 incident with $r \geq K/4$ selected lines and let S be those r lines. At x_0 all these labels are positive; at any other point at most one is positive, since two points have a unique common line. For the unit vector $a = \mathbf{1}_S/\sqrt{r}$,

$$|\sum_j a_j h_j(x)| \geq (r - 2)/\sqrt{r} \geq \sqrt{r}/2 \quad \text{for all } x,$$

since $r \geq 4$. Its Rayleigh quotient is at least $r/4 \geq K/16$, for every μ . This proves (L.9), including marginals concentrated on a single point.

The claimed learner queries $q_j(x, y) = y h_j(x)$ for every line and outputs a hypothesis with maximum answer. The true target has correlation one, hence receives answer at least $1 - \tau$. A maximizing hypothesis has true correlation at least $1 - 2\tau$, and loss at most τ , regardless of the marginal and the oracle. Thus it is one distribution-independent learner.

We prove the fixed- D claim before excluding a common efficient learner. Let $\delta = \epsilon/2$ and sample a length- s sequence from the known D . For any pair of hypotheses at D -distance greater than δ , the probability of identical labels on the sequence is at most $(1 - \delta)^s \leq e^{-\delta s}$. A union bound over fewer than N^2 pairs is at most $N^2 e^{-\delta s} \leq e^{-1} < 1$. Consequently some sequence has the property that hypotheses with identical traces are within D -distance δ . Hardwire one such sequence and one representative hypothesis per trace into the learner. On at most s distinct sampled points there are at most $1 + s + \binom{s}{2}$ traces: the empty trace, at most s singleton traces, and at most one line through

each pair of distinct sampled points. Every trace of size at least two is determined by one such pair. The representatives therefore form a D -cover of the claimed size. Query their label correlations and select the largest answer. A representative within δ of the true target has correlation at least $1 - 2\delta$. With tolerance $\tau = \epsilon/2$, the selected output has correlation at least $1 - 2\delta - 2\tau$ and error at most $\delta + \tau = \epsilon$. The hardwired cover depends on D , so the algorithm is distribution-specific.

To rule out a much cheaper common learner, choose for target line j the distribution

$$D_j(x) = \begin{cases} (2k)^{-1}, & P_{xj} = 1, \\ (2(N-k))^{-1}, & P_{xj} = 0. \end{cases}$$

Take the reference P_0 to be uniform on $X_q \times \{-1, +1\}$. With $c = k/(N-k)$, its residual likelihood ratios satisfy

$$r_j(x, +1) = A_{xj}, \quad r_j(x, -1) = -cA_{xj}.$$

For example the first formula is $2ND_j(x) - 1 = N/k - 1$ on the line and -1 elsewhere; the second is -1 on the line and $N/(N-k) - 1 = c$ elsewhere. Consequently

$$\begin{aligned} \Gamma &= \frac{1+c^2}{2N} A^\top A, \\ \Lambda &= \frac{qN}{2} \left(\frac{1}{(q+1)^2} + \frac{1}{q^4} \right), \\ R &= \frac{2}{q((q+1)^{-2} + q^{-4})} \geq \frac{(q+1)^2}{q} \geq q. \end{aligned}$$

The penultimate inequality uses $q^4 \geq (q+1)^2$ for $q \geq 2$, which follows from $q^2 \geq q+1$. By Theorem 26, $R \leq 4 + 4M$ for $M = m/\tau^2$. Therefore

$$\text{dc}(\mathcal{H}_q) \leq 2q + 3 \leq 2R + 3 \leq 11 + 8M \leq 19M,$$

where the final inequality uses $M \geq 1$. □

M.6 Proof of Theorem 29

Proof of Theorem 29. The grid has spacing $2/(B-1) \leq \tau$, so nearest-grid rounding errs by at most $\tau/2$. The marginal tree has at most B^u leaves: suppress unary nodes and use induction on the number of remaining branching nodes per path. It has at most $(m+1)B^u$ nodes, because every node has a descendant leaf and a depth between zero and m , and assigning each node one such pair is injective when the depth is fixed along a leaf's path. Collect the function b at every internal node and the Boolean output at every leaf. These are the common features $f_1, \dots, f_s$, where $s \leq (m+1)B^u$ and each feature has absolute value at most one.

Fix D, h . Follow the tree by giving the exact constant response or rounding $\mathbb{E}_D a$ to the grid. If at some node $|\mathbb{E}_D h b| > \tau/2$, one of the signed features $b, -b$ has correlation greater than $\tau/2$. Otherwise every answer is a valid SQ response: the difference from the true expectation $\mathbb{E}_D a + \mathbb{E}_D h b$ is at most $\tau/2 + \tau/2$. A valid policy realizing this finite transcript extends to all other histories by exact answers. The deterministic guarantee then gives leaf correlation at least $1 - 2\epsilon > 1/2 \geq \tau/2$. We have proved

$$\forall D, \quad \max_{i, s_i \in \{-1, +1\}} \mathbb{E}_D[h(x) s_i f_i(x)] \geq \tau/2. \quad (\text{M.2})$$

Apply Lemma 21 with instance set X and actions the signed features (i, s_i) . Equation (M.2) gives a convex combination whose coordinate at every x is at least $\tau/2$. Its coefficients therefore define a common-feature linear representation with strict target product at every point. This proves (L.11). For correlational-only queries $a = 0$, hence $u = 0$.

For a randomized learner, fix a seed and instead retain all grid answers to each full query. There are at most B^m leaves. Use all leaf outputs as coordinates of an embedding, padding when needed. This distribution over embeddings depends only on the learner's seed, not on D, h . For fixed D, h , the policy rounding each true expectation to the grid is valid. Its output is one of the leaf features. Selecting that coordinate as w gives

$$\inf_w L_{D,h}(\text{sign}\langle w, \phi_\omega \rangle) \leq L_{D,h}(A_\omega^{O_{D,h}}).$$

Averaging over the seed proves (L.12). No seed is selected after observing D or h . $\square$

M.7 Proof of Proposition 30

Proof of Proposition 30. Let P have lines as rows and points as columns, with value one on incidences. Each row has p ones. Two rows with different slopes intersect once; rows with the same slope and different intercepts are disjoint. Hence

$$PP^\top = pI + J - \text{diag}(J_p, \dots, J_p).$$

Set $A = pP - J$. Direct multiplication using $PJ = pJ$ gives

$$AA^\top = p^2(pI - \text{diag}(J_p, \dots, J_p)), \quad \|A\|_{\text{op}}^2 = p^3.$$

For target line h , use probability $1/(2p)$ on each point of the line and $1/(2p(p-1))$ on each other point; labels are positive exactly on the line. Against the uniform fair-label reference, the likelihood-ratio residuals are $r_h(x, +1) = A_{hx}$ and $r_h(x, -1) = -A_{hx}/(p-1)$. Therefore

$$\Lambda = \frac{p}{2}(1 + (p-1)^{-2}), \quad R = p^2/\Lambda.$$

This proves the ratio and the expected-error consequence.

For the success-probability statement use the fixed, valid reference-following oracle from the proof of Theorem 26. For each fixed learner seed, at most fraction $m/(R\tau^2)$ of targets depart from its reference path. Let c_h be the correlation of the reference leaf with target h . The energy lemma gives $\sum_h c_h^2 \leq \Lambda$. Consequently at most fraction $1/(4\gamma^2 R)$ of targets have $c_h \geq 2\gamma$. The fraction of successful actual outputs is no larger than the sum of these two fractions. Average over the learner seed and use the assumed success probability on each target. This gives $\alpha \leq m/(R\tau^2) + 1/(4\gamma^2 R)$ and the stated inequality. Substitution of t yields $(2/3)p/t^2 - 1/4 = \Omega(\sqrt{p})$ in the cited weak regime. This comparison keeps both tolerance and success criteria explicit. $\square$

M.8 Proof of Theorem 31

Proof of Theorem 31. We prove the lower bound for any bipartite incidence graph whose degrees on both sides are at most $\Delta \geq 1$. Fix arbitrary product weights $\mu \times \nu$ and let p be their total incidence mass. If $p \geq 1/2$, some point has incident row mass at least $1/2$. At most Δ rows meet it, so one row has prior mass at least $1/(2\Delta)$. One of that row's two color classes has point mass at least $1/2$. This gives a monochromatic rectangle of mass at least $1/(4\Delta)$.

If $p < 1/2$, select each point independently with probability $t = 1/(2\Delta)$, obtaining S , and let T be all rows with no neighbor in S . The rectangle $S \times T$ is nonincident. For a nonincident pair (x, h) ,

$$\Pr[x \in S, h \in T] = t(1-t)^{\deg h} \geq t(1-\Delta t) = 1/(4\Delta).$$

The inequality $(1-t)^j \geq 1-jt$ follows by induction. Averaging the rectangle mass gives at least $(1-p)/(4\Delta) > 1/(8\Delta)$. Hence some rectangle has that mass. Taking $\Delta = q+1$ proves the universal lower bound.

For the upper bound choose uniform weights. Let $K = q^2 + q + 1$, $k = q + 1$, and let P be the incidence matrix. The identities $P^\top P = qI + \mathbf{1}\mathbf{1}^\top$ and $P\mathbf{1} = k\mathbf{1}$ give

$$C = P - (k/K)\mathbf{1}\mathbf{1}^\top, \quad \|C\|_{\text{op}} = \sqrt{q}.$$

A nonincident rectangle of sizes a, b satisfies $(k/K)ab \leq \sqrt{q}\sqrt{ab}$, so $ab/K^2 \leq q/k^2$. An incident rectangle has a singleton side and area at most k . Its mass is at most $1/K \leq q/k^2$, since $qK - k^2 = q^3 - q - 1 > 0$. This establishes the claimed upper bound on the infimum. $\square$

M.9 Proof of Theorem 32

Proof of Theorem 32. All integer inner products are odd and nonzero, so the identity embedding proves the dimension claim. Distinct parameter vectors give distinct rows: conditioning on all but coordinate i shows

$$\mathbb{E}_x[x_i \text{sign}\langle w, x \rangle] = w_i \Pr[Z_1 + \dots + Z_{n-1} = 0],$$

where the Z_j are independent fair signs and the probability is positive. The conditional difference of the two signs is two exactly when that even sum is zero, and is zero otherwise. Thus uniform parameters are also a uniform prior on the distinct hypotheses. We first bound every rectangle under this uniform product measure. A negative rectangle gives two subsets of the cube with all cross Hamming distances at least $t = (n+1)/2$. For a positive rectangle, negate one of its sides to get the same separation condition. Negation preserves cardinality.

Let A, B be nonempty subsets separated in this way. For uniform Z on the cube, set $f(Z) = \text{dist}_{\text{Ham}}(Z, A)$ and $s = \mathbb{E}f$. Changing one coordinate changes f by at most one. The elementary bounded-differences calculation proved in Appendix U gives

$$\Pr[f - s \leq -a] \leq e^{-2a^2/n}, \quad \Pr[f - s \geq a] \leq e^{-2a^2/n} \quad (a \geq 0).$$

Writing $\alpha = |A|/2^n$, $\beta = |B|/2^n$, if $0 \leq s \leq t$ then

$$\alpha\beta \leq \exp\{-2(s^2 + (t-s)^2)/n\} \leq e^{-t^2/n} \leq e^{-n/4}.$$

If $s \geq t$, already $\alpha \leq e^{-2t^2/n}$ and $\beta \leq 1$. Empty rectangles have zero mass. Taking the maximum under this one product measure proves the rectangle upper bound.

For learning, set $z = x/\sqrt{n}$, $\gamma = 1/n$, and optimize

$$F(v) = \mathbb{E}_D(\gamma - y\langle v, z \rangle)_+ \quad \text{over } \|v\| \leq 1.$$

The target $v^* = w^*/\sqrt{n}$ has $F(v^*) = 0$. At a current v , a valid subgradient is

$$g(v) = \mathbb{E}[-yz \mathbf{1}_{\{y\langle v, z \rangle < \gamma\}}], \quad \|g(v)\| \leq 1.$$

Query each coordinate using the bounded function $q_i(x, y) = -yx_i \mathbf{1}_{\{y\langle v, x/\sqrt{n} \rangle < \gamma\}}$ and divide its answer by $\sqrt{n}$. The resulting vector is $g(v) + e$ with $\|e\| \leq \tau$, regardless of the coordinatewise answer choices.

Start $v_0 = 0$, take $T = \lceil 100/(\epsilon^2 \gamma^2) \rceil$, step $a = T^{-1/2}$, and update by projection of $v_t - a(g_t + e_t)$ onto the unit ball. For any point v^* in that ball, projection cannot increase distance to v^* . Expanding the squared distance and using convexity yields

$$\frac{1}{T} \sum_{t \leq T} F(v_t) \leq \frac{1}{2aT} + \frac{a}{2}(1 + \tau)^2 + 2\tau \leq \frac{5}{2\sqrt{T}} + 2\tau \leq \epsilon\gamma/2.$$

Here $\|v_t - v^*\| \leq 2$, $\tau = \epsilon\gamma/8 < 1$, and $\sqrt{T} \geq 10/(\epsilon\gamma)$. Jensen's inequality gives the same hinge bound for $\bar{v} = T^{-1} \sum_t v_t$. Every classification mistake of $\text{sign}\langle \bar{v}, z \rangle$ incurs hinge loss at least γ , including a zero score when the true label is negative. Thus its error is at most $\epsilon/2$. There are nT scalar queries. This proves a deterministic, adversarial-oracle guarantee stronger than the stated expected-error requirement. $\square$

M.10 Proof of Theorem 33

Proof of Theorem 33. Take logarithms of Theorem 29, $\text{dc}(H_A) \leq (m+1)(\lceil 2/\tau \rceil + 1)^u$. The high-dimension event gives $\ln \text{dc}(H_A) \geq \ln N - \ln \log M + O(1)$. Under the stated query budget the numerator is at least $\frac{1}{2} \ln N$ for large N . The upper bound is Theorem 9. All constants depend only on the displayed fixed parameters. $\square$

M.11 Proof of Corollary 34

Proof of Corollary 34. The published theorem gives a uniform-product rectangle of the displayed product fraction. For rational row and column distributions, duplicate each row and column according to integer denominator multiplicities. Approximate sign-rank does not increase: duplicate columns use the same embedding, and every distribution on copies pushes forward to a distribution on original columns; duplicate rows require no new representation. Apply the rectangle theorem to this duplicated total matrix. Projecting a monochromatic rectangle to the original indices retains its sign and can only increase its product-weighted mass. Finally approximate arbitrary weights by rational weights. The maximum over the finitely many rectangle product masses is continuous, so the lower bound persists. Apply Theorem 1 with $\rho = \rho_\eta$. $\square$

N Gradient models and the prescribed online process

For comparison, the remaining question fixes $X = \{-1, +1\}^n$ and a fully connected bias-free ReLU network

$$f_\theta(x) = \theta_L \sigma(\theta_{L-1} \sigma(\cdots \sigma(\theta_1 x))), \quad S = \sum_{i=1}^L n_i n_{i-1}.$$

Here $n_0 = n$ and $n_L = 1$. Each layer- i weight is initialized independently with law $N(0, 1/n_{i-1})$. With one fixed step size η , for $t = 0, \dots, T-1$ draw $x^{(t)} \sim D$ independently and set

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_{\theta} \ell(h^*(x^{(t)}) f_{\theta^{(t)}}(x^{(t)})), \quad \ell(z) = \log(1 + e^{-z}).$$

The returned predictor is

$$\hat{h}(x) = \text{sign} \left(\sum_{t=\lceil T/2 \rceil}^T f_{\theta^{(t)}}(x) \right).$$

The architecture, step size and horizon are fixed for the class before D and h^* . Open Question 1 of Feldman et al. [17, Section 2] asks whether a guarantee $\mathbb{E} L_{D, h^*}(\hat{h}) \leq \epsilon < 1/4$ for every D, h^* , with expectation over both initialization and online sampling, forces $\text{dc}(H) \leq CTS$. Their Section 4 also asks for a polynomial bound in (T, S) . The open question is restated in Section 8.

A population-gradient method with S coordinates, T calls, per-example coordinate bound B_g and adversarial Euclidean tolerance δ has an SQ simulation with

$$m = TS, \quad \tau = \delta / (B_g \sqrt{S}).$$

Query each gradient coordinate divided by B_g . After rescaling, each error is at most $\delta/\sqrt{S}$, so the total squared error is at most δ^2 . Induction over the calls proves the simulation. Conversely, the adaptive objective $J_q(t; x, y) = tq(x, y)$ has derivative $\mathbb{E} q$. These observations concern gradient-oracle models. A sampled online gradient need not lie in a prescribed small ball about the population gradient, and its sampling-law guarantee does not imply success against every perturbation in that ball; see also Abbe et al. [2], Feldman et al. [16].

Proposition 35 (A bias-free ReLU representation). *Pad the grid domain to a cube of dimension $n = \lceil \log_2(2N^3) \rceil$, giving every padding point label +1. Every member is represented exactly by a bias-free ReLU network with widths $(n, 2n, N + 1, 1)$ and at most*

$$S = 2n^2 + 2n(N + 1) + (N + 1)$$

weights. Its common dimension complexity is at most $2N + 1$. This proposition is an upper bound, not a claim that S parameters are necessary.

The proof is in Appendix P.1.

A fixed b -bit-per-weight architecture has at most 2^{bS} parameter choices. Representing all 2^N independent labelings of one line requires $bS \geq N$.

O The engineered full-batch simulation

The model used by Abbe et al. [2, Theorem 3d] is a directed acyclic neural network with $n + 1$ inputs, including a constant input, one output, and one trainable weight per edge. Its activation is the piecewise-linear “two-stage ramp”

$$\sigma(z) = \begin{cases} -2, & z < -3, \\ z + 1, & -3 \leq z \leq -1, \\ 0, & -1 < z < 0, \\ z, & 0 \leq z \leq 2, \\ 2, & z > 2. \end{cases}$$

Initialization is a specified polynomial-time computable map $W : \{0, 1\}^r \rightarrow \mathbb{R}^p$, rather than an independent Gaussian law. Labels lie in $\{0, 1\}$ and the loss is $\frac{1}{2}(f(x) - y)^2$. Full-batch updates reuse one iid sample S of size m_s , with a fixed constant step size and

$$\theta_{t+1} = \theta_t - \gamma g_t, \quad g_t \in \lambda \mathbb{Z}^p \cap [-1, 1]^p, \quad \left\| g_t - \frac{1}{m_s} \sum_{(x,y) \in S} [\nabla_{\theta} \frac{1}{2}(f_{\theta_t}(x) - y)^2]_1 \right\|_{\infty} \leq \frac{3\lambda}{4}. \quad (\text{O.1})$$

Here $[\cdot]_1$ denotes coordinatewise clipping to $[-1, 1]$. The guarantee is $\mathbb{E}_{S,R} \sup_{(g_t) \text{ admissible}} L(f_{\theta_{T'}})$: expectation over the reused sample and initialization follows the supremum over all admissible rounded-gradient sequences. The supremum may depend on that entire sample. The constructed trajectories are differentiable and have gradient coordinates bounded by one, so clipping leaves them unchanged.

The imported simulation states that an SQ algorithm with k queries, tolerance τ_0 , r random bits and running time TIME has such a simulation with

$$\lambda = \tau_0/16, \quad T' = 2k, \quad m_s \lambda^2 > C(k \log(1/\lambda) + \log(1/\delta)), \quad p = \text{poly}(k, 1/\tau_0, r, \text{TIME}, 1/\delta), \quad (\text{O.2})$$

whose expected square loss exceeds the SQ algorithm’s by at most δ . The constant C is universal. These are the parameters of the full-batch part of that theorem; the sample is not refreshed at each step.

O.1 Application to the succinct learner

Proof of Corollary 18. Use the deterministic improper order-median learner of Theorem 12 at accuracy $\epsilon/8$. It works for every key with $k = O(n^2)$ queries, tolerance $\Omega(\epsilon)$, and polynomial running time

including query evaluation. Recode its labels and predictions from $\{-1, +1\}$ to $\{0, 1\}$. Its square loss is half its classification loss, hence at most $\epsilon/16$ against every valid response sequence. A dyadic tolerance τ_0 within a factor two of the advertised one preserves this guarantee and the asymptotic bounds.

Give the deterministic source learner one ignored random bit and apply (O.2) with $r = 1$ and $\delta = \epsilon/16$. Its initialization map may ignore this bit as well. The sample size and model size are polynomial in n for fixed c, ϵ , and the expected final square loss is at most $\epsilon/8$. A thresholding mistake implies $|f(x) - y| \geq 1/2$ and therefore square loss at least $1/8$. The expected classification loss is at most ϵ . The PRF sign-rank conclusion is Theorem 14; it is independent of the simulation. The deterministic source learner satisfies the simulation's supremum-over-responses premise without exchanging a supremum and an expectation. $\square$

P Proofs for differentiable-learning comparisons

P.1 Proof of Proposition 35

Proof of Proposition 35. The first hidden layer computes $\sigma(x_i), \sigma(-x_i)$ for every coordinate. These give x_i by subtraction and the constant $c = \sigma(x_1) + \sigma(-x_1) = 1$ on the cube. For every negative point p , a second-layer unit computes

$$\sigma\left(\sum_i p_i x_i - (n-1)c\right).$$

At $x = p$ its value is one; at every other cube point the sum is at most $n-2$ and the unit is zero. Pass c through one additional unit and output c minus twice the sum of the negative-point units. Pad unused units by zero weights. There are at most N negative points, and counting all fully connected weights gives the claimed S . The moment-curve proof in Theorem 4 still applies to the padded domain. Padding preserves every dimension lower bound by restriction; it preserves a rectangle lower bound up to a factor two, because either padding carries at least half the point mass or the original domain does. $\square$

P.2 Proof of Proposition 20

Proof of Proposition 20. Given an oracle function, generate the learner's independent uniform point samples on ℓ_0 and label them by its oracle values at the corresponding incidence encodings. Run the learner without giving it a key. Draw an independent uniform test point on ℓ_0 , query its label, and output one if the predictor agrees. In the PRF case acceptance is at least $1/2 + \gamma(n)$. In the random-function case, conditional on the training transcript, a previously unqueried point has an independent fair label. A fresh test point repeats a training point with probability at most $q(n)/N$. Acceptance is therefore at most $1/2 + q(n)/(2N)$. The simulation and one output evaluation are polynomial-time. Since $N = 2^{(n-1)/3}$, the collision term is negligible, contradicting PRF security for inverse-polynomial γ . $\square$

Q Description length and classical certificates

Theorem 36 (Source-description length alone does not repair the implication). *Fix $0 < \epsilon < 1/4$. Interpret ℓ as the bit length of a finite learner program, including literal hard-coded constants and data, but do not charge running time or generated workspace. There is a computable sequence of incidence-flip classes and learners with*

$$n, \ell, m = O(\log N), \quad 1/\tau = O_\epsilon(1), \quad \text{dc}(H) = \Omega(N/\log N).$$

Consequently no universal bound $\text{dc}(H) \leq \text{poly}(m, 1/\tau, n, \ell)$ holds in this interpretation.

The proof is in Appendix R.1.

Proposition 37 (The classical certificate barrier). *For the orientation in which rows are hypotheses, let*

$$m_{\text{avg}}(A) = \inf_{\mu \in \Delta(X)} \sup_{\text{strict unit-vector representations of } A} \min_h \mathbb{E}_{x \sim \mu} |\langle v_h, u_x \rangle|.$$

Then

$$\sqrt{\text{VC}(H)} \leq m_{\text{avg}}(A)^{-1} \leq \text{rect}(A)^{-1}.$$

In particular, if $\text{rect}(A)^{-1} \leq \text{poly}(n)$, these three classical lower-bound parameters cannot certify superpolynomial sign-rank. The bound concerns the three displayed certificates; actual sign-rank can be larger.

The proof is in Appendix R.2.

R Proofs of the resource barriers

R.1 Proof of Theorem 36

Proof of Theorem 36. Let $K = 2N^3$, $b_N = \lceil \log_2(4K) \rceil$, and $d_N = \lfloor N/(128b_N) \rfloor$. For sufficiently large N , the counting bound in Theorem 4 guarantees a flip matrix of sign-rank greater than d_N : indeed $\log_2(4eK) \leq 2b_N$, so

$$2Kd_N \log_2(4eK) \leq N^4/16 < I.$$

Select the lexicographically first such matrix. This selection is computable. For a proposed matrix A , the predicate $\text{signrank}(A) \leq d_N$ is the truth of the finite real sentence

$$\exists U \in \mathbb{R}^{K \times d_N}, V \in \mathbb{R}^{d_N \times K} \bigwedge_{i,j} A_{ij} \sum_{s=1}^{d_N} U_{is} V_{sj} > 0.$$

Decision of finite polynomial real sentences is the classical real-closed-field decision theorem. A constructive treatment of this decidability result is given by Mahboubi and Cohen [29]. Enumerate the finitely many flip tables and decide the displayed sentence for each until the first failure. The counting argument ensures termination. Small N can be handled by any fixed table.

A single fixed program performs this generation and then runs the finite rational rectangle-mixture algorithm. Encode N in $O(\log N)$ bits. For each fixed ϵ , run at a fixed rational accuracy below ϵ , whose description length is constant with respect to N . Conservative rational ceilings can replace the logarithmic constants in Theorem 1, making every step an ordinary finite computation. The table generation happens before the first SQ; it has no query cost. The learner therefore has $m = O(\log K)$ and constant positive tolerance, while the chosen class has $\text{dc} > d_N = \Omega(N/\log N)$. Its program description has size $O(\log N)$. The generated table occupies far more space, which this interpretation does not charge. This contradicts the displayed polynomial implication. $\square$

R.2 Proof of Proposition 37

Proof of Proposition 37. Fix a point prior μ . Apply the finite separation argument of Lemma 21 to the transposed matrix, obtaining a distribution λ over monochromatic rectangles $S_R \times T_R$ with

$$\forall h, \quad \mathbb{E}_R[\mathbf{1}_{T_R}(h)\mu(S_R)] \geq r, \quad r = \text{rect}(A).$$

In the Euclidean space indexed by rectangles, put

$$a_h(R) = c_R \sqrt{\lambda_R} \mathbf{1}_{T_R}(h), \quad b_x(R) = \sqrt{\lambda_R} \mathbf{1}_{S_R}(x).$$

Their norms are at most one, and their inner product is A_{hx} times the probability that R covers (h, x) . These products may be zero, so this alone is not a strict representation. Add an orthogonal strict unit-vector representation scaled by $\sqrt{\delta}$, and scale the displayed vectors by $\sqrt{1 - \delta}$. Every resulting product is strict with the required sign, and its average absolute value in each row is at least $(1 - \delta)r$. Pad row norms and point norms separately in orthogonal private coordinates to make them one without changing any cross inner products. Letting $\delta \downarrow 0$ proves $m_{\text{avg}}(A) \geq r$.

For the VC bound, suppose v points are shattered and use their uniform point prior. In any strict unit-vector representation, select one row h_σ for each sign pattern $\sigma \in \{-1, +1\}^v$. If the minimum row-average margin is a , then

$$va \leq \sum_{i=1}^v \sigma_i \langle v_{h_\sigma}, u_i \rangle \leq \left\| \sum_{i=1}^v \sigma_i u_i \right\|.$$

Averaging the square of the last norm over independent fair signs gives v , because cross terms cancel. Some sign pattern has norm at most $\sqrt{v}$, so $a \leq 1/\sqrt{v}$. Take the supremum over representations and the infimum over priors. For $v = 0$ the inequality is immediate. This proves the chain, including the strict-sign correction. $\square$

R.3 Proof of Proposition 19

Proof of Proposition 19. Condition on the sampled point indices, their observed labels, and the learner's random coins. The label at any unobserved point of ℓ_0 remains an independent fair bit. The learner's prediction there has conditional error $1/2$. A fixed point is unobserved with probability $(1 - 1/N)^T$. Average over the N points to get the first inequality. The second follows by induction from $(1 - z)^T \geq 1 - Tz$ for $0 \leq z \leq 1$. Rearranging proves the sample requirement. For the final claim, $T > N/2$ and $S \geq 2$ give $TS \geq N$, whereas $\text{dc}(H_A) \leq 2N + 1 \leq 3N$ for every table. $\square$

S Comparison with randomized statistical dimension

The rectangle theorem also bounds Feldman's randomized statistical dimension. The main learner does not use this characterization. For the search task $\mathcal{Z}_\eta$, an instance is a realizable labeled distribution $P_{D,h}$; an allowed solution is a predictor g with loss at most η . Fix any reference distribution P_0 on $Z = X \times \{-1, +1\}$, not necessarily realizable or fair-labeled. For a law $\mathcal{P}$ on predictors, put

$$\mathcal{B}(\mathcal{P}) = \{j : \Pr_{g \sim \mathcal{P}}[g \text{ is } \eta\text{-good for } j] < 1/4\}.$$

A query distribution covers this bad set at level $1/d$ when every j in it is distinguished from P_0 by more than κ with probability at least $1/d$. Taking the best $\mathcal{P}$ and query law, and the worst P_0 , is the covering formulation of Feldman's $\text{RSD}_{\kappa_1}(\mathcal{Z}_\eta, \kappa, 1/4)$.

Theorem 38 (Rectangle bound for randomized statistical dimension). *Suppose H is finite and nonempty, $\rho \geq 1$, and $\text{rect}(H) \geq 1/\rho$. Let $L = \lceil \rho \ln(4/\eta) \rceil$ and $\kappa = \eta/(4L)$, where $0 < \eta < 1/4$. Then*

$$\text{RSD}_{\kappa_1}(\mathcal{Z}_\eta, \kappa, 1/4) \leq 4L.$$

This statement allows arbitrary target-dependent marginals and any reference P_0 .

Proof. Fix a finitely supported prior π on instances. It induces a prior ν on hypotheses. Let $\mathcal{Q}$ be the finite query family $q_{S,c}(x, y) = \mathbf{1}_S(x)\mathbf{1}_{y \neq c}$. Set

$$\beta = \max_{q \in \mathcal{Q}} \Pr_{j \sim \pi} [|P_j q - P_0 q| > \kappa].$$

Peel L rectangles using the prior-averaged example marginal restricted to the current unpeeled set U , and the unchanged hypothesis prior ν . Empty residual mass permits immediate completion. The rectangle condition gives both

$$\nu(T) \geq r, \quad \mathbb{E}_j D_j(S) \geq r \mathbb{E}_j D_j(U).$$

Suppose first $\beta < r$. On every instance with target in T , the mismatch query has expectation zero. Therefore $P_0 q \leq \kappa$: otherwise a prior mass at least r would be distinguished. Apart from a prior mass at most β , we then have $P_j q \leq 2\kappa$.

After L peels, the expected residual mass is at most $(1 - r)^L \leq \eta/4$. Markov's inequality shows that at least half the prior puts residual mass at most $\eta/2$. Outside a union of exceptional sets of total mass at most $L\beta$, each of the L wrong-color masses is at most 2κ . The predictor assembled from the rectangle colors and an arbitrary residual label consequently has loss at most $2L\kappa + \eta/2 = \eta$ for prior mass at least $1/2 - L\beta$. If $\beta \geq r$, then $L\beta \geq L/\rho \geq \ln(4/\eta) > 1/2$, so the following inequality remains true trivially:

$$\max_g \Pr_{\pi}[g \text{ good}] + L \max_{q \in \mathcal{Q}} \Pr_{\pi}[q \text{ distinguishes}] \geq 1/2. \quad (\text{S.1})$$

Apply Lemma 21 to the action pairs (g, q) , whose payoff on instance j is

$$\mathbf{1}_{g \text{ good for } j} + L \mathbf{1}_{q \text{ distinguishes } j}.$$

It gives laws $\mathcal{P}, \mathcal{Q}'$ satisfying

$$\Pr_{g \sim \mathcal{P}}[g \text{ good for } j] + L \Pr_{q \sim \mathcal{Q}'}[q \text{ distinguishes } j] \geq 1/2 \quad \text{for every } j.$$

On $\mathcal{B}(\mathcal{P})$, the second probability is greater than $1/(4L)$. This proves the required coverage.

The continuum of possible marginals creates no compactness gap in this minimax step. Both the predictor set and $\mathcal{Q}$ are finite. An instance is relevant only through the finite vector of Boolean good-solution and distinguishing-query indicators. There are finitely many such types. Choose one representative of each realized type and apply the finite argument to them. Its inequalities then hold for all original instances. No discretization in total variation and no factor polynomial in $|X|$ is introduced. $\square$

Comparison with Feldman's characterization. In Feldman [15, Theorem 4.8], the randomized upper bound, in the notation used here, is

$$\text{RQC}(\mathcal{Z}, \text{STAT}(\kappa/3), \alpha - \delta) = O\left(d \frac{R_{\text{KL}}(\mathcal{D})}{\kappa^2} \log \frac{R_{\text{KL}}(\mathcal{D})}{\kappa\delta}\right), \quad d = \text{RSD}_{\kappa_1}(\mathcal{Z}, \kappa, \alpha). \quad (\text{S.2})$$

The displayed asymptotic expression uses the usual positive-radius conventions; a ceiling and a maximum with one handle degenerate values. Here $R_{\text{KL}}(\mathcal{D}) = \inf_Q \sup_{P \in \mathcal{D}} \text{KL}(P \| Q)$. Taking uniform Q on Z proves $R_{\text{KL}} \leq \ln |Z|$: expand $\text{KL}(P \| Q) = \ln |Z| + \sum_z P(z) \ln P(z)$ and use $\ln P(z) \leq 0$. Thus the relevant domain term is $\ln(2|X|)$, not $|X|$ or the logarithm of a discretized instance count. The deterministic theorem has a different set-cover logarithm; it must not be substituted here.

For clarity, the mechanism behind (S.2) does not depend on a hidden finite-instance enumeration. Maintain a reference Q starting uniform on Z . A query separated from the true P by κ supplies a

bounded direction with a signed expectation gap of order κ . For $f \in [-1, 1]$ and update $Q'(z) \propto Q(z)e^{\lambda f(z)}$, direct expansion gives

$$\text{KL}(P\|Q') - \text{KL}(P\|Q) = -\lambda \mathbb{E}_P f + \log \mathbb{E}_Q e^{\lambda f} \leq -\lambda(\mathbb{E}_P f - \mathbb{E}_Q f) + \lambda^2/2.$$

The final inequality follows by twice differentiating $\log \mathbb{E}_Q e^{\lambda f}$: its second derivative is a variance of a variable in $[-1, 1]$, at most one. Choosing λ proportional to κ decreases KL by $\Omega(\kappa^2)$. There are therefore $O(\ln |Z|/\kappa^2)$ updates. At each reference, sampling $O(d \ln(T/\delta))$ covering queries prevents an undetected bad instance over T updates, by $(1 - 1/d)^s \leq e^{-s/d}$ and a union bound. This proves the parameter dependence and extends the argument to all realizable marginals here.

At fairness $1/4$ and constant $\delta < 1/4$, this yields constant success probability. Fresh repetitions and a single loss query to verify each candidate amplify success with $O(\ln(1/\delta'))$ repetitions and loss slack $O(\kappa)$. This reasoning remains valid for history-dependent oracles by conditioning on earlier runs. The verifiable-search Theorem 4.11 in Feldman [15] is stated using a different statistical dimension over references admitting no good solution; its dimension must not be identified automatically with the fairness- $1/4$ quantity above. In either route verification does not erase the R_{KL}/κ^2 term of that generic simulation. The direct learner of Theorem 1 gives the sharper bound without it.

T A finite-bit convex program for the cap implementation

This appendix proves the weakly polynomial implementation needed in Theorem 9. It does not require a claim that an exact radial-isotropic matrix has rational entries. Approximate Forster computation is established prior work; see Diakonikolas et al. [11] for stronger algorithms. We verify the particular general-position convex program and the bit bounds needed here.

Let $k > d$ and let $v_1, \dots, v_k \in \mathbb{Q}^d$ be in linear general position. Assume the numerator and denominator of every coordinate have absolute value at most 2^b , with nonzero denominators. Let Δ_I be the determinant of the rows indexed by a d -subset I . Every Δ_I is nonzero. Cauchy–Binet gives the convex function

$$f(t) = \log \det \left(\sum_i e^{t_i} v_i v_i^\top \right) - \frac{d}{k} \sum_i t_i = \log \sum_{|I|=d} \Delta_I^2 e^{\sum_{i \in I} t_i} - \frac{d}{k} \sum_i t_i. \quad (\text{T.1})$$

The identity follows by applying the determinant expansion to the product of the $d \times k$ column matrix with entries $e^{t_i/2} v_i$ and its transpose; only distinct d -subsets remain. Differentiating the log-sum-exp expression shows that its Hessian is the covariance matrix of the incidence vector $\mathbf{1}_I$ under the displayed positive weights. It is positive semidefinite and has operator norm at most $\mathbb{E} \|\mathbf{1}_I\|^2 = d$. Also $\|\nabla f\| \leq 2\sqrt{d}$. Thus f is convex, d -smooth, and invariant under adding a constant to all coordinates.

We give a polynomial radius bound on the zero-sum gauge. Write $a_- = \min_I \log \Delta_I^2$ and $a_+ = \max_I \log \Delta_I^2$. For $t_1 \geq \dots \geq t_k$ with $\sum t_i = 0$, put $s = t_1 + \dots + t_d$. The average of the largest $d-1$ of $t_2, \dots, t_k$ is at least their overall average; hence

$$s \geq t_1 \frac{k-d}{k-1} \geq \frac{t_1}{k-1}.$$

Likewise the largest d of $t_1, \dots, t_{k-1}$ have average at least $-t_k/(k-1)$, giving $s \geq -dt_k/(k-1)$. Therefore $s \geq \|t\|_\infty/(k-1)$, and

$$f(t) \geq a_- + \frac{\|t\|_\infty}{k-1}, \quad f(0) \leq a_+ + \log \binom{k}{d}.$$

Sublevel sets on the gauge are compact. A minimizer exists and has infinity norm at most $(k-1)(a_+ - a_- + \log \binom{k}{d})$.

The determinant bounds are explicit. A common denominator for a $d \times d$ determinant is the product of its d^2 coordinate denominators, so

$$2^{-bd^2} \leq |\Delta_I| \leq d!2^{bd}.$$

Consequently the integer

$$R = 1 + (k-1)(2bd^2 + 2bd + 2d\lceil \log_2(d+1) \rceil + d\lceil \log_2(k+1) \rceil)$$

exceeds the minimizer radius. Work on the rational polytope $Q = \{t : \sum t_i = 0, |t_i| \leq 2R\}$.

For completeness, a deliberately conservative projected-subgradient implementation suffices. Set

$$\delta = \frac{d}{512k^2}, \quad a = \frac{\delta}{36d}, \quad \xi = \frac{\delta}{32Rk}, \quad T = \left\lceil \frac{4R^2k}{a\delta} \right\rceil.$$

At each step compute a rational approximation $\tilde{g}$ of ∇f with Euclidean error at most ξ , and project $t - a\tilde{g}$ onto Q . The projection is rational and polynomial-time: it is $\text{clip}(z_i - \lambda, -2R, 2R)$ in each coordinate, with λ chosen to make the sum zero. The sum is a continuous piecewise-linear decreasing function; sorting its $2k$ breakpoints and solving on the containing interval finds λ . The projection inequality follows directly from its normal-cone signs: for every $u \in Q$, $\langle z - \Pi_Q z, u - \Pi_Q z \rangle \leq 0$, so $\|\Pi_Q z - u\| \leq \|z - u\|$.

Let t^* be a minimizer with $\|t^*\| \leq R\sqrt{k}$, and start at zero. The standard distance expansion, supplied here to account for error, gives

$$\frac{1}{T} \sum_{s < T} (f(t_s) - f(t^*)) \leq \frac{R^2k}{2aT} + \frac{a}{2}(2\sqrt{d} + \xi)^2 + 4R\sqrt{k}\xi.$$

Indeed convexity bounds $f(t_s) - f(t^*)$ by $\langle \nabla f(t_s), t_s - t^* \rangle$, the gradient error contributes at most $4R\sqrt{k}\xi$, and squared distances telescope. Since $\xi \leq \sqrt{d}$, each of the three terms is at most $\delta/8$. Convexity gives $f(\bar{t}) - \min f < \delta$ for the rational average $\bar{t}$.

Smoothness on the full space implies

$$\|\nabla f(\bar{t})\|^2 \leq 2d(f(\bar{t}) - \min f) < \frac{d^2}{256k^2} :$$

apply the quadratic upper bound to $\bar{t} - \nabla f(\bar{t})/d$ and compare with the global minimum. Put $S = \sum_i e^{\bar{t}_i} v_i v_i^\top$. Differentiating the first expression in (T.1) yields

$$\frac{15d}{16k} < \ell_i := e^{\bar{t}_i} v_i^\top S^{-1} v_i < \frac{17d}{16k}.$$

For any factor T_0 with $T_0 S T_0^\top = I$, the normalized row covariance satisfies

$$\frac{16}{17d} I \preceq \frac{1}{k} \sum_i \frac{T_0 v_i v_i^\top T_0^\top}{\|T_0 v_i\|^2} \preceq \frac{16}{15d} I.$$

To see this, write each summand as $e^{\bar{t}_i} T_0 v_i v_i^\top T_0^\top / \ell_i$ and use the preceding bounds together with their unnormalized sum I .

All calculations can be performed to the required accuracy with polynomially many bits, rather than an unspecified exact-real oracle. For fixed d , the number of terms $\binom{k}{d}$ in (T.1) is polynomial. The gradient is a ratio of sums of the nonnegative numbers $\Delta_I^2 \exp(\sum_{i \in I} t_i)$. Their exponent arguments have absolute value at most $2dR$, and their rational coefficients have polynomial bit length by the determinant bounds. Exponentials on this bounded range admit rational interval approximations

with bit complexity polynomial in $R, b, k, \log(1/\xi)$: for nonnegative arguments use the Taylor series and its positive tail bound, and for negative arguments take reciprocals. The elementary inequality $j! \geq (j/e)^j$, obtained by integrating $\log x$, bounds the number of Taylor terms by a polynomial in the argument bound and target precision. Positive sums have no cancellation; the denominator has the explicit lower bound $2^{-2bd^2} e^{-2dR}$. Compute each gradient coordinate to error at most $\xi/(2k)$ and round it to one fixed dyadic grid of spacing at most $\xi/(2k)$. The total coordinate error is at most ξ/k , giving the required Euclidean error. These grid values have polynomial bit length independent of the iteration index. Exact rational projection introduces only the denominators of the fixed step size and a divisor at most k at each iteration, so projection and averaging retain polynomial bit length over the polynomial number of iterations. Thus the preceding inexact-gradient implementation has a polynomial bit-operation bound for fixed d .

Finally, approximate T_0 by an invertible rational matrix T . This also requires only polynomial precision. The bounds

$$\lambda_{\max}(V^\top V) \leq kd 2^{2b}, \quad \lambda_{\min}(V^\top V) \geq \frac{2^{-2bd^2}}{(kd 2^{2b})^{d-1}}$$

follow from trace, determinant, and Cauchy–Binet. Since $|\bar{t}_i| \leq 2R$, multiplying these bounds by $e^{\pm 2R}$ bounds the eigenvalues of S . Their logarithms have polynomial size. Interval Cholesky factorization followed by inversion, approximating each square root by rational bisection, consequently computes T_0 to $2^{-q(k,b,R)}$ operator error in polynomial time for a sufficiently large fixed polynomial q . This last assertion can also be checked directly: Cholesky pivots are bounded below by the smallest eigenvalue, all divisions have polynomial logarithmic conditioning, and there are only $O(d^3)$ operations. If $\|a - b\| \leq \|a\|/2$, then

$$\|a/\|a\| - b/\|b\|\| \leq 2\|a - b\|/\|a\|.$$

The displayed eigenvalue bounds and the nonzero rational input coordinates therefore give an explicit polynomial precision at which the normalized covariance changes by less than $1/(8d)$. Its lower eigenvalue remains above $3/(4d)$. For rational T , this final statement is independently decidable by rational arithmetic: the covariance is the sum of $Tv_i v_i^\top T^\top / (v_i^\top T^\top T v_i)$, and positivity above $3I/(4d)$ is checked using leading principal minors. This completes the finite-bit construction.

SQ replies themselves need no unlimited precision in the computational corollary. Request half the advertised tolerance and round the reply to a rational grid of spacing at most that tolerance. The effective error remains within the advertised tolerance. This changes only an absolute constant and uses $O(\log(1/\tau))$ reply bits. Each cap test is a fixed-degree rational inequality; the unpeeled set is represented by the negations of earlier peeling tests. No table-dependent advice beyond the stated input is hidden in this computation.

U The cube concentration calculation used for non-necessity

Let f on the uniform n -cube change by at most one when one coordinate is flipped. Reveal the coordinates successively and write the Doob differences as D_i . Conditional on the first $i-1$ coordinates, the two conditional expectations after revealing coordinate i differ by at most one: couple the remaining coordinates identically and use the bounded-difference hypothesis. Thus D_i is conditionally centered and lies in an interval of length at most one.

For any centered random variable Z in an interval of length one, let $F(\lambda) = \log \mathbb{E} e^{\lambda Z}$. Differentiation of the finite sum gives $F''(\lambda)$ equal to the variance under the exponentially tilted law. A variable $Y \in [a, b]$ has variance at most $(b-a)^2/4$: from $(Y-a)(b-Y) \geq 0$ obtain $\text{Var}(Y) \leq (\mathbb{E}Y - a)(b - \mathbb{E}Y) \leq (b-a)^2/4$. Hence $F'' \leq 1/4$, and integrating twice using $F(0) = F'(0) = 0$ yields $F(\lambda) \leq \lambda^2/8$ for either sign of λ .

Iterating conditional expectation gives

$$\mathbb{E} \exp\{\lambda(f - \mathbb{E}f)\} \leq \exp(n\lambda^2/8).$$

Markov's inequality and $\lambda = 4a/n$ imply $\Pr[f - \mathbb{E}f \geq a] \leq e^{-2a^2/n}$. Apply the same calculation to $-f$ for the lower tail. All distributions here are finite, so differentiation and conditional expectations require no limiting justification.

V Additional geometric bounds

Proposition 39 (Independent constant rectangle bound). *Let c_d be the normalized surface measure of the cap of Section 5.1 on S^{d-1} , $d \geq 2$. Every matrix F of sign-rank at most d satisfies $\text{rect}(F) \geq c_d/(4d)$. If F has no negative 2×2 , every monotone flip A satisfies $\text{rect}(A) \geq c_d/(8d)$. In particular, for $d = 6$ this is greater than 2^{-18} .*

Proof. First take a uniform prior on $k > d$ row copies. Perturb their representing vectors, if necessary, so every d are independent while preserving the finitely many strict signs. The full minimization argument in Theorem 27 produces an invertible transform whose normalized row vectors are isotropic. Apply the inverse transpose to the point vectors and normalize; all signs persist.

For each uniform random direction u , the two sets $T_+(u) = \{v : \langle u, v \rangle > \alpha\}$ and $T_-(u) = \{v : \langle u, v \rangle < -\alpha\}$ have combined prior mass at least $1/(2d)$. The rectangles $C_u \times T_+(u)$ and $C_u \times T_-(u)$ have constant signs. For any fixed point marginal μ , rotational invariance gives $\mathbb{E}_u \mu(C_u) = c_d$, because each unit point lies in a random cap with probability c_d . Therefore the expected sum of their product masses is at least $c_d/(2d)$, and some single rectangle has mass at least $c_d/(4d)$.

Apply Lemma 22, repeating all row copies further so their number exceeds d . Representations of identical copies may differ, but their signs agree with the original entries; projection therefore preserves monochromaticity. A negative rectangle has a singleton side under the additional assumption on F . Split its other side according to the new signs in A ; one color retains at least half its product mass. Positive rectangles survive unchanged. This proves the second assertion.

For the numerical estimate at $d = 6$, project the cap onto its last five coordinates. Its image is the five-dimensional ball of radius $\alpha = 1/\sqrt{12}$, and the surface-area Jacobian is at least one. The volume of the unit five-ball is $8\pi^2/15$ and the area of S^5 is π^3 , obtained by polar integration of the Gaussian integral. Thus

$$c_6 \geq \frac{8}{15\pi 12^{5/2}} > \frac{8}{15 \cdot 4 \cdot 576} = \frac{1}{4320}, \quad \frac{c_6}{48} > \frac{1}{207360} > 2^{-18}.$$

This independent constant, rather than the sharper imported one, is already sufficient for Theorems 1–5 to refute every polynomial dimension implication. $\square$

Proposition 40 (Uniform-product rectangles in a projective plane). *For the point–line incidence sign matrix of $\text{PG}(2, q)$, let $K = q^2 + q + 1$ and $k = q + 1$. Under uniform distributions on both sides, the largest monochromatic-rectangle mass lies between*

$$\frac{1}{8(q+1)} \quad \text{and} \quad \frac{q}{(q+1)^2}.$$

It is therefore $\Theta(1/q)$.

Proof. Let P be the incidence matrix. As proved in Theorem 28, $P\mathbf{1} = k\mathbf{1}$ and $P^T P = qI + \mathbf{1}\mathbf{1}^T$. Thus $C = P - (k/K)\mathbf{1}\mathbf{1}^T$ has operator norm $\sqrt{q}$. A nonincident rectangle with side sizes a, b satisfies

$$\frac{k}{K}ab = |\mathbf{1}_S^T C \mathbf{1}_T| \leq \sqrt{q}\sqrt{ab},$$

so its mass is at most q/k^2 . An all-incident rectangle has a singleton side and size at most k . Its mass k/K^2 is also at most q/k^2 : use $K \geq qk$ and $q^3 \geq k$ for $q \geq 2$ to obtain $qK^2 \geq q^3k^2 \geq k^3$.

For the lower bound, choose $s = \lfloor K/(2k) \rfloor$ lines. Their union has at most $sk \leq K/2$ points, leaving at least $K/2$ points incident with none of them. The resulting rectangle has mass at least $s/(2K)$. Since $K/(2k) \geq 1$, $s \geq K/(4k)$, proving the lower bound. $\square$

W Related work and imported formulations

SQ learning and dimension. Statistical-query learning accesses bounded expectations rather than individual examples [28]. Feldman’s randomized search characterization supplies a general dimension of this access model [15]; our use of it is confined to Appendix S. The question of Feldman et al. [17] asks whether one learner serving every marginal must admit a common low-dimensional linear representation. Our rectangle learner answers negatively without invoking the characterization. Euclidean representation complexity originates in Ben-David et al. [7]; probabilistic and approximate versions were developed by Chornomaz et al. [10] and Kamath et al. [26]. The exact and averaged variants require the distinct reductions in Theorem 5 and Lemma 6. In particular, uniform approximation on the sparse random matrix would not prove the averaged-dimension lower bound.

Correlational queries and the converse direction. The source note records a margin-complexity characterization of distribution-independent weak correlational learning [17, Section 3], citing Feldman [13]. Strong correlational learning is more restrictive: Feldman [14] proves hardness for conjunctions at vanishing target error. These two accuracy regimes are distinct. General SQ halfspace algorithms use rescaling and marginal information [6, 12]. The rescaling simulation depends polynomially on dimension, inverse accuracy and the logarithm of the inverse margin; finite-grid margin bounds control the latter through coordinate bit complexity [21]. The negative result here concerns reversing the implication without requiring a low-dimensional or bounded-margin representation.

Prior random-feature bounds. Karchmer and Malach [27] study approximate dimension complexity, random features and distribution-free differentiable learning. The source note states that their claimed upper bounds do not hold, attributing the correction to a 2026 personal communication from those authors [17, Section 3]. The probabilistic statement in Corollary 7 fixes one embedding law before both the target and marginal. Under these quantifiers it rules out an unrestricted polynomial bound from common SQ learnability to the nondegenerate averaged dimension. This comparison is about the precisely stated implication, rather than an alternative marginal-dependent random-feature distribution.

Sign-rank, margin and geometry. Forster’s normalization method provides spectral lower bounds [18]. The construction and the separation between sign-rank, VC dimension, average margin and rectangle ratio are due to Hatami et al. [24], building on Alon et al. [5]. We import their grid construction and Warren’s general strict-pattern theorem [33]; we derive the application-specific counting and probabilistic lower bounds. The homogeneous-rectangle theorem of Alon et al. [4] is used only to sharpen the inverse rectangle constant to 2^{15} . Our cap proof supplies an independent dimension-dependent constant. The minimax rectangle-mixture idea occurs in the average-margin argument of Hatami et al. [24, Theorem 3.1]; the new step here is its adversarial-oracle peeling/elimination analysis and bounded query budget. Approximate Forster transforms are prior algorithmic work [11]; Appendix T gives the fixed-dimensional implementation needed here.

The finite-field separation of 2017. Feldman [15, Theorem 7.7] studies affine lines on $\mathbb{F}_p^2$ and proves a distribution-independent weak-learning lower bound of $t/2 - 1$ queries at tolerance $1/t$, where $t = (p/32)^{1/4}$, for the stated accuracy and success-probability regime. Theorem 7.8 there gives a separately fixed-distribution learner with $O(1/\epsilon^2)$ queries of tolerance $\epsilon^2/13$. For the affine matrix, Proposition 30 applies the joint certificate to his affine class. At constant expected error it gives $m/\tau^2 = \Omega(p)$, hence $m = \Omega(p)$ at constant tolerance. The proposition also supplies the exact success-probability comparison at his tolerance. These are quantitative refinements of the hardness side; his class is not a small-complexity distribution-independent learner and therefore is not the present counterexample. Our real-grid monotone flips lie on the easy side of that universal-marginal requirement.

Approximate sign-rank, May 2026. Bindua et al. [8, Theorem 1.3] extend the large-rectangle property to approximate sign-rank and partial matrices. Their approximate rank fixes one *deterministic* column embedding and, for each row and each marginal, permits a new separating vector with error at most η . It is not the random-embedding quantity dc^η used here. Their theorem supplies row and column fractions $d^{-C_\eta d}$ and $d^{-C_\eta d^2}$, respectively, with C_η stated as $O(\log(1/(1 - 2\eta)))$, for fixed $0 < \eta < 1/2$; one may enlarge the constant and use $d \geq 2$. Corollary 34 converts this published geometric theorem into a common SQ learner with the displayed parameters. For partial matrices their rectangle is homogeneous on nonstar entries, possibly containing stars; we do not silently regard it as a full monochromatic rectangle of an arbitrary completion.

Index and list replicability, June 2026. Blondal et al. [9] prove $\text{Ind}_{\mathbb{Z}_2}(H) \leq 2 \text{LR}(H) - 1$ and show that projective-plane incidence classes have list replicability at most three and index at most five despite unbounded sign-rank. Our joint-distribution theorem shows that those same projective-plane classes are SQ-hard when a common learner must serve every marginal. Thus bounded list replicability or bounded index is not the missing SQ upper-bound criterion. Their separation concerns different parameters and does not resolve the question answered here. Earlier and concurrent topological bounds [19, 25], and the total Hamming-distance upper bounds of Göös et al. [22], likewise do not give the sought efficient explicit high-sign-rank selection for the constant-rectangle flip family.

Concurrent response-rank result. The version-specific comparison is given in the introduction.

Learning versus communication and differentiable learning. The converse communication construction in Theorem 24 is Hatami et al. [24, Proposition 3.14]; its composition with RECTIFY yields the SQ sufficient condition and the halfspace boundary case. The proper completion and the order-median succinct learner exploit the real incidence geometry rather than invoking a generic protocol-to-learning conversion. The differentiable-learning simulations of Abbe et al. [2] use engineered models and initialization and charge the simulated computation in their network-size bounds. They do not preserve the remaining prescribed benign-SGD algorithm. The hidden-key bounds in Section 7 explicitly distinguish uniform key-independent learners from class-dependent architecture choices.

Neural dynamics and staircase structure. Malach et al. [30] compare gradient learning with tangent kernels under specified model conditions. The staircase analyses of Abbe et al. [1] and Abbe et al. [3] identify hierarchical structures learnable by particular regular-network dynamics under their distributional assumptions. Their conclusions concern those dynamics and data regimes. Corollary 18 instead composes a common SQ learner with the full-batch simulation of Abbe et al. [2]; its engineered architecture and initialization retain the simulation’s computational cost. The prescribed Gaussian-initialized online-SGD question therefore remains a separate problem.

X Experiments and reproducibility

All experiments use CPU computation. The public key is

8959fe4a6b404807275400d0a95800e62d0b
119c2f8e0061d0ff5d241b229c78

with the two lines concatenated. The experimental flip bit is the low bit of the first byte of HMAC-SHA256 on the key and the message consisting of `SQDC4`, a zero byte, and the unsigned 64-bit big-endian tuple (N, a, b, x, y) . Off-incidence entries remain positive. The random-search seed is 202609104. This specifies the experimental keyed-hash instantiation. The cryptographic theorems separately use the family and ambient encoding in Section 6.

Oracle and distribution access. The learners receive only the entry evaluator and SQ answers; the simulator alone knows the chosen target and rational distribution. Every query records its support mask, exact expectation, returned rational answer and public control state. The adversarial policy chooses a tolerance endpoint using the complete public transcript and the current query. For threshold decisions it chooses an endpoint that favors premature stopping, ambiguous median comparisons, a high-error candidate acceptance, or continued search when available, according to the explicit priority rules in the scripts. The cap marginal policy cannot inspect a private row side or color not revealed by the query. The policy is the explicit public-history endpoint adversary implemented in the scripts. Exact-oracle runs are reported separately in the raw outputs.

Structured distributions include uniform template points, actual negative points, one heavy negative point with a small tail, and mixed positive/negative supports. The searches use random restarts followed by coordinate changes of rational weights, with support at most 20. The order-median sweep uses three restarts and four mutations; the cap sweep uses two restarts and three mutations. Reporting uses fresh learner seeds. Every recorded error is recomputed against the actual target, and every recorded proper output is checked for class membership.

Succinct order-median learner. The script `seeded_fast.py` executes the proper algorithm of Theorem 12 for $N = 2, \dots, 8, 16, 32$. It does not construct a truth table or enumerate all domain points inside the learner. The simulator represents D by its finite support. The stated accuracy is $\epsilon = 1/10$, and the exact-rational experiments use effective tolerance $\zeta = \epsilon/4096 = 1/40960$. The finite-bit theorem requests raw tolerance $\zeta/2$ to reserve the other half for response rounding; these runs have no additional rounding error.

N	Runs	Largest error	Most queries	Max. wall (ms)	Slope-search runs
2	16	0.0010000000	8	0.235	0
3	36	0.0010000000	11	0.425	0
4	36	0.0143149285	12	0.168	0
5	36	0.0243217961	14	0.394	0
6	36	0.0010000000	28	0.393	2
7	36	0.0010000000	15	0.252	0
8	36	0.0010000000	16	2.993	0
16	36	0.0010000000	52	0.915	12
32	36	0.0010000000	89	1.368	16

Error and wall-time entries are rounded upward. Wall time includes the exact finite-support oracle, not the cost of a physical statistical oracle. Raw output also records simulator time separately. The entries are maxima from the fresh median sweep archived under `audits/current/rerun_results/fast`. All 304 runs passed 4,798 exact query-answer checks; all 304 outputs were proper. The 30 slope-search runs specifically exercise the order comparison used to resolve the all-light case.

The small grid sweep cannot reach the large-star sampling branch because its conservative threshold is at least $128/\epsilon = 1280$. A separate unit test therefore uses $N = 4096$ and the point $p = (1, N + 1)$, whose star has 4096 lines. Exactly 1968 of these keyed rows are negative at p , exceeding the required local quarter fraction. On 20 learner seeds, a distribution putting mass .7 at p and .3 at a positive outside point produced a proper zero-error answer. Full transcripts are saved. The test generates and verifies this star without constructing the full $N = 4096$ table.

Keyed Proper RECTIFY. The script `run_keyed_caps.py` runs the proper cap learner on the same keyed family for $N = 2, \dots, 8$, using only the public-history endpoint oracle. It uses outer accuracy $1/10$, rectangle-phase accuracy $1/40$, $\rho = 2^{20}$, and tolerance $1/23226329280$. A numerical radial-isotropic optimizer proposes a finite catalog of directions and rectangles. Before that catalog is used, the integer truth table verifies every rectangle and the exact pointwise coverage of the discrete mixture. Thus the implemented coverage guarantee does not rely on rounding a numerical covariance residual in the favorable direction.

N	Runs	Largest error	Most rounds	Most queries	Max. wall (s)
2	8	.0062500001	63	133	.010
3	8	.0062500001	75	159	.031
4	8	.0085570466	198	409	.105
5	8	.0062500001	393	798	.203
6	8	.0062500001	645	1308	.462
7	8	.0062500001	2881	5779	1.446
8	8	.0062500001	749	1521	3.838

The 56 recorded runs passed 28,262 exact query-answer checks. Their 499 catalogs passed 231,150 monochromatic-rectangle checks and 228,684 exact pointwise coverage checks. Every returned hypothesis belonged to the class. The entire keyed-cap experiment, including search, took 63.35 seconds in the recorded environment. Query types and the peel, elimination and rejection counts appear in the aggregate CSV and full transcripts. The much smaller query count of the order-median method is expected: it uses the grid order rather than sampling from a conservative generic rectangle law.

Subgrid counts and elementary checks. The following exact counts enumerate the independent incidence positions. The rank threshold from Lemma 13 is zero at all these small scales, so its low-rank event is empty. The table records finite construction counts and the zero rank threshold at these scales.

k	$K_k = 2k^3$	I_k	D_k	$k^4/2$
2	16	23	0	8
3	54	126	0	40.5
4	128	412	0	128
5	250	1025	0	312.5
6	432	2151	0	648

The script `algebra_and_stress.py` passed 280 exact rational endpoint checks for the new median, split and orientation inequalities. It also exhausts all label patterns and all sample-index sequences for $N \in \{2, 3, 4\}$ and $T \in \{0, 1, 2, 3\}$: memorization with a constant fallback on unseen points achieves exactly $\frac{1}{2}(1 - 1/N)^T$ on average in each of the 12 cases. The affine certificate script additionally checks the incidence and Gram identities over five prime fields and the corresponding success-probability substitution. The experiments concern statistical-query algorithms.

Later-result checks. The electronic ledger for Appendix B contains 77 rows across the nine rechecked results. The fresh exact checks cover 162 endpoint inequalities, 4,080 nonmonotone median paths, 1,000 prefix-suffix identities, 1,000 fixed-bit star samplers, 3,487 ambient-encoding comparisons, 5,000 response-grid cases, 17,476 halfspace color reductions and 720 unseen-label bounds. Separate floating-point checks cover the logarithmic Warren exponents, PRF parameter arithmetic, packing exponents and hinge constants. For $N = 2, \dots, 8$, the same-slope construction in Appendix G is checked to fit N^2 target rows exactly.

The median sweep, endpoint, unseen-label, large-star, affine and parameter scripts were rerun from fresh directories for this release. The keyed-cap table records the preserved cap sweep. The repository keeps both datasets under distinct paths. Run `python tools/run_checks.py` from the repository root for the fresh checks and median sweep; add `-caps` to repeat the keyed-cap sweep. The exact commands, dependency versions and output manifests are supplied with the source.